27 Pathophysiology of Acute Kidney Injury

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CHAPTER OUTLINE

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PATHOPHYSIOLOGY OF CLINICAL ACUTE KIDNEY INJURY

The three major pathophysiologic categories—namely, prerenal, intrinsic, and postrenal (obstructive)—provide a framework for understanding the mechanisms of acute kidney injury (AKI).

PRERENAL ACUTE KIDNEY INJURY

Prerenal azotemia is the most common cause of AKI, accounting for approximately 40% to 55% of all cases.[1-3] It results from kidney hypoperfusion due to reductions in actual or effective arterial blood volume (EABV—the volume of blood effectively perfusing the body organs). Common conditions causing true hypovolemia include hemorrhage (traumatic, gastrointestinal, surgical); gastrointestinal losses (vomiting, diarrhea, and nasogastric suction); renal losses (overdiuresis and diabetes insipidus); and "third space" fluid accumulation (e.g., pancreatitis and hypoalbuminemia). In addition, cardiogenic shock, septic shock, cirrhosis, hypoalbuminemia, and anaphylaxis all are pathophysiologic conditions that decrease EABV, independent of total body volume status, resulting in reduced renal blood flow. Prerenal azotemia reverses rapidly if renal perfusion is restored because, by definition, the integrity of the renal parenchyma and cell viability has remained intact. However, severe and prolonged hypoperfusion may result in tissue ischemia and cell death, leading to acute tubular necrosis (ATN). Therefore prerenal azotemia and ischemic ATN represent different points on the spectrum of renal hypoperfusion manifestations.

Prerenal azotemia has also been divided into volume-responsive and volume-nonresponsive forms. The former is easy to comprehend, whereas the latter is less straightforward. In volume-nonresponsive forms, additional intravenous volume is of no help in restoring kidney perfusion and function. Disease processes such as congestive heart failure, liver failure, and sepsis may not respond to intravenous fluids because markedly reduced cardiac output and/or reduced total vascular resistance prevent improved kidney function.

True or effective hypovolemia causes a decrease in mean arterial pressure that activates baroreceptors and initiates a cascade of neural and humoral responses, leading to activation of the sympathetic nervous system and increased production of catecholamines, especially norepinephrine. There is increased release of antidiuretic hormone, mediated primarily by hypovolemia, resulting in vasoconstriction, water retention, and urea back diffusion into the papillary interstitium. In response to volume depletion or states of decreased EABV, there is increased intrarenal angiotensin II (Ang II) activity via activation of the renin-angiotensin-aldosterone system (RAAS). Ang II is a potent vasoconstrictor that preferentially increases efferent arteriolar resistance, preserving glomerular filtration rate (GFR) in the setting of decreased renal perfusion, hence increasing the filtration fraction, through the maintenance of glomerular hydrostatic pressure. In addition, Ang II increases proximal tubular sodium absorption through a combination of alterations in hydrostatic forces in the peritubular capillaries and through direct activation of sodium-hydrogen exchangers. During severe volume depletion, Ang II activity is even greater, leading to afferent arteriolar constriction, which reduces renal plasma flow, GFR, and the filtration fraction, and markedly augments proximal tubular sodium reabsorption in an effort to restore plasma volume.[4] Ang II has also been shown to have direct effects on transport in the proximal tubule through receptors located in the proximal tubule. It has also been postulated that the proximal tubule can produce Ang II locally. Hence under conditions of volume depletion, Ang II stimulates a larger fraction of tubular transport, whereas volume expansion will blunt this response.[5-9]

Renal sympathetic nerve activity is significantly increased in prerenal azotemia. In the setting of hypovolemia, adrenergic activity independently constricts the afferent arteriole and changes the efferent arteriolar resistance through Ang II. α_1-adrenergic activity primarily influences kidney vascular resistance, whereas renal nerve activity is linked to renin release through β-adrenergic receptors on renin-containing cells. In contrast, α_2-adrenergic agonists primarily decrease the glomerular ultrafiltration coefficient via Ang II. Although vasodilation might be expected in response to acute removal of adrenergic activity, a transient increase in Ang II is actually seen, maintaining GFR and renal blood flow. Even after subacute renal denervation, renal vascular sensitivity to Ang II increases due to robust upregulation of Ang II receptors. Hence complex effects on renin-angiotensin activity occur within the kidney secondary to increased renal adrenergic activity during prerenal azotemia.[10,11]

All these systems work together and stimulate vasoconstriction in musculocutaneous and splanchnic circulations. They inhibit salt loss through sweat, as well as the stimulation of thirst, collectively causing retention of both salt and water to maintain blood pressure and preserve cardiac output and cerebral perfusion. Concomitantly, there are compensatory mechanisms to preserve glomerular perfusion.[12] Autoregulation is achieved by stretch receptors in afferent arterioles that cause vasodilation in response to reduced perfusion pressure. Under physiologic conditions, autoregulation works only to a mean systemic arterial blood pressure of 75 to 80 mm Hg. Below this level, the glomerular ultrafiltration pressure and GFR decline abruptly. Renal production of prostaglandins, kallikrein, kinins, and nitric oxide (NO) is increased, contributing to vasodilation.[13,14] Nonsteroidal antiinflammatory drugs (NSAIDs), by inhibiting prostanoid production, worsen kidney perfusion in patients with hypoperfusion. Selective efferent arteriolar constriction, a result of Ang II, helps preserve the intraglomerular pressure and hence the GFR. Angiotensin-converting enzyme (ACE) inhibitors inhibit the synthesis of Ang II and therefore disturb this delicate balance in patients with severe reductions in EABV, such as severe congestive heart failure, or bilateral renal artery stenosis and, in these settings, can worsen prerenal azotemia. On the other hand, high levels of Ang II, as seen in circulatory shock, cause constriction of both afferent and efferent arterioles, negating its protective effect.

Although these compensatory mechanisms minimize the progression toward AKI, they, too, are overcome in states of severe hypoperfusion. Renovascular disease, hypertensive nephrosclerosis, diabetic nephropathy, and older age predispose patients to prerenal azotemia[15] at lesser degrees of hypotension.[15] Prerenal azotemia also predisposes patients to radiocontrast media–induced AKI and interventions such as anesthesia and surgery result in further decreases in renal blood flow. Therefore it is imperative to diagnose prerenal azotemia promptly and initiate effective treatment because it is a potentially reversible condition that can lead to ischemic ATN and/or nephrotoxic AKI if therapy is delayed or the severity of the underlying condition increases. In patients with advanced liver disease and portal hypertension, hepatorenal syndrome represents an extreme form of prerenal disease, characterized by peripheral and splanchnic vasodilation, with intense intrarenal vasoconstriction unresponsive to volume resuscitation.[16-18] AKI can also result from abdominal compartment syndrome, characterized by a marked elevation in intraabdominal pressure, resulting in a clinical presentation with features similar to those of prerenal AKI.[19,20] Using laser microdissection to isolate specific domains of the kidney, followed by RNA sequencing and gene expression profiling in distinct animal models of volume-responsive versus intrinsic AKI, has revealed that different signal transduction pathways were operative in the two contrasting models. Volume-responsive genes rapidly reverse with volume resuscitation, whereas intrinsic AKI genes did not change. These results suggest that volume-dependent AKI is not an attenuated form of intrinsic AKI.[21]

INTRINSIC ACUTE KIDNEY INJURY

DISEASES OF LARGE VESSELS AND MICROVASCULATURE

Total occlusion of the renal artery or vein is an uncommon event but can be seen in certain scenarios such as trauma, instrumentation, thromboemboli, thrombosis, and dissection of an aortic aneurysm. Renal artery stenosis is a slow, chronic process, with or without evidence of declining GFR, and rarely presents as an acute event. Renal vein thrombosis has classically and frequently been associated with hypercoagulable states, including nephrotic syndrome, particularly when associated with membranous nephropathy. An atheroembolic source should be considered in patients who present with AKI after instrumentation with angiography, arteriography, thrombolysis, aortic surgery, or even after blunt trauma or acceleration-deceleration injury.[22] Atheroembolic plaques in the aorta or other larger arteries may become disrupted, and fragments may become trapped in smaller renal arteries, leading to hypoperfusion and an intense inflammatory reaction, akin to vasculitis. Other organs may also be affected, leading to gastrointestinal ischemia, peripheral gangrene, livedo reticularis, and acute pancreatitis. Patients frequently develop fevers and exhibit eosinophilia, an elevated erythrocyte sedimentation rate, and hypocomplementemia, which sometimes help in differentiating this condition from other simultaneous insults (e.g., transient hypotension and/or radiocontrast media administration).

Renal artery thrombosis is usually a posttraumatic or postsurgical complication, especially in the transplantation setting, but can also occur in other hypercoagulable states, such as antiphospholipid antibody syndrome.[23-25] Diseases affecting the large vessels, including Takayasu arteritis,[26] and small vessels, including polyarteritis nodosa, necrotizing granulomatous vasculitis, hemolytic-uremic syndrome, thrombotic thrombocytopenic purpura, and malignant hypertension, are generally termed *vasculitides*. They tend to occlude vessels by fibrin deposition, along with platelets. Endothelial cell damage leads to an inflammatory response in the renal microvasculature (and in other organs), leading to reduced macrovascular and microvascular blood flow and tissue ischemia, sometimes giving rise to superimposed ATN. One should keep in mind the intricate relationship among these inflammatory vasculitides and subsequent ischemic injury because even though the origin of these disease processes is located at a site distant from the tubules, the final result is often ATN if not treated early. Hence, virtually any disease that compromises blood flow within the renal microvasculature can induce AKI.

DISEASES OF THE TUBULOINTERSTITIUM

Ischemic and septic ATN are the most common causes of intrinsic AKI. These are discussed extensively in later sections of the chapter on ATN. Other disorders of the tubulointerstitium causing AKI, such as acute allergic interstitial nephritis, drug-induced tubular toxicity, and endogenous toxins, are presented in the following sections.

Interstitial disease

Acute interstitial nephritis (AIN) results from an idiosyncratic hypersensitivity response to different pharmacologic agents, most commonly to antibiotics (e.g., methicillin and other penicillins, cephalosporins, sulfonamides, and quinolones), NSAIDs (e.g., ibuprofen and naproxen),[27] chemotherapeutic agents,[28] and proton pump inhibitors.[29] Although evidence supports the role of vancomycin as a tubule toxin, most biopsy reports have described AIN.[30] The expanding list of newer targeted agents has led to more cases of acute tubulointerstitial injury. Serine-threonine protein kinase inhibitors such as vemurafenib and dabrafenib target a protein called B-Raf and have been associated with acute and chronic tubulointerstitial damage.[31] Other agents that cause acute tubulointerstitial injury include checkpoint inhibitors such as ipilimumab, nivolumab, and pembrolizumab. There are three major types of immune checkpoint inhibitors used in clinical practice: anti-programmed cell death receptor-1 (PD-1) antibodies, anti-programmed cell death ligand-1 (PD-L1) antibodies, and anti-cytotoxic T lymphocyte–associated antigen-4 (CTLA-4) antibodies.[32] PD-1 is expressed on T cells, and PD-L1 is expressed on normal and cancer cells. Binding of PD-1 ligand to its receptor (PD-L1) blocks immune elimination by T cells. Cancer cells evade immune surveillance in part by this mechanism. Similarly, CTLA-4 is another checkpoint protein expressed on T cells that leads to self-tolerance, and its blockade leads to various forms of autoimmune injury including acute interstitial injury.[33]

Other conditions such as leukemia, lymphoma, sarcoidosis, bacterial infections (e.g., *Escherichia coli*), and viral infections (e.g., cytomegalovirus) can also cause AIN, leading to AKI. Systemic allergic signs such as fever, rash, and eosinophilia are often present in antibiotic-associated AIN but are not usually present in NSAID-related AIN, in which lymphocytes tend to predominate.[34] The presence of inflammatory infiltrates within the interstitium is the key hallmark of AIN. These inflammatory infiltrates are often patchy and present most commonly in the deep cortex and outer medulla. Interstitial edema is typically seen with the infiltrates, and sometimes patchy tubular necrosis may be present in close proximity to areas with extensive inflammatory infiltrates.[27] The composition of cells in the interstitial infiltrate can suggest the cause of AIN. The presence of frequent eosinophils suggests a diagnosis of drug-induced AIN, whereas a high number of neutrophils favor bacterial infection, and a high number of plasma cells is dominant in immunoglobulin G4 (IgG4)–related tubulointerstitial nephritis and in renal allografts infected with polyomavirus.[35] Most cases of AIN are probably induced by extrarenal antigens being produced by drugs or infectious agents that may be able to induce AIN by the following: 1. binding to kidney structures; 2. modifying the immunobiology of native renal proteins; 3. mimicking renal antigens; or 4. precipitating as immune complexes and hence serving as the site of antibody- or cellular-mediated injury.[36] This reaction is triggered by many events including activation of complement and release of inflammatory cytokines by T cells and phagocytes. In other cases, loss of tolerance from checkpoint inhibitors leads to immune-mediated inflammation and injury.[28]

Tubular disease—exogenous nephrotoxins

Nephrotoxic ATN is the second most common cause of intrinsic AKI. We shall briefly review the common drug nephrotoxicities in the context of AKI. The kidneys are vulnerable to toxicity due to the high blood flow, and they are the major elimination and metabolizing routes of many of these nephrotoxins. Furthermore, because of the medullary tonicity, the concentration of drugs within the tubular lumen increases along the nephron, exposing the tubules to toxic levels for a more prolonged exposure time. Several well-known therapeutic agents, such as amphotericin B, vancomycin, aminoglycosides, acyclovir, indinavir, cidofovir, foscarnet, pentamidine, and ifosfamide, can directly cause acute tubular injury and associated AKI.

Radiocontrast media–induced nephropathy. The administration of iodinated contrast material may be associated with AKI. Historically, when AKI occurred after administration of iodinated contrast material, it was referred to as contrast-induced nephropathy (CIN); however, because it is often not possible to exclude other causes of AKI, the term contrast-associated AKI (CA-AKI) has been more commonly used. The incidence of CA-AKI is controversial and varies from <1% to >30% with little consensus.[37] In some patients with moderate to advanced CKD, the risk of CA-AKI may be as high as 50%. Other risk factors include diabetes mellitus, effective arterial volume depletion, use of high-osmolality contrast media, advanced age, proteinuria, and anemia.[38,39]

Unlike many other forms of intrinsic tubular injury, radiocontrast medium–induced AKI is usually associated with urinary sodium retention and fractional excretion of sodium (FE_{Na}) of <1%. AKI resulting from iodinated contrast media is typically nonoliguric and rarely requires dialysis. However, requirements for renal support, prolonged hospitalization, and increased mortality are associated with (although not necessarily caused by) this condition.

The pathophysiology of CA-AKI likely consists of combined hypoxic and toxic renal tubular damage associated with renal endothelial dysfunction and alterations in the microcirculation.[40,41] The administration of radiocontrast media causes vasoconstriction and markedly affects renal parenchymal oxygenation, especially in the outer medulla, as documented in various studies where the cortical PO_2 declined from 40 to 25 mm Hg and the medullary Po_2 fell from 26 to 30 mm Hg to 9 to 15 mm Hg.[41-43] Radiocontrast media injection leads to an abrupt but transient increase in renal plasma flow, GFR, and urinary output.[44] This effect is due to the hyperosmolar radiocontrast medium–enhancing solute delivery to the distal nephron and leads to increased oxygen consumption by enhanced tubular sodium reabsorption. Video microscopy has shown that radiocontrast media markedly reduce inner medullary papillary blood flow, even to the extent of near-cessation of red blood cell (RBC) movement in papillary vessels, associated with RBC aggregation within the papillary vasa recta.[45] In isolated vasa recta from rats and humans contrast medium applied to the lumen has led to constriction and enhanced vasa recta responses to Ang II.[46,47] However, it should be noted that there may be different patterns of response possibly related to the type, volume, and route of radiocontrast medium administration. Numerous neurohumoral mediators may contribute to the changes in renal microcirculation caused by radiocontrast medium injection. Intrarenal NO synthase activity, NO concentration, plasma endothelin, adenosine, prostaglandins, and vasopressin are all thought to play a role in altering the cortical and medullary microcirculation after radiocontrast medium injection. Mechanical factors may also play a role because radiocontrast media increase blood viscosity and may affect the flow in the complex, low-pressure medullary microcirculation.[43] An increased plasma viscosity after radiocontrast medium administration can interfere with blood flow, particularly under the hypertonic conditions of the (inner) renal medulla, where the plasma viscosity is already increased as a result of hemoconcentration. Indeed, several animal studies have shown a correlation between experimental CIN and viscosity of the radiopaque compound.[48,49]

Evidence also suggests direct tubular toxicity from radiocontrast media. Radiocontrast media has direct toxic effects on proximal tubular cells (PTCs) in vitro.[50] Radiocontrast media (e.g., diatrizoate and iopamidol) induced a decline in tubule K^+, adenosine triphosphate (ATP), and total adenine nucleotide contents. At the same time, there was a decrease in the respiratory rate of the tubules and an increase in intracellular Ca^{2+} content. These changes were more pronounced with the very high-osmolality ionic compound diatrizoate than the lower osmolality nonionic iopamidol. Importantly, cytotoxic effects were aggravated by hypoxia, indicating interactions between direct cellular mechanisms and vasoconstriction-mediated hypoxia.[51] Radiocontrast media cause release of tubular marker enzymes, ultrastructural changes, and cell death in both Madin-Darby canine kidney (MDCK) and porcine kidney proximal tubule epithelial (LLC-PK_1) cells.[52] Radiocontrast medium–induced critical medullary hypoxia may lead to formation of reactive oxygen species (ROS), with subsequent membrane and DNA damage. A vicious cycle of hypoxia, free radical formation, and further hypoxic injury may be activated after radiocontrast medium exposure. Clinically, CA-AKI presents as an acute decline in the GFR within 24 to 48 hours of administration, with a peak serum creatinine concentration usually occurring in 3 to 5 days and return to baseline within 1 week, although return to baseline serum creatinine concentration may take longer in patients with moderate to advanced CKD. Existing CKD, diabetic nephropathy, advanced age, congestive heart failure, volume depletion, and coincident use of NSAIDs also increase the risk for CIN.

Unlike the first-generation contrast media, high-osmolar contrast media (HOCM; osmolality of 1500–1800 mOsm/kg), low-osmolar contrast media (LOCM; osmolality of 600–850 mOsm/kg), and isosmolar contrast media (IOCM; osmolality of 270–320 mOsm/kg) are associated with a lower incidence of contrast-induced AKI.[53-55] However, whether the incidence of contrast-induced AKI is lower in IOCM than LOCM is still debated.[53-56] One factor that may mitigate against the theoretic value of IOCM versus LOCM is that IOCM is a dimer and has a higher viscosity than LOCM. Viscosities for HOCM, LOCM, and IOCM are 0.00275, 0.00525, and 0.0114 pascal-second (Pa·s) at 37°C, respectively.[30] According to Poiseuille's law, the greater the viscosity, the greater the resistance to fluid flow and the greater the shear stress on the vascular endothelium. Following intravenous injection, contrast medium becomes diluted in the bloodstream, the viscosity and osmolality are reduced, and therefore nonkidney organs are exposed to low concentrations of contrast medium. In the kidney, however, because of the increase in medullary osmolality, the concentration of contrast medium increases in the peritubular capillary and, following filtration, the concentration of contrast rises in the tubule lumen. The consequence is that 1. the distal tubules are exposed to increasing concentration and viscosity of contrast medium and 2. the tubule flow rate decreases, leading to prolonged exposure to contrast, potentially enhancing direct tubule nephrotoxicity.[30,57] Furthermore, when infused in animals, medullary PO_2 is lower when IOCM is utilized when compared with LOCM.[38]

Aminoglycoside nephrotoxicity. The nephrotoxicity of aminoglycosides has best been characterized for gentamicin, a polar drug excreted by glomerular filtration. It is thought that cationic amino groups (NH_3^+) on the drug bind to anionic phospholipid residues on the brush border of PTCs, and the drug is then internalized by endocytosis. Although the precise cellular mechanisms responsible for renal accumulation of aminoglycosides have not been fully elucidated, binding to the endocytic complex formed by megalin and cubilin at the apical surface of PTCs appears important.[58,59] The complete elimination of aminoglycoside uptake in mice deficient in megalin suggests that this is the major pathway responsible for renal aminoglycoside accumulation.[59] Chloride transporters including cystic fibrosis transmembrane conductance regulator (CFTR) and the ClC-5 have been implicated because mice lacking functional CFTR *(Cftr$^{\Delta F/\Delta F}$)* or deficient in the Cl^-/H^+ exchanger *(Clcn5$^{Y/-}$)* have decreased kidney accumulation of gentamicin.[60] A three-dimensional (3D) model has described the complexity between megalin and gentamicin. Gentamicin binds to megalin with low affinity and exploits the common ligand-binding motif using the indole side chain of amino acids Trp-1126 and the negatively charged residues of Asp-1129, Asp-1131, and Asp-1133.[61] Once endocytosed, aminoglycosides inhibit endosomal fusion. They are also directly trafficked to the Golgi apparatus and, through retrograde movement, to the endoplasmic reticulum (ER). From the ER, gentamicin moves into the cytosol in a size- and charge-dependent manner.[62] Once in the cytosol, either from the ER[62] or via lysosomal rupture, aminoglycosides distribute to various intracellular organelles and mediate organelle-specific toxicity, such as mitochondrial dysfunction.[62,63] Gentamicin acts on mitochondria, activating the intrinsic pathway of apoptosis, disrupting ATP production,[64,65] and producing hydroxyl radicals and superoxide anions.[66,67] Also, delivery to the ER via retrograde transport from the Golgi apparatus allows for the binding of aminoglycosides to the 16S rRNA subunit,[68] resulting in a reduction of protein synthesis[69,70] and altering posttranslational protein folding.[71] The number of cationic groups on the molecules determines the facility with which these drugs are transported across the cell membrane and is an important determinant of toxicity.[72,73] Neomycin is associated with the most nephrotoxicity, gentamicin, tobramycin, and amikacin are intermediate, and streptomycin is the least nephrotoxic.

Receptor-associated protein (RAP) and cilastin, known to associate with megalin and block proximal tubule endocytosis, reduce gentamicin nephrotoxicity.[74,75] Risk factors for aminoglycoside nephrotoxicity include the use of high or repeated doses or prolonged therapy, CKD, volume depletion, diabetes, advanced age, and the coexistence of renal ischemia, hypotension, sepsis or use of other nephrotoxins.[76-78]

Vancomycin nephrotoxicity. Vancomycin was discovered, developed, and approved by the U.S. Food and Drug Administration (FDA) in 1958[79,80] and was found to be active against most gram-positive organisms including penicillin-resistant staphylococci.[80] Initial formulations of the drug were dubbed "Mississippi mud" due to the brown color presumably due to impurities, which were thought to be responsible for nephrotoxicity and ototoxicity.[80-82] Reformulation resulted in a drug that had improved purity and was called *vancomycin* (from the word vanquish).

The incidence of vancomycin nephrotoxicity is variable, ranging from as low as 0% to >40%, and those with vancomycin-associated nephrotoxicity are more likely to have higher trough levels and prolonged duration of treatment.[83] Durations of 7 to more than 15 days of treatment are associated with nephrotoxicity.[84-86] With modern formulations of vancomycin, the variability in nephrotoxicity is due to concomitant drug use, severity of illness, and variability in the definition of AKI. Daily total dose of vancomycin of more than 4 g and supratherapeutic levels (>30 mg/L) are a risk factor for nephrotoxicity.[87,88] Vancomycin-associated nephrotoxicity is thought to be more common when combined with antipseudomonal β-lactams. In a retrospective matched cohort study of patients receiving vancomycin and cefepime (588 patients) or vancomycin and piperacillin-tazobactam (3605 patients), the unadjusted incidence of AKI was 12.6% versus 21.4%, respectively ($P < .0001$). Vancomycin and piperacillin-tazobactam were associated with a more than twofold increase in the risk of AKI, after matching for severity of illness.[85] In non-intensive care unit (ICU) pediatric patients, serum trough levels of vancomycin >24.35 μg/mL and concomitant use of piperacillin-tazobactam predicted nephrotoxicity and longer hospital stay.[89]

In a patient who received vancomycin without coadministration of an aminoglycoside, the biopsy showed obstructive tubular casts composed of noncrystal nanospheric vancomycin aggregates associated with uromodulin.[90] In eight additional patients with AKI associated with high vancomycin levels, vancomycin-associated casts were also found. These findings, which were reproduced in mice given vancomycin, demonstrate a link between vancomycin and AKI.

Cisplatin nephrotoxicity. Treatment with cisplatin (cis-platinum), a platinum-based chemotherapeutic agent, is commonly associated with nephrotoxicity. The pathophysiologic mechanisms of cisplatin-induced tubular damage are complex and involve a number of interconnected factors, such as accumulation of cisplatin mediated by membrane transport, conversion into nephrotoxins, DNA damage, mitochondrial dysfunction, oxidative stress, inflammatory response, activation of signal transducers and intracellular messengers, autophagy, cell cycle regulation, and activation of cell death pathways[91] (Fig. 27.1).

The S3 segment of the proximal tubule in the corticomedullary region is the primary target of cisplatin to AKI in rats. In contrast, more distal sites may be affected in humans; however, glomeruli remain unaffected. Cisplatin flux in renal tubular cells occurs through transporters in a basolateral to apical direction. The solute carrier family 22 member 2 (SLC22A2, also known as organic cation transporter 2 [OCT2]), which is expressed on the basolateral membrane of proximal convoluted tubules, and SLC31A1 (also known as the copper transport protein 1 [CTR1]) expressed on proximal and distal tubules are the main transporters mediating cisplatin uptake.[92] Other transporters mediating uptake may include SLC22A6, SLC22A8.[93] SLC47A1 (also known as multidrug and toxin extrusion 1 [MATE1]), expressed on the apical membrane of the proximal tubule cells, is responsible for the efflux of cisplatin.[92,94] When cisplatin was administered to *Mate1*$^{-/-}$ mice, blood urea nitrogen (BUN) and creatinine levels were higher than in *Mate1*$^{+/+}$ mice. Furthermore, cisplatin levels were higher in plasma and in kidneys of *Mate1*$^{-/-}$ mice compared with *Mate1*$^{+/+}$ mice, suggesting that MATE1 mediates the efflux of cisplatin and could contribute to cisplatin nephrotoxicity.[94] The differential expression of SLC22A2, which is predominantly expressed in kidneys but only a few types of human tumor cells, provides the rationale for specifically targeting this transporter.[91]

The effects on the kidneys of low but frequent doses of cisplatin given once a week for 4 weeks were studied. Mice who received multiple doses of cisplatin had increased kidney fibrotic markers including fibronectin, transforming growth factor (TGF)-β, and α-smooth muscle actin (SMA), as well as interstitial fibrosis.[95] These studies in mice support observations in humans showing that adult patients with cancer who received multiple doses of cisplatin experience small but permanent declines in the estimated GFR (eGFR).[96,97]

Regulated necrosis contributes to cisplatin-induced AKI. In cisplatin-treated mice, necroptosis contributes to pathogenesis of cell death; receptor-interacting serine-threonine kinase (RIPK1), RIPK3, and mixed-lineage kinase domain–like protein (MLKL) expression were increased and deletion of *Ripk3* or *Mlkl* or pharmacologic inhibition of RIPK1 attenuated kidney injury.[91,98-100] Pyroptosis, an inflammatory mode of regulated cell death, requires the formation of plasma membrane pores through Gasdermin D (GSDMD). Following cisplatin treatment, GSDMD is upregulated and mice deficient in *Gsdmd* are protected from cisplatin-induced AKI.[101] Dysregulation of glutathione and iron metabolism by cisplatin leads to ferroptosis and AKI. A pharmacologic inhibitor of ferroptosis, namely Ferrostatin-1 (Fer-1), was reported to attenuate AKI.[102]

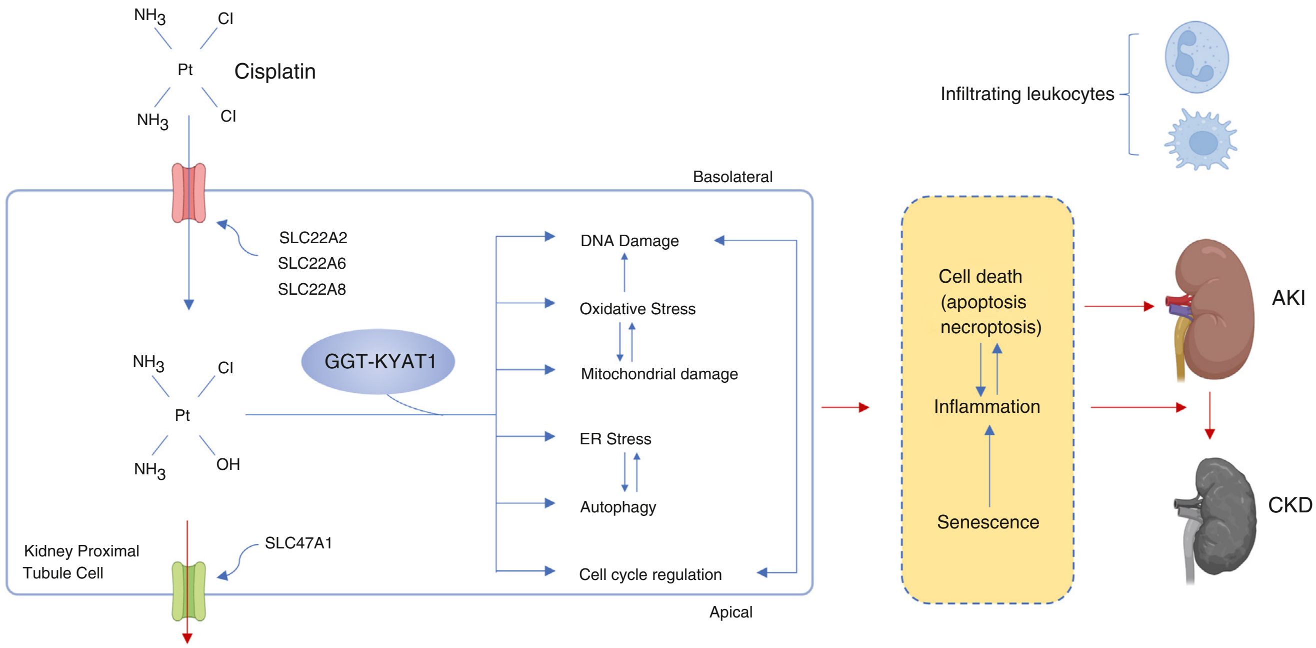


Fig. 27.1 The pathophysiology of cisplatin nephrotoxicity. Uptake of cisplatin in the PTCs is mediated by solute carrier family 22 member 2 (SLC22A2), SLC22A6, and SLC22A8 and its extrusion is mediated by SLC47A1 (MATE1). Upon entering PTCs, cisplatin is hydrated and activated to the toxic form by a γ-glutamyl transpeptidase *(GGT)* and kynurenine aminotransferase 1 *(KYAT1)*-dependent pathway. Cisplatin accumulation to toxic levels in PTCs induces DNA damage, mitochondrial damage, oxidative stress, endoplasmic reticulum *(ER)* stress, autophagy, and cell cycle regulation. Ultimately, these processes lead to cell death, acute kidney injury, and chronic kidney disease. (Created in BioRender. Okusa M. 2025. https://BioRender.com/bbf7fdx.)

Tubular disease—endogenous nephrotoxins

Myoglobin and hemoglobin. Myoglobin and hemoglobin are the endogenous toxins most commonly associated with ATN. Myoglobin, a 17.8-kDa heme protein released during skeletal and cardiac muscle injury, is freely filtered and causes red-brown urine, with a dipstick result positive for heme in the absence of RBCs in the urine. Intravascular hemolysis results in circulating free hemoglobin, which, when excessive, is filtered, resulting in hemoglobinuria, hemoglobin cast formation, and heme uptake by PTCs. The uptake in the proximal tubule may be mediated via endocytosis by megalin and cubilin. Megalin knockout mice have reduced accumulation of injected myoglobin in tubule cells and reduced nephrotoxicity.[103] The heme center of myoglobin may directly induce lipid peroxidation and, in addition, the liberation of free ferrous iron, depending on the redox potential, can promote hydroxyl radical formation by the Haber-Weiss (Fenton) reaction, resulting in the oxidation of lipids, proteins, and nucleic acids.[104,105] Iron is an intermediate accelerator in the generation of free radicals. Studies have suggested that there is increased formation of H_2O_2 in rat kidney models of myohemoglobinuria.[106] The subsequent hydroxyl (OH^-) radical plays a vital role in oxidative stress–induced AKI through mechanisms discussed in detail later in this chapter. In response, heme protein induces the heme-degradative enzyme, heme oxygenase, and increased synthesis of ferritin. Ferritin, a major factor in sequestering free iron,[107] is made up of two types of 24 subunits, heavy chain and light chain. It is the ferritin heavy chain (FtH) that has ferroxidase activity necessary for iron incorporation and to limit toxicity. Proximal tubule-specific, *FtH*-knockout mice (*FtH*$^{PT-/-}$ mice) have significant mortality in myoglobin-induced AKI, indicating the protective role of proximal tubule FtH in AKI.[108] Various iron chelators such as deferoxamine and other scavengers of ROS, such as glutathione, provide protection against myohemoglobinuric AKI.[109] Similarly, endothelin antagonists also prevent hypofiltration and proteinuria in rats that have undergone glycerol-induced rhabdomyolysis.[110] NO supplementation may be beneficial by preventing heme-induced renal vasoconstriction perhaps, in part, because heme proteins scavenge NO.[111,112] Finally, precipitation of myoglobin with Tamm-Horsfall protein (THP) associated with shed PTCs leads to cast formation and tubular obstruction, which is enhanced in acidic urine.[113] In human studies, volume expansion and perhaps alkalization of urine to limit cast formation are the preventive measures generally used because none of the experimental agents used in animal studies has been convincingly beneficial. This emphasizes the multifactorial nature of these conditions. It is unlikely that a single agent will be beneficial in this setting.[114]

Immunoglobulin light chains

Direct tubule toxicity. Excessive immunoglobulin light chains, produced in diseases such as multiple myeloma, are filtered, absorbed, and then catabolized in proximal tubule cells and can induce proximal tubulopathy.[115] The concentration of light chains leaving the proximal portion of the nephron depends on both the concentration of light chains in the glomerular filtrate and the capacity of the proximal tubule to reabsorb and catabolize them. Certain light chains can be directly toxic to the proximal tubules themselves.[116]

In the proximal tubules, free light chains (FLCs) are reabsorbed by binding to the proximal tubule heterodimeric complex consisting of megalin and cubilin.[117-119] Accumulation of light chains in the endosomes and lysosomes of the proximal tubule leads to cellular desquamation and fragmentation, vacuolization, and focal loss of the brush border.[120] Mechanisms for tubule toxicity may include blocking of transport of glucose, amino acids, or phosphate.[115] FLCs generate hydrogen peroxide,[121] which leads to the production of chemokines and cytokines,[121-124] with nuclear translocation of nuclear factor-κB (NF-κB), suggesting that light chain endocytosis leads to the production of inflammatory cytokines through activation of NF-κB.[125] Monoclonal FLC also promotes apoptosis through apoptosis signal-regulating kinase (ASK1), also called mitogen-activated protein kinase kinase kinase 5 (MAP3K5).[124] Subsequent inflammation leads to tubulointerstitial fibrosis.[126]

Cast nephropathy. Once the capacity for proximal tubule uptake is overwhelmed, a light chain load is presented to the distal tubule, where, on reaching a critical concentration, the light chains aggregate and coprecipitate with THP and form characteristic light chain casts.[127] FLCs bind to specific sites on THP through the CDR3 domain of FLC, leading to their coprecipitation in the lumen of the distal nephron and tubule flow.[128] There are critical determinants of the binding site between CDR3 and FLCs. This realization led to the therapeutic development of a cyclized competitive peptide. This peptide inhibited binding of FLCs to THP and was effective in inhibiting intraluminal cast formation and AKI.[128] Some studies have shown that light chains, in the amount seen in plasma cell dyscrasia patients, are capable of catalyzing the formation of hydrogen peroxide in cultured HK-2 cells. Hydrogen peroxide stimulates the production of monocyte chemoattractant protein-1 (MCP-1), a key chemokine involved in monocyte or macrophage recruitment to PTCs.[121]

Any process that reduces the GFR, such as volume depletion, hypercalcemia, or NSAIDs, will accelerate and aggravate light chain cast formation. It has been proposed that acutely reducing the presented light chain load by plasmapheresis or dialysis using high-cutoff membranes might be beneficial in limiting cast formation and reducing the extent of AKI in certain select patients, allowing for the initiation of chemotherapy to decrease bone marrow–dependent light chain formation.[129,130]

Uric acid. Tumor cell necrosis following chemotherapy can release large amounts of intracellular contents such as uric acid, phosphate, and xanthine into the circulation, potentially leading to AKI. Acute uric acid nephropathy with intratubular crystallization leading to obstruction and interstitial nephritis is not seen as commonly as it was in the past, mainly due to the prophylactic use of allopurinol or rasburicase before chemotherapy to lower the serum uric acid concentration acutely.

POSTRENAL ACUTE KIDNEY INJURY

Postrenal azotemia occurs from obstruction of the ureters, bladder outlet, or urethra. AKI from ureteric obstruction requires that the blockage occur bilaterally at any level of the ureters or unilaterally in a patient with a solitary functioning kidney or CKD. Ureteric obstruction can be intraluminal or external. Bilateral ureteric calculi, blood clots, and sloughed renal papillae can obstruct the lumen, whereas external compression from a tumor, blood vessels, lymph nodes, or hemorrhage can block the ureters as well. Fibrosis of the ureters intrinsically or from the retroperitoneum can narrow the lumen to the point of complete luminal obstruction. The most common cause for postrenal azotemia is structural or functional obstruction of the bladder neck. Prostatic conditions, therapy with anticholinergic agents, bladder stones, and a neurogenic bladder can all cause postrenal AKI. Relief of the obstruction usually causes prompt return of the GFR if the duration of obstruction has not been excessive. The rate and magnitude of functional recovery are dependent on the extent and duration of the obstruction.[131]

AKI resulting from obstruction usually accounts for less than 5% of cases, although in certain settings (e.g., transplantation), it can be as high as 6% to 10%. Clinically, patients can present with pain and oliguria, although these are neither specific nor sensitive. Because of the availability of retroperitoneal imaging using ultrasonography or computed tomography (CT), the diagnosis is usually straightforward, although, on occasion, a volume-depleted patient or a patient with severe reduction in the GFR may not show hydronephrosis on radiologic assessment. Because the GFR is typically not affected early in the course of obstructive AKI, volume repletion can increase the sensitivity of diagnosis by increasing the GFR and urine production into the ureter, leading to dilation of the ureter proximal to the obstruction, and enhancing ultrasonographic visualization. Early diagnosis and prompt relief of obstruction remain key goals in preventing long-term parenchymal damage because the shorter the period of obstruction, the better the chances for recovery and favorable long-term outcomes.

PATHOPHYSIOLOGY OF ACUTE KIDNEY INJURY

OVERVIEW OF THE PATHOPHYSIOLOGY OF ACUTE KIDNEY INJURY

AKI is a summation of temporally activated systems that together result in inflammation, endothelial cell injury, activation of cell death pathways, tubular obstruction, back leak, altered glomerular hemodynamics, and loss of the GFR. Within the kidney, mechanisms pertaining to microvascular compartments, innate immunity, and ATN result in temporary, partial, or permanent loss of the GFR. Furthermore, it is becoming widely accepted that AKI is a systemic process that affects a number of organs, leading to the high morbidity and mortality seen in patients with AKI. Systemic responses to AKI may influence the extent of injury. Hemodynamic alterations (e.g., decrease in cardiac output, low blood pressure, and vasoconstriction) may initiate AKI or exacerbate intrinsic microenvironmental mechanisms of AKI. Systemic immunologic mechanisms of proinflammatory or antiinflammatory conditions may affect AKI, and neural mechanisms may attenuate AKI. Thus the complexity of the pathogenesis of AKI requires careful understanding of its molecular mechanisms through defining important targets in humans and testing in relevant models of AKI (Fig. 27.2).[132-137]

ACUTE TUBULAR NECROSIS

EPITHELIAL CELL INJURY

Although all segments of the nephron may undergo injury during an ischemic insult, the major and most commonly injured epithelial cell involved in AKI related to ischemia, sepsis, and/or nephrotoxins is the PTC. Of the three segments (S1–S3), the S3 segment of the proximal tubule in the outer stripe of the medulla is the cell most susceptible to ischemic injury for several reasons.[138] First, it has limited capacity to undergo anaerobic glycolysis due to its dependence on fatty acid oxidation (FAO) as the major source of energy. Second, due to its unique primarily venous capillary regional blood flow, there is marked hypoperfusion and congestion in this medullary region after injury that persists, even though cortical blood flow may have returned

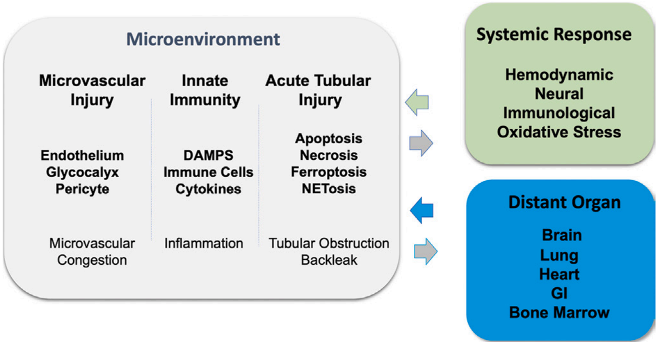


Fig. 27.2 Overview of pathophysiology of acute kidney injury (AKI). In the kidney, mechanisms pertaining to microvascular compartment, innate immunity, and ATN result in temporary, partial, or permanent loss of the glomerular filtration rate. Systemic responses may influence the extent of injury. Hemodynamic alterations may initiate AKI or exacerbate intrinsic microenvironmental mechanisms of AKI. Systemic immunologic mechanisms of proinflammatory or anti-inflammatory conditions may affect AKI, and neural mechanisms may attenuate AKI (see text for details).

to near-normal levels after ischemic injury. Endothelial cell injury and dysfunction are primarily responsible for this phenomenon, often referred to as the “extension phase” of AKI.[139] The other major epithelial cells of the nephron involved are those of the medullary thick ascending limb located more distally. Cells of the S1 and S2 segments are usually involved in toxic nephropathy due to their high rates of endocytosis, leading to increased cellular uptake of the toxin. PTCs and thick ascending limb of loop of Henle (TAL) cells are involved as sensors, effectors, and injury recipients of AKI stimuli.

PTC injury and dysfunction during ischemia or sepsis lead to a profound drop in the GFR through afferent arteriolar vasoconstriction, mediated by tubular glomerular feedback and proximal tubular obstruction. This phenomenon, along with tubular backleak, leads to a fall in the effective GFR[140,141] (Fig. 27.3 and eFig. 27.1).

Morphologic changes

The classic histologic hallmark of ATN was described in a landmark study by Oliver and coworkers,[142] in which individual nephrons from autopsy specimens of patients dying of acute renal failure (ARF) were microdissected. Portions of glomerular ultrafiltrate had become sequestered in tubules that were obstructed (necrotic PT cells), which suggested that filtrate leaked back through damaged tubular walls and entered the interstitium, which caused it to become edematous.[142] Early on, there is the loss of the apical brush border of the PTCs. Microvilli disruption and detachment from the apical cell surface result in the formation of membrane-bound blebs. This process, which occurs early in the pathogenesis, is followed by release into the tubular lumen. Patchy detachment and subsequent loss of tubular cells result in areas of tubular basement being denuded and exposed. Focal areas of proximal tubular dilation, along with the presence of distal tubular casts, are also major pathologic findings in ATN.[143] The sloughed tubular cells, brush border vesicle remnants, and cellular debris in combination with THP form the classic muddy brown granular casts. These distal casts have the potential to obstruct the tubular lumen. Frank

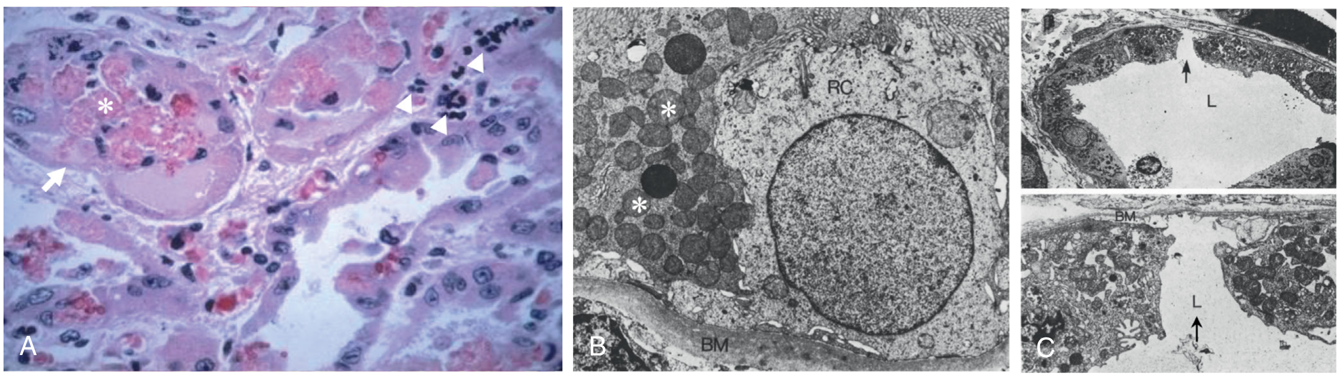


Fig. 27.3 Morphology of human acute tubular necrosis. (A) Human biopsy specimens reveal significant proximal tubular cell damage, with intraluminal accumulation of apical membrane fragments and detached cells *(*),* thinning of proximal tubular cells to maintain monolayer tubule integrity *(arrow),* and dividing cells and accumulation of white cells within the microvascular space in the peritubular area *(arrowheads).* (B) Electron micrograph of a regenerating epithelial cell. Shown are small fragmented mitochondria *(*).* (C) Electron micrograph of renal epithelial cell showing nonreplacement site *(black arrow)* that morphologically supports the concept of back leak in the pathophysiology of AKI. (A, Courtesy M. Venkachatalam; B and C from Olsen TS, Olsen HS, Hansen HE. Tubular ultrastructure in acute renal failure in man: epithelial necrosis and regeneration. *Virchows Arch A Pathol Anat Histopathol.* 1985;406(1):75–89.)

necrosis itself is inconspicuous and is restricted to the highly susceptible outer medullary regions. Alternatively, features of apoptosis are more commonly seen in proximal and distal tubular cells. Glomerular epithelial cell injury in ischemic, septic, or nephrotoxic injury is not typically seen, although some studies have shown thickening and coarsening of foot processes, including podocyte-specific molecular and cellular changes.[144] The future morphologic course of the tubular cell alterations varies according to the type and extent of injury (see "Supplement on Cytoskeletal and Intracellular Structural Changes").

Regulated cell death pathways. Regulated cell death pathways can be grouped into either a caspase-controlled cell death system (apoptosis, necroptosis, and pyroptosis) or a lipid peroxidation–autoxidation-controlled necrosis system (ferroptosis; Fig. 27.4).[145]

It is not dependent on caspase activation but results from a rise in intracellular calcium and the activation of membrane phospholipases.[146,147] Functionally, severe ATP depletion results first in mitochondrial injury, with subsequent arrest of oxidative phosphorylation, depletion of energy stores, and robust formation of ROS. This in turn mediates further cellular injury. In some cases, defined regulated molecular pathways may lead to necrotic cell death, a process referred to as *necroptosis*.[148] Differentiation between necrosis and necroptosis requires that necroptosis-dependent cell death involves the receptor-interacting protein kinase 3 (RIPK3).[149]

Apoptosis. Cells undergoing sublethal or less severe injury have the capacity for functional and structural recovery if the insult is interrupted. Cells that suffer a more severe (or lethal) injury undergo apoptosis or necrosis. Apoptosis is an energy-dependent, programmed cell death due to both intrinsic (mitochondrial) and extrinsic activation of caspases resulting, either of which results in the condensation of nuclear and cytoplasmic material, forming apoptotic bodies. These apoptotic bodies, which are plasma membrane bound, are rapidly phagocytosed by macrophages and neighboring viable epithelial cells[150] (eFig. 27.2).

The caspase family of proteases is an important initiator and effector of apoptosis.[151,152] Both the intrinsic (mitochondrial) and extrinsic (death receptor) apoptotic pathways are activated in human AKI. Specifically, activation of procaspase-9 primarily depends on intrinsic mitochondrial pathways regulated by the Bcl-2 family of proteins, whereas that of procaspase-8 results from extrinsic signaling via cell surface death receptors, such as Fas and their ligand FADD (Fas-associated protein with death domain). There is also considerable crosstalk between the intrinsic and extrinsic pathways. The other group of caspases—3, 6, and 7 are effector caspases, which are more abundant and catalytically robust, cleaving many cellular proteins and resulting in the classic apoptotic phenotype.

Caspase activation in epithelial cells occurs due to ischemic and other cytotoxic insults, whereas inhibition of caspase activity is protective against such injury in cultured and in vivo renal epithelial tubular AKI.[153,154] Several pathways including the intrinsic (Bcl-2 family, cytochrome *c*, and caspase-9), extrinsic (Fas, FADD, and caspase-8), and regulatory (p53 and NF-κB) pathways appear to be activated during ischemic renal tubular cell injury. It has also been

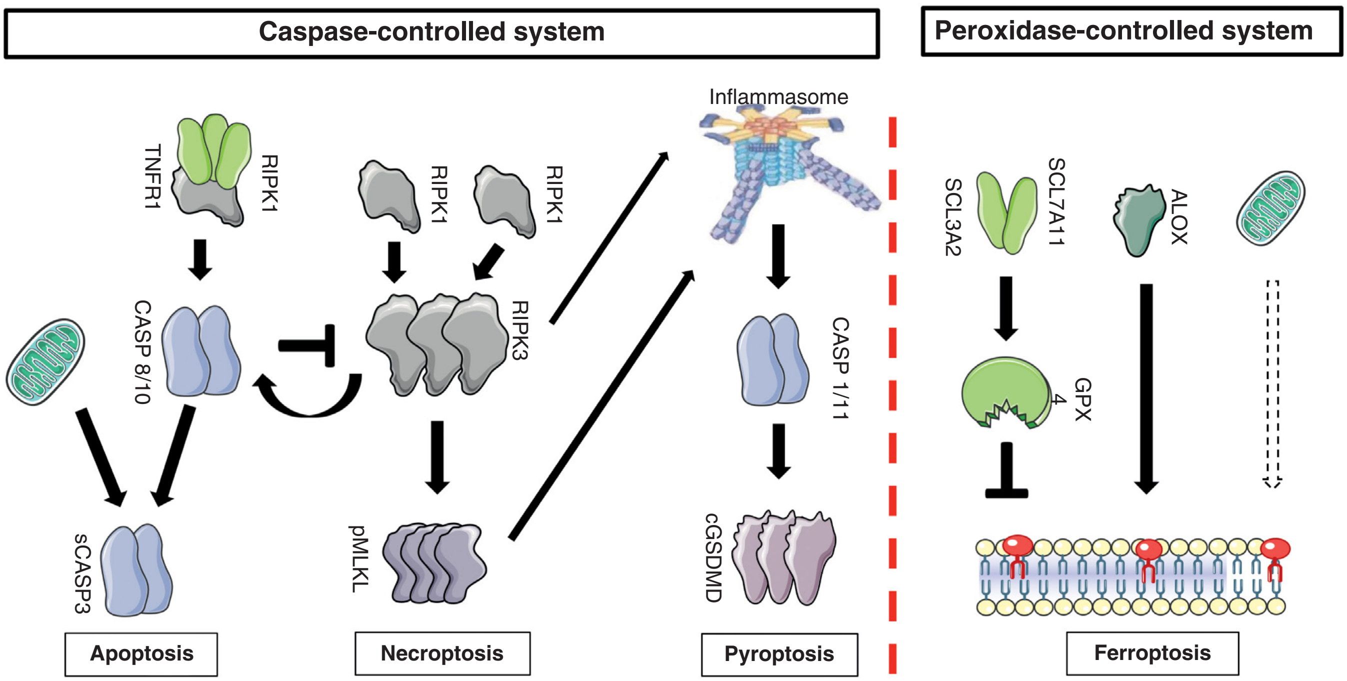
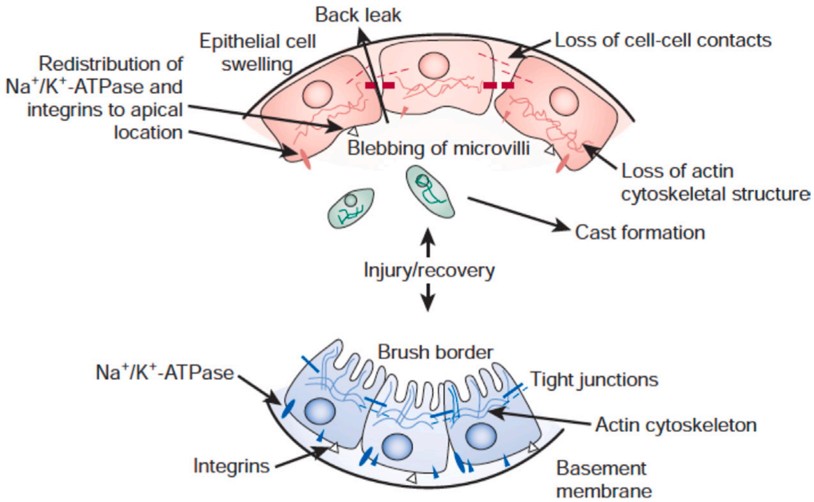
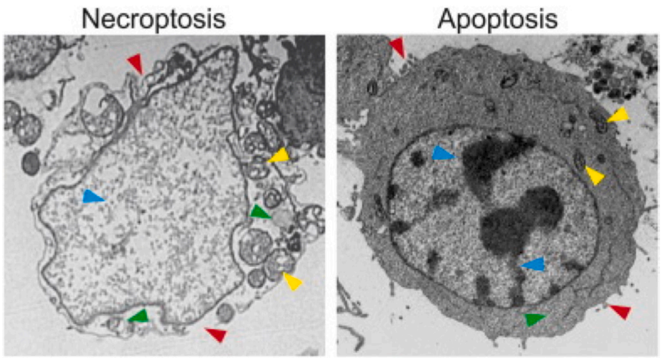


Fig. 27.4 Overview of the pathways of regulated cell death. In general, two systems may be best differentiated when regulated cell death is considered. The caspase-controlled system includes apoptosis, necroptosis, and pyroptosis. In contrast, the peroxidation-controlled system of ferroptotic cell death functions entirely independently of the caspase-controlled network. (From Tonnus W, Gembardt F, Latk M, et al. The clinical relevance of necroinflammation highlighting the importance of acute kidney injury and the adrenal glands. *Cell Death Differ.* 2019;26[1]:68–82.)



e-Fig. 27.1 Overview of sublethally injured tubular cells. Sodium-potassium adenosine triphosphatase (Na^+-K^+-ATPase) pumps are normally located at the basolateral membrane. In sublethal ischemia, the pumps redistribute to the apical membrane of the proximal tubule. On reperfusion, the pumps reverse back to their basolateral location. (From Sharfuddin A, Sharfuddin AA, Sandoval RM, Berg DT, et al. Soluble thrombomodulin protects ischemic kidneys. *J Am Soc Nephrol.* 2009;20(3):524–534.)



e-Fig. 27.2 Morphologic features of necroptosis and apoptosis. HT29 colon cancer cells treated with an anticancer drug for 48 hours were analyzed by transmission electron microscopy. The cell undergoing necroptosis shows plasma membrane rupture and permeabilization, compared with the intact plasma membrane, with blebbing in the apoptotic cell *(red arrowheads).* The necroptotic cell exhibits cytoplasm swelling and vacuolization, which are absent in the apoptotic cell *(green arrowheads).* The necroptotic cell has swelled mitochondria, in contrast to those in the apoptotic cells *(yellow arrowheads).* The necroptotic cell also lacks the condensed and fragmented nuclei seen in the apoptotic cell *(blue arrowheads).* (From Chen D, Yu J, Zhang L. Necroptosis: an alternative cell death program defending against cancer. *Biochim Biophys Acta.* 2016;1865[2]:228–236.)

shown that the balance between cell survival and death depends on the relative concentrations of the proapoptotic (Bax, Bcl-2–associated death promoter [Bad], and Bid) and antiapoptotic (Bcl-2 and Bcl-xL) members of the Bcl-2 family of proteins. Overexpression of proapoptotic or relative deficiency of antiapoptotic proteins may lead to the formation of mitochondrial pores. Conversely, the inhibition of such pore formation may occur with the opposite imbalance.[155-157]

Other proteins that have been shown to play a significant role in the apoptotic pathways include NF-κB and p53.[158,159] The kinase-mediated pathways such as ERKs and c-Jun N-terminal kinases (JNKs) are responsible for mediating cellular responses involved in apoptosis, survival, and repair through their interaction with other signals from growth factors, such as hepatocyte growth factor, insulin-like growth factor-1, epidermal growth factor, and vascular endothelial growth factor (VEGF).[160,161] These independent mechanisms can inhibit proapoptotic proteins such as Bad and activate the antiapoptotic transcription of cyclic adenosine monophosphate response element binding (CREB) factors. There is rapid delivery of small interfering RNA (siRNA) to PTCs in AKI; targeting siRNA to minimize p53 production leads to a dose-dependent attenuation of apoptotic signaling and kidney function, suggesting potential therapeutic benefit for ischemic and nephrotoxic kidney injury.[162] In vivo, microRNA-24 (miR-24) also regulates the HO-1 and H2A histone family, member X. Overall, these results indicate that miR-24 promotes renal ischemic injury by stimulating apoptosis in endothelial and tubular epithelial cells (TECs).[163]

A novel Ying Yang 1-KIM1-death receptor 5 (YY1-KIM1-DR5) axis has been identified in the progression of AKI.[164] The transcription factor YY1, a transcriptional repressor of KIM1, is downregulated upon AKI, leading to KIM1 secretion. KIM1 binds to DR5 and activates the caspase cascade inducing renal cell apoptosis and exacerbating AKI. Blocking the KIM1-DR5 interaction with rationally designed peptides exhibits kidney protection from AKI. Considering the various pathways available for blockade or modulation, the therapeutic implications of targeting apoptosis in preventing epithelial cell injury are significant. However, it is likely that the "window" to avert lethal injury and prevent cells from progressing to necrosis is in the early initiating apoptotic phases.

Numerous studies have shown that apoptosis lead to a rise in intracellular calcium through impairment of calcium ATPases, whereas inhibition of Na^+-K^+-ATPase activity resulting from ATP depletion potentiates calcium entry into the cell via the sodium-calcium exchanger. Increased cytosolic calcium causes further mitochondrial injury and cytoskeletal alterations.[165] This chain of events results in the downstream activation of proteases such as calpain and phospholipases. Phospholipases such as phospholipase A2 cause direct hydrolytic damage to membranes and also release toxic free fatty acids. They also cause release of eicosanoids that have vasoactive and hemokinetic activities, resulting in an intense surrounding inflammatory response. Calpain activation mediates plasma membrane permeability and hydrolysis of the cytoskeletal proteins.[166,167] Finally, there is release of lysosomal enzymes and proteases that degrade histones, resulting in accessibility of the endonucleases to the entire segment, typically seen as the smear pattern on gel electrophoresis, in contrast to the typical ladder pattern seen in apoptosis.[168]

Necrosis and necroptosis. Epithelial cell necrosis is a passive nonenergy-dependent process that develops secondary to severe ATP depletion from toxic or ischemic insult and is independent of caspase activation. In necrosis, there is cellular and organelle swelling, with loss of plasma membrane integrity and release of unprocessed intracellular contents, including cellular organelles, highly immunogenic proteins such as ATP, HMGB1, double-stranded DNA, and RNA components, also referred to as damage-associated molecular patterns (DAMPs), a concept known as *necroinflammation*.[169] Differentiation between necrosis and necroptosis requires that necroptosis-dependent cell death involves the receptor-interacting protein kinase 3 (RIPK3) (see Fig. 27.4).[149] Although tubular necrosis was thought to be accidental, work done over the past 2 decades has revealed several pathways of genetically determined and regulated necrosis in which receptor-interacting serine-threonine protein kinase 1 (RIPK1),[170,171] RIPK3[172] and its substrate, MLKL, have been directly implicated in the regulation of the novel cell death pathway known as necroptosis.[173] This activation of unique signaling paradigms forms the basis for differentiating between necrosis and necroptosis. The signaling pathway that triggers necroptosis includes the engagement of death receptors in the presence of caspase inhibition, stimulation of Toll-like receptors, signaling through interferons, with the activation of kinase RIPK3, and phosphorylation of pseudokinase MLKL. Phosphorylation of MLKL by RIPK3 leads to a molecular switch mechanism that induces plasma membrane rupture.[174] Demonstration of a protective effect on kidney function with the use of necrostatin-1, an inhibitor of RIPK1, suggests that necroptosis occurs in ischemic AKI.[175] Necroptosis also occurs in the model of cisplatin-induced AKI, as demonstrated by the significant protection of kidney function when using RIPK3 and MLKL-deficient mice.[98] Peptidylprolyl isomerase F (PPIF) and phosphorylated MLKL have been shown to be localized to injured tubules in human kidney biopsies of oxalosis-related AKI. Further, mice lacking PPIF or MLKL or given an inhibitor of mitochondrial permeability displayed attenuated oxalate-induced AKI.[176]

Despite significant advances in necroptosis research in experimental systems, clinical studies on necroptosis inhibitors for the treatment of acute and chronic kidney disease are lacking. Few reports thus far could detect necroptosis in human ischemic kidney biopsies. One study of 10 transplanted living-donor kidneys, biopsied 1 hour after transplantation, identified phospho-MLKL staining on immunohistochemistry.[177] A later study of kidney allografts with AKI identified phospho-MLKL in 10% to 15% of tubule cells. Interestingly, screening a panel of clinical plasma membrane channel blockers identified that an anticonvulsant drug, phenytoin, blocked upstream necrosome formation by attenuating RIPK1 kinase activity in vitro, due to the hydantoin scaffold that is also present in necrostatin-1. Phenytoin also protected mice from renal ischemia-reperfusion induced (IRI) and tumor necrosis factor-α (TNF-α)-induced systemic inflammatory response syndrome (SIRS).[178] These findings suggest that drugs used for other indications could be used in the future to target necroptosis for the benefit of patients with AKI.

Ferroptosis. Ferroptosis is a nonapoptotic form of regulated cell death that is iron dependent and characterized by increased lipid peroxidation resulting from lack of activity of the lipid repair selenoprotein enzyme glutathione peroxidase 4 (GPX4).[179-181] This leads to the accumulation of lipid-based ROS including lipid hydroxyperoxides. This form of iron-dependent cell death is distinct from other forms of cell death such as apoptosis, unregulated necrosis, and necroptosis because it mediates cell death in a noncell autonomous and synchronized manner, providing a potential explanation for nephron loss during AKI.[182] Erastin and RSL3 are ferroptosis-inducing compounds that induce cell death in the absence of apoptotic features.[179,183-185] Ferroptosis plays a key role in folic acid–induced AKI,[186] resulting in increased inflammation. Ferroptosis is also importantly regulated by heme oxygenase-1 (HO-1), an enzyme discussed earlier that mediates the breakdown of heme.[187] Immortalized proximal tubule cells from HO-1$^{+/+}$ mice were much more susceptible to erastin or RSL3 than cells from HO-1$^{-/-}$ mice. Iron supplementation decreased cell viability further in HO-1$^{-/-}$ compared with HO-1$^{+/+}$ cells. Finally, ferrostatin (a ferroptosis inhibitor), deferoxamine (an iron chelator), or *N*-acetyl-L-cysteine (a glutathione replenisher) attenuate erastin-induced ferroptosis[187] (see Fig. 27.4).

Unbiased transcriptomics studies have identified novel targets for intervention in AKI. Charged multivesicular body protein 1a (CHMP1A) and dipeptidase 1 (DPEP1) regulate ferroptosis in tubular cells via altering cellular iron trafficking, and their deletion attenuated injury in cisplatin, folic acid, and UUO-induced kidney injury.[188] Silencing the repressor element 1-silencing transcription factor (REST), a master regulator of gene repression under hypoxia, is upregulated in AKI patients, mice, and RTECs. Downregulation of REST attenuated ferroptosis in primary RTECs and AKI-to-CKD transition.[189] Loss of ferroptosis suppressor protein 1 (Fsp1) or the targeted manipulation of the active center of the selenoprotein glutathione peroxidase 4 (Gpx4cys/−) sensitizes kidneys to tubular ferroptosis. Linkermann and colleagues[190] generated Nec-1f, which simultaneously targeted RIPK1 and ferroptosis. These findings warrant additional studies on the role of ferroptosis in AKI.

Pyroptosis. This highly inflammatory form of regulated cell death requires caspase 1,4, 5 (caspase 11 in mice) for activation[145,191] (Fig. 27.4). Intracellular damage and release of DAMPs activate NLRP3 inflammasome leading to caspase activation and cleavage of pro–interleukin-1 β (IL-1β) and pro–IL-18 to their mature forms. Pyroptosis executioner, GSDMD, is activated, releasing the N-terminal fragment (GSDMD-NT),[145,192] which oligomerizes and forms a membrane pore on the cell membrane,[193] leading to cell swelling, osmotic lysis, and release of IL-1β and IL-18 and other intracellular contents.[145,192] Unlike apoptosis, pyroptosis results in plasma membrane rupture and release of DAMPs, which activate innate immunity and recruitment of immune cells.[194]

Multiple cell death pathways are involved in the regulation of successful and maladaptive repair; pyroptosis and NLRP3 inflammasomes are intertwined with AKI and fibrogenesis. Genes associated with pyroptosis (*Casp1, Casp3, Casp4, Il1b, Il18,* and *Gsdmd*) and ferroptosis (*Ptgs2, Chac1, Acsl4, Slc7a11,* and *Hmox1*) were highly enriched in PT cells following long IRI.[195] Attempts have been made to identify novel modulators of pyroptosis in AKI. During sepsis, inhibition with disulfiram or genetic deletion of GSDMD abrogated neutrophil activation formation, reducing multiorgan dysfunction and lethality.[196] Another inhibitor, dapansutrile, was also found to inhibit the NLRP3 inflammasome and subsequent activation of IL-1β. The dsDNA sensor, absent in melanoma 2 (AIM2), forms an inflammasome and induces GSDMD cleavage, which activates pyroptosis. In a mouse model of rhabdomyolysis-induced AKI, AIM2 deficiency delayed recovery and worsened fibrosis in the kidney. Thus macrophage pyroptosis serves in an immunoregulatory capacity and determines the healing process in rhabdomyolysis-induced AKI.[197]

CCAAT/enhancer-binding protein β (C/EBPβ), a critical regulator of the immunosuppressive environment in cancers, upregulates mitochondrial transcription factor A (TFAM) and induces the secretion of IL-1β and IL-18 in a model of sepsis-induced AKI, suggesting that an interaction between C/EBPβ and TFAM facilitated pyroptosis via NLRP3/caspase-1 signal.[198] Other endogenous molecules, such as ketone bodies (e.g., β-hydroxybutyrate), were shown to protect against ischemic tissue injury via increasing the expression of forkhead transcription factor O3 (FOXO3), an upstream regulator of pyroptosis.[199] Understanding cell death pathways can lead to specific interventions to preserve apoptosis and avoid inflammation while blocking the more inflammatory pathways such as pyroptosis, considered the most proinflammatory of the regulated pathways.[145] During the COVID-19 pandemic, remdesivir (RDV, GS-5734), a broad-spectrum antiviral nucleotide prodrug, alleviated AKI by specifically inhibiting NLRP3 inflammasome activation in macrophages.[200]

Autophagy. Autophagy is an essential mechanism for normal homeostasis, disease pathogenesis, and aging in kidneys.[201-204] Fig. 27.5 reviews the three forms of autophagy—macroautophagy, microautophagy, and chaperone-mediated autophagy.

Autophagy serves a protective function in renal tubular cells during ischemia-reperfusion injury; it precedes the appearance of apoptotic cells, and suppression of autophagy exacerbates renal injury, in part by inhibiting apoptosis.[205,206] The importance of autophagy has been demonstrated through the generation of proximal tubule-specific ATG5 or ATG7 knockout mice.[207-209] The absence of ATG5 in proximal tubule cells led to the accumulation of deformed mitochondria and cytoplasmic inclusions or protein aggregates, [207]more proinflammatory cytokine production, and increased Ang II-induced phosphorylation and NF-κB activation,[210] whereas overexpression of ATG5 attenuated activation of NF-κB and protected against renal inflammation.[210] Similar results were observed in proximal

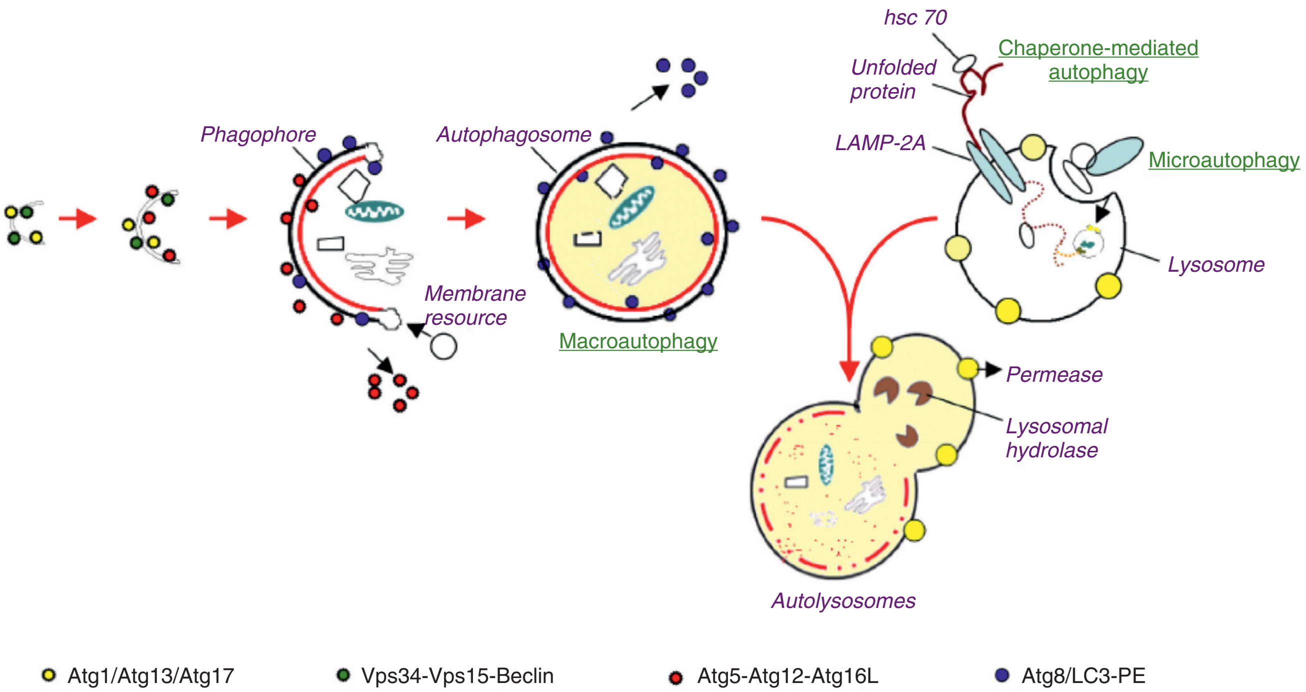


Fig. 27.5 Autophagy. The forms of autophagy are macroautophagy, microautophagy, and chaperone-mediated autophagy. Macroautophagy starts with the de novo formation of a cup-shaped isolation double membrane that engulfs a portion of cytoplasm. Microautophagy involves the engulfment of cytoplasm instantly at the lysosomal membrane by invagination, protrusion, and separation. Chaperone-mediated autophagy is a process of direct transport of unfolded proteins via the lysosomal chaperonin hsc70 and LAMP-2A. All forms of autophagy subsequently lead to the degradation of intra-autophagosomal components by lysosomal hydrolases. *PE,* Phosphatidylethanolamine. (From Periyasamy-Thandavan S, Jiang M, Schoenlein P, Dong Z. Autophagy: molecular machinery, regulation, and implications for renal pathophysiology. *Am J Physiol Renal Physiol.* 2009;297[2]:F244–256.)

tubule-specific ATG7 knockout mice.[211] In contrast, the lack of ATG5 in distal tubules does not cause significant alterations in kidney function.[208] Importantly, renal ischemia-reperfusion injury was exaggerated in proximal tubule-specific ATG5 or ATG7 knockout mice.[207,208,211] A number of AKI models demonstrate changes in the autophagy pathway. For instance, autophagy activation was also detected in cisplatin-induced AKI in mice.[212] $ATG7^{-/-}$ mice demonstrated worse kidney function and injury with cisplatin treatment.[211] Chloroquine, an inhibitor of autophagy, blocked lysosomal degradation of microtubule associated protein LC3 in autophagosomes and worsened kidney function in a cisplatin-induced model of AKI.[211] Similarly, autophagy was induced in the septic AKI model of LPS treatment of mice, and inhibition by chloroquine or tubular ATG7 ablation worsened LPS-induced AKI.[213]

Mitophagy is a specialized form of autophagy, a cellular process that selectively targets and degrades damaged or dysfunctional mitochondria. Mediators of mitophagy are implicated in AKI: Pink1 and/or Park2-knockout mice had greater accumulation of damaged mitochondria, tubular cell death, ROS generation and inflammation, and worsening AKI outcomes. Of interest, PINK1/Parkin were downregulated in human CKD kidney samples and TGF-β1–treated human renal macrophages suggesting targets for intervention.[214] Providing further evidence for a role of mitophagy in AKI, the induction of BNIP3-mediated mitophagy also provided renoprotection during ischemic AKI.[215] Knockdown of prohibitin 2 (PHB2), a newly identified intracellular receptor of mitophagy that participates in the removal of damaged mitochondria, attenuated mitophagy leading to increased AKI, whereas overexpression of PHB2 ameliorated mitochondrial dysfunction and NLRP3 activation, highlighting PHB2 as a new target in AKI.[216]

Caloric restriction mimics ischemic preconditioning (IPC) and stimulates the SIRT, SGK1–FOXO3a–(HIF1α) axis to prevent a decrease in peroxisome proliferator-activated receptor gamma coactivator 1-alpha (PGC1α), leading to renoprotection. IPC induced autophagy in renal tubular cells in mice, and the protection was abolished by pharmacologic inhibitors of autophagy or PTC-specific deletion of Atg7.[217] Treatment with Beclin1 peptide induced autophagy and protected against IRI. In sepsis-induced AKI, bone marrow–derived stem cells inhibited inflammation and upregulated expressions of Parkin and SIRT1, promoting mitophagy.[218]

However, autophagy might also have a profibrotic role as chronic hypoxia after severe ischemic AKI; Atg5 deletion in the S3 segment inhibited tubular senescence, attenuated interstitial fibrosis, and improved kidney function. Chronic hypoxia after AKI activated FOXO3 in a HIF1α-dependent manner, and Foxo3 deletion attenuated autophagy and transition to CKD.[219] Genetic deficiency in tubular autophagy diminished FGF2 in TGFB1/TGF-β1-treated renal tubular cells, and this effect was suppressed by FGF2 neutralizing antibody and reduced fibrosis. In humans, renal biopsies from post-AKI patients showed higher levels of autophagy and FGF2 in tubular cells that significantly correlated with renal fibrosis.[220] The inhibition of mTOR could also induce persistent activation of autophagy and impaired proliferation.[221] These findings indicate that autophagy is a dynamic process that is critical for survival of tubular cells initially after AKI and later stages may regulate fibrosis productive renal repair.

Thus taken together, these collective data, from both pharmacologic studies and genetically deficient mice, have provided strong evidence for a renoprotective role of autophagy. In contrast, dysregulated autophagy can result in increased renal degeneration, leading to progressive kidney disease, as characterized by interstitial fibrosis.[209,222]

Innate immunity and inflammation

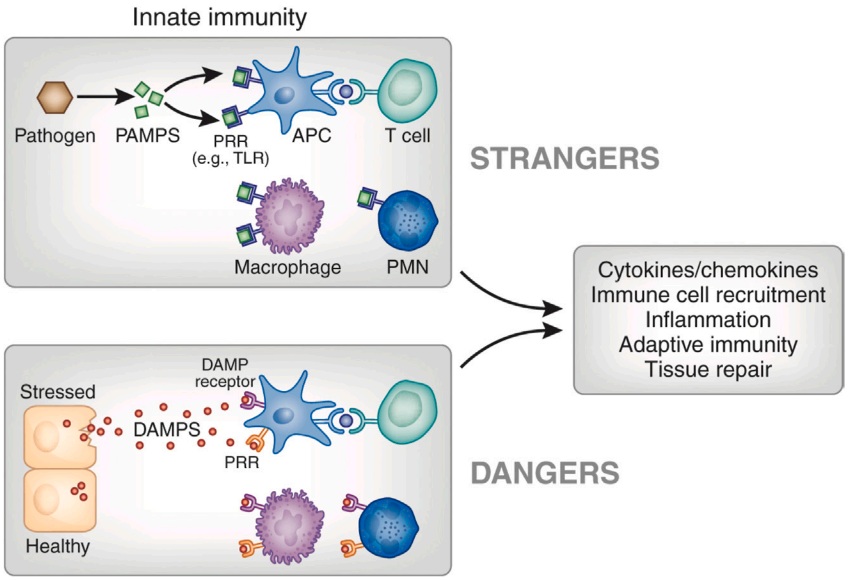
Kidney microenvironment. The interstitial microenvironment, that region between the basement membranes of epithelial cells and peritubular capillaries, is a reactive compartment containing mononuclear phagocytes, interstitial fibroblasts, and pericytes, as well as various soluble mediators.[223] Leukocytes, including but not limited to macrophages, NKT cells, B lymphocytes, and T lymphocytes, contribute to kidney IRI.[224-229] Selective depletion, knockout mouse models, and specific blockade have shown that all of these cells interact and contribute to tubular injury at various phases.[224,226-229]

Mononuclear phagocytes in acute kidney injury. The mononuclear phagocytic system consists of bone marrow–derived macrophages and dendritic cells,[230] which overlap in both functional characteristics and surface biomarkers.[231] Macrophages are resident tissue phagocytic cells that function generally to clear dying cells and produce cytokines and growth factors.[232,233] Although the mononuclear phagocytic system is well known to be activated in response to foreign pathogens,[234] the microenvironment responds to endogenous molecules released from dying cells following tissue injury or to changes produced by conditions, such as hypoxia, ischemia, or other forms of sterile inflammation.[235,236] Matzinger proposed the danger model to explain exceptions to the classical role of immune responsiveness to foreign antigens.[237] Dendritic cells are activated by DAMPs or pathogen-associated molecular patterns (PAMPs)[235] and are key initiators of the innate immune system following ischemia-reperfusion (eFig. 27.3). Dendritic cells, a resident population of bone marrow–derived cells, and macrophages form a network between the basement membranes of tubular epithelial and peritubular endothelial cells.[223,238] Although dendritic cells and macrophages are often considered distinct cell types with characteristic functions, they share considerable overlap in cell surface markers and function[230] (Fig. 27.6).

Dendritic cells have enormous plasticity and can be antiinflammatory or proinflammatory.[239,240] They contribute early in the course of IRI-induced activation of NKT cells and the IL-17/IL-23 signaling pathway.[241-243] Depletion of dendritic cells, using transgenic mice expressing the human diphtheria toxin receptor ([DTR]; human heparin-binding epidermal growth factor–like growth factor in CD11c$^+$ cells (CD11c-DTR mouse), demonstrates their importance in activating the innate immune response in AKI.[244] Diphtheria toxin (DT)-treatment depleted CD11c-DTR mice had significantly less injury than CD11c-DTR mice in response to IRI compared with those mice pretreated with a catalytically inactive mutant DT (mDT),[245] strongly supporting the concept that dendritic cells contribute to the early innate response in IRI. Dendritic cells are the earliest producers of IL-6, TNF-α, MCP-1, and RANTES (regulated on activation, normal T cell–expressed and secreted).[246] In addition to initiating the recruitment of inflammatory cells, dendritic cells also participate in recovery via IL-10 production[247] and can induce tolerance. Tolerogenic dendritic cells are functionally immature, express inadequate positive or enhanced negative costimulatory signals and reduced proinflammatory cytokines, and can generate immune tolerance by inducing T cell anergy or deletion or by induction or expansion of regulatory T (Treg) cells.[248,249] Adenosine 2A receptor (A_2AR)-induced tolerized dendritic cells suppressed NKT cell activation in vivo and attenuated kidney IRI.[250] However, mature dendritic cells also can promote tolerance.[248,251] Both immature and mature dendritic cells can prime Treg cells thus preventing autoimmunity. In contrast to IRI, dendritic cell ablation in DTR mice increased injury in a cisplatin-induced model of AKI, which demonstrates the importance of dendritic cells for tissue protection in a distinct model of kidney injury.[252] Thus these studies have revealed disparate roles of tissue-resident dendritic cells in AKI and suggest that the interstitial microenvironment created by different pathogenic circumstances, such as cisplatin toxicity, IRI, or endogenous molecules (e.g., PAMPS, DAMPS, and cytokines, autacoids such as adenosine), may differentially regulate dendritic function.[239] Finally, dendritic cells migrate away from the kidney via the lymphatic system to present antigens and regulate adaptive lymphocytic responses.[253] Thus dendritic cells serve at the crossroads of communication between the epithelium and endothelium, regulating both innate and adaptive immunity, self-tolerance, and tissue injury and repair.

Macrophages are phagocytic innate immune cells that contribute to host defenses and, based on surface markers, five distinct populations have been identified.[254,255] Tissue-resident macrophages are derived from a heterogeneous population of bone marrow-derived monocytes.[233,256-258] They are characterized by low surface expression of chemokine receptor 2 (CCR2), Gr-1, and Ly6C and high surface expression of the fractalkine receptor CX3CR1.[233,259] These monocytes migrate into normal tissue and differentiate into resident dendritic cells and macrophages.[258] In contrast, macrophages that infiltrate inflamed tissue have a phenotype characterized by high surface expression of CCR2, Ly6C, and Gr-1 and low surface expression of CX3CR1. It is likely that the microenvironment in tissue determines macrophage phenotype. TNF-α, IL-4, and IL-15 skew monocyte differentiation toward a dendritic cell phenotype,[260-262] whereas interferon-gamma (IFN-γ) and IL-6 direct monocyte differentiation toward a macrophage phenotype.[263,264]

Improving the ability to enhance clearance of debris (efferocytosis) by professional phagocytes, as well as tubular cells themselves, improves AKI outcomes. The junctional adhesion molecule-like protein (JAML) is upregulated in AKI kidneys and its deletion in macrophages, but not tubular cells, worsened AKI via regulation of efferocytosis through a macrophage-inducible C-type lectin-dependent mechanism.[265]



e-Fig. 27.3 Danger and stranger models. Infections of pathogenic bacteria or viruses cause the release of pathogen-associated molecular patterns *(PAMPs)* that bind to pattern recognition receptors *(PRRs)*, such as Toll-like receptors *(TLRs),* on immune cells and stimulate an innate immune response that is accompanied by inflammation and activation of adaptive immunity, and eventually processes to resolve the infection and allow for tissue repair. The danger model recognizes that similar events occur when cells are stressed or injured and that necrotic cells release molecules normally hidden within the cell. In the extracellular space, these damage-associated molecular patterns *(DAMPs)* can bind to PRRs or to specialized DAMP receptors to elicit an immune response by promoting the release of proinflammatory mediators and recruiting immune cells to infiltrate the tissue. The immune cells that participate in these processes include, for example, antigen-presenting cells *(APCs)*, such as dendritic cells and macrophages, as well as T cells and neutrophils (polymorphonuclear leukocytes *[PMNs]*). DAMPs may also stimulate adaptive immunity and participate in autoimmune responses and tissue repair. A wide variety of intracellular and extracellular molecules function as DAMPs when released from cells (see Table 27.1). The functions of such a diverse group of molecules may not yet be fully elucidated; it is unknown whether different DAMPs have specific roles, whether specific functions are elicited in different cell types or conditions, or even whether immune responses to DAMPs can be distinguished from those of PAMPs. (From Rosin DL, Okusa MD. Dangers within: DAMP responses to damage and cell death in kidney disease. *J Am Soc Nephrol.* 2011;22[3]:416–425.)

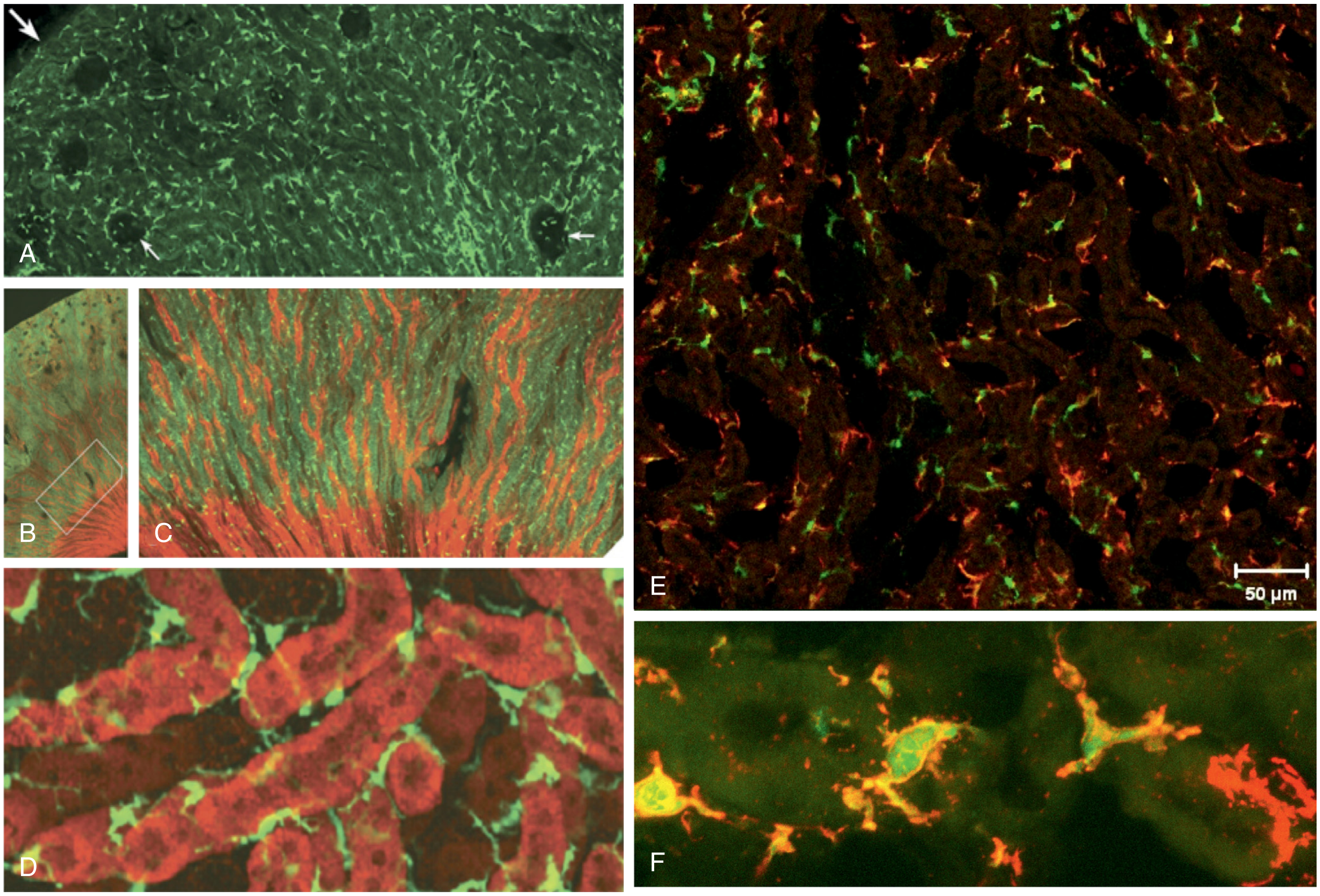


Fig. 27.6 Heterogeneity of mononuclear phagocytes within the microenvironment. Mononuclear phagocytes within the microenvironment, identified as CX3CR1+/GFP+ cells (labeled *green*), in the kidney are shown in panels A to D, and the heterogeneity of mononuclear phagocytes is shown in panel E. (A) A heterogeneous population of CX3CR1+/GFP+ mononuclear phagocytes in superficial cortex includes dendritic cells and macrophages and reside from the cortex and medulla. CX3CR1+ GFP+ cells extend to the edge of the cortex *(large arrow)* and populate the entire interstitial space abundantly. The Bowman capsule is encased by CX3CR1+/GFP+ cells *(small arrows).* Also notable are CX3CR1+ GFP+ cells that lie within each glomeruli. (B) Rhodamine-conjugated peanut agglutinin *(red fluorescence)* highlights distal tubules and collecting ducts, capsule *(upper left)* to the papilla *(lower right).* (C) Magnification of the boxed area in panel B showing CX3CR1+ GFP+ cells in the interstitium of the medulla, including transition into pyramidal tracks. Note the same spatial regularity as in the cortex. (D) Three-dimensional rendering of stellate-shaped CX3CR1+ GFP+ cells within the interstitium demarcated by tubular segments in the medulla *(red fluorescence).* (E) Kidney sections from the medulla of CX3CR1+/GFP+ mice. GFP is expressed mainly on monocyte-macrophages and dendritic cells, and many CX3CR1+GFP+ green fluorescing cells are seen in the cortex; most IA+ cells *(red label)* are seen in the medulla. Apparent in this image is the heterogeneity of CX3CR1+/GFP+–only cells, IA+–only cells, and dual labeling with CX3CR1+/GFP+ and IA+, which in the latter case represents dendritic cells in the medulla. (F) Higher magnification, Z-stack projection image of five optical slides at 0.69-mm intervals. These results demonstrate the heterogeneity of mononuclear phagocytes within the kidney microenvironment. (A–D from Soos TJ, Sims TN, Barisoni L, et al. CX3CR1+ interstitial dendritic cells form a contiguous network throughout the entire kidney. *Kidney Int.* 2006;70[3]:591–596 and E, F from Li L, Huang L, Sung SS, et al. The chemokine receptors CCR2 and CX3CR1 mediate monocyte/macrophage trafficking in kidney ischemia-reperfusion injury. *Kidney Int.* 2008;74[12]:1526–1537.)

On the other hand, boosting the efferocytosis of tubular cells, but not myeloid cells, using overexpression of chimeric efferocytic receptors alleviated inflammation in models of AKI.[266] Likewise, inhibition of retinoic acid receptors in PTECs protected against AKI that involved increased Kim-1 dependent efferocytosis, which enhanced dedifferentiation, proliferation, and metabolic reprogramming of PTECs.[267] Treatment with geniposide, an herbal compound, also improved outcomes in experimental AKI by promoting the ability of macrophages to phagocytose NETs via AMPK-PI3K/Akt signaling.[268]

Although controversial and challenged by recent studies, the original classification of M1 and M2 macrophages does provide an important functional framework.[269] M1 macrophages, also referred to as “classic macrophages,” are activated by IFN-γ and LPS, and express high levels of inducible nitric oxide synthase (iNOS). M1 macrophages have high microbicidal activity through the production of proinflammatory cytokines, such as TNF-α, IL-6, and IL-12 via signal transducer and activator of transcription 1 (STAT1), NF-κB, and ROS.[255] M2 macrophages, referred to as “alternatively activated macrophages,” are activated by IL-4 or IL-13 and participate in tissue repair and resolution of inflammation through insulin-like growth factor 1 (IGF-1), mannose receptor 1 (MRC1/CD206), and arginase-1 (Arg1) following STAT6 activation.[270]

Evidence for these disparate macrophage roles can be assessed in vivo through the administration of liposomal clodronate. Liposomal clodronate is engulfed by macrophages and clodronate, which is released from the liposome through action of lysosomal phospholipases in the macrophage, becomes toxic and kills cells by apoptosis.[271] The functional role of macrophages was determined by liposomal clodronate depletion studies that focused on the time course for macrophage depletion (early vs. late). Depletion of kidney and spleen macrophages using liposomal clodronate before renal IRI prevented AKI, and adoptive transfer of macrophages reconstituted AKI.[224] However, depletion of macrophages 3 to 5 days after IRI slowed tubular cell proliferation and repair,[255] suggesting that M1 macrophages exhibiting a proinflammatory phenotype are important for injury, whereas M2 macrophages exhibiting an antiinflammatory phenotype are important for tissue repair.[255,272,273]

Following IRI, the proinflammatory monocyte entering kidney tissue is activated to a proinflammatory macrophage through the infiltration of polymorphonuclear leukocytes, T cells, and NKT cells.[225,241,274] Resident dendritic cells activate NKT cells to produce INF-γ and other inflammatory mediators such as ROS, semaphorin-3A, DAMPs, and TNF-α, which conditions the microenvironment and favors the proinflammatory macrophage phenotype.[255] These proinflammatory macrophages then produce proinflammatory cytokines, which leads to further tissue injury. Whether polarization of macrophages from M1 to M2 involves the recruitment of circulating cells or reprogramming is unclear.[255,275] Studies by Lee and colleagues[276] have supported the concept that macrophage polarization occurs in situ. The investigators injected labeled M1 macrophages at the time of injury and examined labeled macrophages that infiltrated the injured kidneys. Most of the labeled cells maintained an M1 phenotype; however, when labeled M1 macrophages were injected 3 days after injury, most of the macrophages expressed an M2 phenotype. These studies support the concept that macrophages may be reprogrammed in situ[255,275,277] and that the kidney microenvironment provides important conditioning cues.[239]

Macrophage phenotype is also regulated by mediators of inflammation. Treatment with a synthetic C-reactive protein (CRP) was shown to improve AKI and clear infection in the cecal-ligation and puncture (CLP)-induced septic AKI model that was associated with the M2 skewing of macrophages 24 hours after AKI.[278] Similarly, stimulation of PGE_2 receptor or E-type prostanoid receptor 4 (EP4) induced MafB expression in macrophages to induce an M2 phenotype, and deletion of Cox2 or EP4 following AKI led to increased renal fibrosis.[279] Conversely, M2 phenotype macrophages are associated with fibrosis in a PTEN- and Irf4-dependent manner in late stages after AKI[280] and deletion of Arg1, a marker of M2 macrophages, attenuated renal dysfunction, and reduced expression of IL-6 and IL-1.[281]

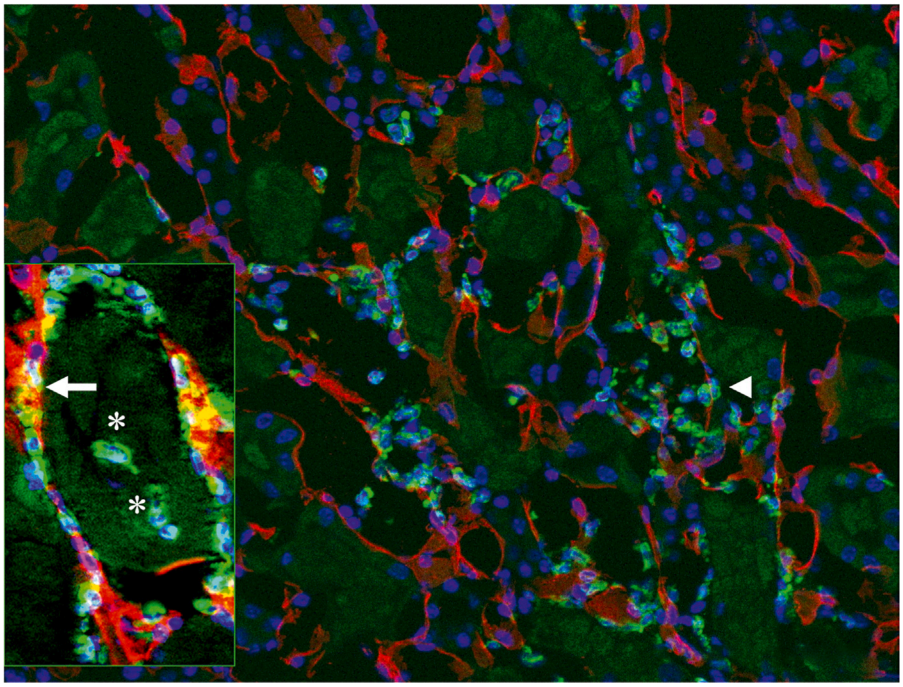
Early inflammation is classically characterized by the margination of leukocytes to the activated vascular endothelium via interactions between selectins and ligands that allows rolling and later firm adhesion, followed by transmigration.[282,283] A number of potent mediators are generated by the injured proximal TEC, including proinflammatory cytokines such as TNF-α, IL-6, IL-1β, MCP-1, IL-8, transforming growth factor-β (TGF-β), and RANTES.[250,284] TLR2 is an important mediator of endothelial ischemic injury, and TLR4 plays a similar role in animal models of both ischemic and septic injury,[285] especially in PTC.[286] In septic AKI, F4/80 (EGF-like module-containing mucin-like hormone receptor-like 1 or EMR1) deletion in macrophages led to higher kidney IL-6 levels that were ameliorated by anti-IL-6 therapy. It was hypothesized that the F4/80^{hi} macrophages express IL-1R antagonist to restrict IL-6 production and limit septic AKI.[287]

The neuroimmune axis[288] regulates AKI; pulsed ultrasound[289,290] and vagus nerve stimulation[291,292] activate the cholinergic antiinflammatory pathway and are renoprotective. This stimulation activated α-7 nicotinic receptor (α7nAChR) expressed in splenocytes and peritoneal macrophages to attenuate AKI, and adoptive transfer of activated splenocytes or peritoneal macrophages reduced IRI in recipient mice.[289,292,293] This protection was abrogated in the absence of α7nAChR or upon splenectomy and was dependent on the expression of hairy and enhancer of split-1 (Hes1), a basic helix-loop-helix DNA-binding protein, in macrophages.[289,292,293]Administration of GTS-21, an α7nAChR-specific agonist, reduced the production of TNF-α in macrophages [294] and in cisplatin-induced AKI.[295]

Thus a diverse set of data indicate that the kinetics and localization of macrophages during and after AKI may dictate the role of macrophage phenotype. While the early infiltration of monocyte-derived M1 phenotype macrophages may be proinflammatory, their later skewing to an M2 phenotype or the kidney-resident macrophages may play a protective role in AKI. An atlas of the murine kidney-resident macrophage system in the normal and injured kidney identifies several clusters of macrophages; the largest clusters in the papilla near the collecting ducts may drive fibrosis and those in the cortex around the PTC are enriched for iron-handling transcripts. Induction of AKI disrupted localization of all the clusters, which was not fully restored even 28 days after the injury; persistent macrophage-disrupted localization as a result of severe injury may dictate increased susceptibility to AKI-to-CKD transition.[296]

Polymorphonuclear leukocytes (neutrophils) and NETosis in acute kidney injury. Neutrophils accumulate early in ischemic injury in animal models.[245] Both IL-17A and IFN-γ are produced by neutrophils and may positively regulate neutrophil transmigration to the injured kidney following kidney IRI[241] (eFig. 27.4). Although neutrophils are seen early in rodent models of AKI, whether they play a pathogenic role remains controversial because inhibition or depletion studies of neutrophils variably leads to either protection[282,297,298] or lack of protection.[299] Blockade of neutrophil function or neutrophil depletion provides only partial protection against injury. Vascular adhesion protein-1 (VAP-1) is an adhesion molecule[300,301] associated with inflammatory conditions. VAP-1 is expressed primarily in pericytes, and a specific VAP-1 inhibitor, RTU-1096, attenuated renal IRI and decreased neutrophil infiltration. The protective effect of VAP-1 inhibition was absent in neutrophil-depleted rats, suggesting an important role of neutrophil infiltration.[302]

NETosis is a specific form of regulated neutrophil death in which neutrophils release neutrophil extracellular traps (NETs), which are extracellular structures composed of



e-Fig. 27.4 Localization of neutrophils to interstitial and marginated compartments in the kidney. Immunofluorescence staining of kidney outer medulla using antibodies to 7/4 (*green*, neutrophils) and CD31 (*red,* vascular endothelium). Nuclei are depicted by DAPI labeling *(blue).* Neutrophils in both sides of the vascular endothelial wall. *Inset,* Z-stack image (7.0 μm) of 12 optical slices of the kidney at 0.6-μm intervals. Shown are kidney neutrophils in the interstitium and peritubular capillaries *(arrows)*; neutrophils that have transmigrated into the lumen of the tubule *(*)* are shown in the inset. (From Awad AS, Rouse M, Huang L, et al. Compartmentalization of neutrophils in the kidney and lung following acute ischemic kidney injury. *Kidney Int.* 2009;75[7]:689–698.)

chromatin and histones that enable immobilization and killing of bacteria.[303] Infiltrating neutrophils undergoing NETosis contribute to organ damage in ischemic AKI. In the IRI model, infiltrating neutrophils undergo NETosis, leading to the release of cytotoxic DAMPs, such as histones, which exacerbate TEC injury and interstitial inflammation.[304-308] Neutrophil peptidyl arginine deiminase-4 (PAD4) plays an important role in the formation of NETs.[309-311] In CKD patients, SARS-CoV-2 led to the induction of vasculopathy leading to AKI via greater NETosis and microthromboses.[312,313] In sepsis-induced AKI in mice and humans, colocalization of NETs and fibroblast growth factor–inducible molecule 14 (Fn14) was seen. Simultaneous PAD4 deletion and Fn14 inhibition alleviated AKI.[314] These findings suggest that NETosis is an important early target for intervention in diverse forms of AKI.

Adaptive immunity and inflammation

T cells in acute kidney injury. Early work by Burne-Taney and coworkers[299,315,316] demonstrated that T lymphocytes contribute importantly to renal IRI. However, conventional CD4+ T cells are thought to play an obligatory role in antigen-specific, cognate immunity that requires 2 to 4 days for naïve T cell processing, a time course that cannot explain the rapid, innate immune response following IRI. Studies have shown that RTEC may directly stimulate the proliferation of TCR double negative (DN) T cells and not peripheral CD4 and CD8 T cells are stimulated via this mechanism.[317] Infiltration of T cells and the presence of T cell cytokines (IL-2, IL-10, and TNF-α) was markedly increased in the urine of patients who developed AKI associated with immune checkpoint inhibitors (AKI-ICI).[318] T cells can be identified in kidneys several weeks after AKI despite recovery of renal function.[319] In a unilateral IRI model, without contralateral nephrectomy, there was a second wave of T cell and neutrophil infiltration around day 14 that was accompanied by higher expression of tubular injury genes and decreased proportion of tubules when compared with mice that had contralateral nephrectomy.[320]

Natural killer T (NKT) cells are a T cell sublineage[321] and, in mice, NKT cells express an invariant T cell receptor TCRα chain, Vα14-J18. In contrast to conventional T cells, the NKT cell TCR does not interact with peptide antigen presented by classic major histocompatibility complex (MHC) class I or II; rather, it recognizes glycolipids presented by the class I–like molecule, CD1d. Upon activation of NKT cells, vigorous cytokine secretion occurs within 1 to 2 hours, including both Th1-type (IFN-γ, TNF) and Th2-type (IL-4, IL-13) cytokines.[321-327] The rapid response by NKT cells following activation can amplify and regulate the function of dendritic cells, Treg cells, NK and B cells, and conventional T cells and thus link innate and adaptive immunity.[328-333] In AKI, NKT cells participate in the early innate immune response to kidney IRI.[225] NKT-produced IFN-γ was found as early as 3 hours following IRI,[225] supporting previous work demonstrating that CD4+ T cell IFN-γ production is responsible for kidney injury.[334,335] These data demonstrate the central role of IFN-γ from CD4+ T and/or NKT cells in the pathogenesis of renal IRI.

T cell receptor+CD4−CD8− DN T cells may be protective post AKI.[336] DN T cells are present in both mouse and human kidneys and are enriched for the expression of *Kcnq5, Klrb1c, Fcer1g,* and *Klre1*.[337] Mouse kidneys are also enriched for CD4+ T cells expressing the checkpoint molecule T cell immunoreceptor with immunoglobulin and ITIM domain (TIGIT) after IR injury, but the population was restricted to T/natural killer cell subsets in patients with AKI. Kidney Treg cells also expressed TIGIT at baseline that was reduced post-IRI. In another study, deletion of Kelch-like ECH-associated protein 1 (Keap1) in CD4+ T cells increased the expression of Nuclear factor erythroid2-like 2 (Nrf2) target genes, and adoptive transfer of Keap1-KO CD4+ T cells before IRI was protective in IRI in T cell–deficient nu/nu mice.[338]

The gut microbiota influences T cell phenotype and its consequence for AKI, as colonization of mice with post-AKI microbiota worsened AKI outcomes and microbiota depletion with antibiotics protected against IRI[339] or accelerated recovery,[340] along with lower kidney Th1, Th17 cells, and TNF-α expressing double negative T cells and higher Treg, Foxp3+CD8+ T cells.

Treg cells[228,229] also play a role in ischemic AKI.[341,342] In a murine model of ischemic AKI, there is significant trafficking of Treg cells into the kidneys after 3 and 10 days.[343] Anti-CD25 antibodies increased IRI-induced tubule damage, tubular proliferation, immune cell infiltration, and cytokine production.[343,342] FoxP3 (forkhead box P3)+ Treg cell–deficient mice accumulated more inflammatory leukocytes after renal IRI than mice containing Treg cells, and cotransfer of isolated Treg cells and Scurfy lymph node cells (Treg deficient lymph node cells) significantly attenuated IRI-induced renal injury and leukocyte accumulation.[342] These studies demonstrate that Treg cells infiltrate kidneys after IRI and promote repair, likely through modulation of proinflammatory cytokine production of other T cell subsets.[343]

The capacity of Treg cells to protect mice from IRI depends on IL-10 production, adenosine production through CD73, expression of the adenosine 2A receptor, and programmed cell death 1 (PD-1) on the cell surface.[229] PD-1, a negative costimulatory molecule expressed on T lymphocytes, monocytes, dendritic cells, and B cells,[344,345] is indispensable for Treg function. Administration of monoclonal antibodies to PD-L1 or a genetic deficiency of PD-1 on Tregs exacerbated impaired kidney function and ATN after subthreshold ischemia.[346] Stremska and coworkers[347] synthesized a hybrid cytokine, IL233, which combines IL-2 and IL-33. Tregs require IL-2 for homeostasis and upregulate IL-33R, which promotes their recruitment and activation of innate lymphoid cells (ILC2s) in response to IL-33.[348] Collectively, ILCs are the latest family of innate immune cells to be discovered, originating from common lymphoid progenitors (CLPs). In response to pathogenic tissue damage, ILCs play a crucial role in immunity by secreting signaling molecules and regulating both innate and adaptive immune cells. Primarily residing in tissues, ILCs are found in both lymphoid (immune-associated) and nonlymphoid tissues and are rarely present in the blood. ILC2 cells, or type 2 innate lymphoid cells, are a subset of innate lymphoid cells. Unlike ILCs as a broader category, ILC2 cells specifically belong to the lymphoid lineage and are derived from common lymphoid progenitors. These cells do not possess antigen-specific B or T cell receptors due to the absence of recombination-activating genes. Administration of the novel hybrid cytokine, IL233, increased endogenous Tregs in blood and spleen and prevented IRI more efficiently than a mixture of IL-2 and IL-33 administered separately. IL233 also increased the proportion of ILC2s in blood and kidneys, and adoptive transfer of ILC2s protected mice from IRI.[347] Thus the many barriers to cell-based therapy may be overcome through this hybrid cytokine, which increases endogenous Tregs. We need a greater understanding of the role of ILCs, especially the ILC2 subtype, in AKI. We argue that the rapid translation to human studies would be a major advancement in the AKI field.

B lymphocytes. Studies to evaluate the role of B cells and the B1 subset on ischemia-induced AKI have remained controversial, with some studies showing that B cells may be deleterious. Burne-Taney and associates showed that μMT mice, which lack B cells and all immunoglobulins, are protected from AKI despite similar levels of infiltrating granulocytes and macrophages in the postischemic kidneys of wild-type (WT) mice.[226] Sensitivity to ischemia-induced AKI was restored after replenishing these mice with WT serum, but not B cells, thus indicating that a serum factor enhanced the cytotoxic effect of infiltrating granulocytes and macrophages on the ischemic tubular cells. However, Renner and colleagues,[349] using the same μMT mice, could not show that these mice were protected from ischemia-induced AKI. In fact, they had more severe injuries than control WT mice. Additionally, Lobo and associates[350] failed to show protection from ischemia-induced AKI when WT mice were acutely depleted of B cells (with anti-CD20) and not immunoglobulins. More studies are needed to study the role of B cells and immunoglobulins in AKI; it may be preferable to deplete B cells or subsets of B cells acutely to study their role. μMT mice have other immune deficiencies, including low Tregs and lack of T cell receptor (TCR) diversification, which may have contributed to more severe injury in the experiments by Thurman and colleagues.[351,352] Deletion of CD157/Bst1 (a modulator of immune response), mainly found in B cells and neutrophils, attenuated migration of neutrophils and renal IRI.[353] BAFF (B cell activating factor) also plays a proinflammatory role in AKI, and deletion of BAFF can protect kidneys from IRI. Surprisingly, BAFF-receptor KO worsened AKI. The basis for this paradox is not fully understood.[354]

The role of the B1 subset of B cells in murine models of ischemia has also been controversial. Natural IgM produced by B1 cells is deleterious in ischemia-induced injury of murine skeletal muscle, cardiac muscle, and bowel.[352,355,356] Conversely, Lobo and colleagues[357] showed that natural IgM protects WT kidneys from ischemic AKI by inhibiting the innate inflammation that occurs after ischemic injury. Such observations demonstrate that the mechanisms that cause ischemic injury in the kidney may be different from those of other organs.

Inflammation

Altered endothelial cell function mediates inflammation, and ischemia increased the expression of leukocyte adhesion molecules, such as P-selectin, E-selectin, intercellular adhesion molecule (ICAM), and B7-1. Consequently, strategies to pharmacologically block or genetically ablate the expression of these endothelial cell surface proteins can protect animals in models of ischemic or septic AKI.[298]

Whether released from the endothelium or the epithelial cell, numerous cytokines exert a concerted biochemical effort to augment the inflammatory response to ischemic or septic injury.[227] Toll-like receptors (TLRs) are a crucial family of receptors that serve as a first-line defense, especially against microbes. They have the ability to recognize both invading pathogens and endogenous danger signals released from dying cells and damaged tissues. TLRs play a vital role in bridging innate and adaptive immunity. Cultured mouse tubular cells, when stimulated with LPS, upregulate TLR2, TLR3, and TLR4 and secrete CC-chemokines such as CC motif chemokine ligand 2 (CCL2)/MCP-1 and CCL5/RANTES, suggesting that tubular TLR expression might be involved in mediating interstitial leukocyte infiltration and tubular injury during bacterial sepsis.[358] In CLP rat model of sepsis, the expression of TLR4 is markedly increased in proximal and distal tubules, glomeruli, and the renal vasculature.[359] TLR2 and TLR4 are constitutively expressed on renal epithelium, their expression is enhanced following renal IRI and their genetic deletion is protected from renal IRI[286,360] and cisplatin-induced AKI.[361] In this later study using bone marrow chimeras, renal parenchymal TLR4, rather than hematopoietic-derived TLR4, mediated cisplatin-induced nephrotoxicity, which is consistent with other studies[286] and clearly demonstrates the important role TLR plays in AKI.

DNA microarray analysis has shown that pentraxin 3 (PTX3), a protein that can be induced in vascular enothelial cells, is upregulated in ischemic kidneys of TLR4-sufficient mice predominantly on the peritubular endothelia of the outer medulla of the kidney and plasma, when compared with TLR4-deficient mice. Knockout of PTX3 or endothelium-specific deletion of MyD88 ameliorated ROS and AKI.[362,363] Thus the endothelial PTX3/TLR4/MyD88 pathway plays a pivotal role in the pathogenesis of ischemic AKI.[363]

The role of glomerular endothelial injury in AKI is unclear. Studies in a mouse model of LPS-induced sepsis have shown decreased abundance of endothelial surface layer heparin sulfate proteoglycans and sialic acid, leading to albuminuria, likely reflecting altered glomerular filtration permselectivity and decreased expression of podocyte VEGF. LPS treatment decreased GFR, caused ultrastructural alterations in the glomerular endothelium, and lowered the density of glomerular endothelial cell fenestrae. These LPS-induced effects were diminished in TNF receptor 1 (TNFR1) knockout mice, suggesting the role of TNF-α activation of TNFR1. Of interest, intravenous administration of TNF also led to a decreased GFR and loss of glomerular endothelial cell fenestrae, increased fenestrae diameter, and damage to the glomerular endothelial surface layer. Thus glomerular endothelial injury, mediated by higher TNF-α and lower VEGF levels, extends the development and progression of AKI and albuminuria in the LPS model of sepsis in the mouse.[364]

In addition to immune cells responding and sensing DAMPs and PAMPs (see next paragraph and following section on Complement), PTCs function as a sensor of both self and nonself; DAMPs and PAMPs serve as recognition signals for pattern recognition receptors (PRRs) such as TLR4.[365] Proximal tubule TLR4 is upregulated and migrates to the apical domain in response to LPS in S1 PTCs, which are the earliest segments of epithelial cell postglomerular filtration.[359] Interestingly, the S1 cell internalizes and processes LPS via TLR4 receptors, which is inducible with preexposure to LPS but is protected from injury by upregulated defense mechanisms, including heme oxygenase-1 (HO-1) and sirtuin 1 (SIRT1), two cytoprotective proteins. However, S2 to S3 PTCs undergo oxidative injury with minimal uptake of LPS, implying communication, crosstalk, and coregulation between the segments following LPS exposure.[366] This injury is dependent on CD14, likely due to peroxisomal disruption, perhaps mediated by TNF-α, and the PTC injury was independent of systemic cytokines. Treatment with TLR4 agonist PHAD (3-deacyl 6-acyl phosphorylated hexaacyl disaccharide) preserved kidney function in a dose-dependent manner after IRI-AKI.[367]

Epithelial TLR4 and MyD88 mediate ischemic injury.[286] The relative contribution of epithelial versus hematopoietic TLR4 to kidney damage following IRI was assessed using bone marrow chimeras in which TLR4$^{-/-}$ mice were engrafted with WT hematopoietic cells (and vice versa). Both hematopoietic and parenchymal TLR4 contributed to kidney injury, although the effect was more pronounced when TLR4 was expressed only on parenchymal cells. These results suggest that TLR4 signaling on hematopoietic and intrinsic kidney cells contributes to kidney damage, but a more significant role is played by intrinsic kidney cells. Similar results with TLR2$^{-/-}$ mice and chimeric mice suggest that epithelial TLR2 also plays a prominent role during ischemic injury.[368] TLR2 recognizes a wide range of microbial components, including lipoproteins, peptidoglycans, and lipoteichoic acids from bacteria, as well as components from fungi, viruses, and parasites. Cytokine and chemokine production was reduced and white blood cell infiltration was minimized in chimeric mice using antisense therapy. These studies suggest that renal-associated TLR2 is an important initiator of inflammatory responses that lead to renal injury.

Tamm-Horsfall protein (THP; also known as "uromodulin"), a heavily glycosylated protein uniquely produced in the kidney by TALs, modulates kidney innate immunity and inflammation during kidney injury.[365,369,370,371] THP is secreted primarily from the apical membrane of TAL into the urine but is also released basolaterally toward the interstitium and circulation to inhibit tubular inflammatory signaling.[372] THP knockout mice compared with WT controls subjected to kidney IRI showed increased S3 injury and necrosis,[370,373] neutrophil infiltration in the outer medulla, and expression of TLR4 and CXCL2 by S3 segments.[370,373] After IRI, intracellular uromodulin trafficking shifted; THP was redirected from the apical membrane to the basolateral compartment and membrane of the S3 segment and the interstitium despite unchanged epithelial polarity[374] where a putative receptor for uromodulin is expressed.[370] THP$^{-/-}$ mice had worse injury and impaired transition of kidney macrophages toward an M2 healing phenotype, suggesting that interstitial THP may regulate not only mononuclear phagocyte number but also plasticity and phagocytic activity.[372] A significant increase in uromodulin expression was shown in the kidney at the onset of recovery, which was concomitant with the suppression of tubular-derived cytokines and chemokines such as MCP-1, supporting the concept that the protective crosstalk mediated by uromodulin may be important in modulating recovery from AKI.[374]

COMPLEMENT

The complement system is part of the host defense machinery that protects against microbial invasion after injury.[375] The complement system participates in the pathogenesis of a wide range of kidney diseases (IRI, sepsis, and cisplatin),[376-379] as well as in rare kidney diseases of children including atypical hemolytic-uremic syndrome and C3 glomerulopathy.[380] The reactivity and specificity of the complement system are accomplished via a series of circulating pattern recognition proteins (PRPs) that sense PAMPs and initiate the complement cascade (see "Supplement" later). During ischemic injury, CR5a expression is markedly upregulated on proximal tubule epithelial cells and interstitial macrophages. C5a is elevated in rodent models of sepsis, and blocking it or its receptor improves survival in sepsis.[381]

Several complement regulatory proteins are expressed on the surface of kidney cells to inactivate complement convertase and protect tissue cells from complement-mediated damage in the pathogenesis of IRI-induced AKI.[377,382] Global deficiency of factor B, factor H, and Crry reduced IRI-induced AKI. In addition to AKI, increased expression of complement C3 fragments and anaphylatoxin receptors C3aR and C5aR1 is found in PDGFRβ-positive pericytes and immune cells isolated from fibrotic kidneys. Inhibition of complement activation by global deletion of C3 reduces tubulointerstitial fibrosis[378] that was associated with reduced neutrophil activation[383]and NETosis.[384] Activation of C5a induced aberrant methylation in chromatin regions involved in cell cycle, DNA damage, and Wnt signaling in RTEC and increased IL-6, MCP-1, and CTGF expression. Treatment with complement inhibitor (C1-Inh) attenuated this activation.[385] Anti-C5 antibody after IRI ameliorated complement activation and tubular necrosis.[386] Given the increased awareness of the role of complement activation, not only with AKI but also in models of CKD, it is likely that emerging therapies aimed at inhibiting complement activation could be used in future studies to reduce the evolution of AKI and/or the progression of CKD in adults and children.

INTRACELLULAR MECHANISMS

Heat shock proteins

Much of the previous discussion has been on proteins or mechanisms that promote injury. However, there are protective mechanisms that allow cells to defend against numerous stresses. The role of heat shock proteins in AKI is discussed later in "Supplement."

Reactive oxygen species

Oxidative stress plays an important role in the pathogenesis of AKI and progressive kidney disease. Low-level ROS can function as signaling molecules for cellular proliferation and vascular homeostasis under healthy conditions; however, pathologic increases in ROS generation during IRI by mitochondria, nicotinamide adenine dinucleotide phosphate (NADPH) oxidases, or inflammatory cells can aggravate tissue injury. Therefore reducing oxidative stress is considered an important therapeutic strategy to ameliorate loss of kidney function.[387]

ROS such as OH^-, peroxynitrite ($ONOO^-$) (also a member of the reactive nitrogen species, derived from NO), and hypochlorous acid (HOCl) are generated in epithelial cells during ischemic injury by catalytic conversion. ROS can damage cells via peroxidation of lipids in plasma and intracellular membranes. ROS can also destabilize the cytoskeletal proteins and integrins required to maintain cell-cell adhesion, as well as extracellular matrix. ROS can also have vasoconstrictive effects by scavenging NO.[388]

Myoinositol oxygenase (MIOX), a proximal tubule enzyme, participates in oxidant injury by increasing the generation of ROS in kidney tissue.[389,390] MIOX gene disruption ameliorates toxic PTC injury. The Keap1-Nrf2 system allows cells to sense and respond to oxidative stress; it is a master regulator of cytoprotective genes. Nrf2 is a transcription factor that can bind antioxidant response elements in target gene regulatory regions. In kidney IRI, Nrf2 signaling is upregulated in response to injury, and Nrf2 null mice developed worse disease.[391] Bardoxolone methyl ameliorates IRI, in part, through Nrf2.[392]

Given the known adverse effects of using pharmacologic enhancers of the Keap1/Nrf2 pathway,[393] one study used genetically altered mice that express low levels of Keap1 protein.[394] Reduction in levels of this inhibitor of Nrf2 signaling leads to enhanced Nrf2 target transcription. Nrf2 enhancement was protective in the model of ischemia-mediated AKI, as well as in the model of unilateral ureteral obstruction. Decreased levels of Nrf2 in hypoxia were prevented by inhibition of mitochondrial complex I, which protected mice from AKI demonstrating that Nrf2 and HIF1α interact to provide optimal metabolic and cytoprotective responses in ischemic AKI.[395] These results underscore the importance of developing drugs that selectively target increased Nrf2 transcription to ameliorate AKI.

Iron, ferritin, and heme oxygenase

Iron plays a central role in fundamental biological functions, such as in mitochondrial respiration and DNA repair, yet it plays a detrimental role in the pathophysiology of various diseases.[396] Circulating iron is mostly transferrin bound. A small amount bound to low-molecular-weight chelates, referred to as *labile* or *catalytic iron,* is available in biological systems and is thought to be responsible for kidney injury.[397] Although iron often undergoes cyclic oxidation and reduction, via ferric and ferrous ion generation, respectively, to perform its normal functions, this redox activity generates free radicals, leading to lipid peroxidation or generation of superoxide radicals through reaction with hydrogen peroxide (Haber-Weiss reaction).[398] The toxicity of catalytic iron to macromolecular components of the cell and its causal role in mediating disease have been demonstrated through the protection achieved by iron chelators.[66,397,399-402] Heme iron is derived primarily from erythrocyte turnover and is the major contributor to iron cycling in the body.[396]

Kidney tissue iron content increases after AKI,[403-406] suggesting that regulators of iron metabolism such as hemojuvelin and hepcidin could be used as therapeutic targets for the treatment of AKI. IRI causes an increase in serum iron and kidney nonheme iron levels, and hepcidin administered 24 to 48 hours before IRI significantly reduced IRI-induced apoptosis, oxidative stress, and inflammatory cell infiltration.[404] Hepcidin also provides protection in the hemoglobin-mediated AKI model.[407] Ferritin, the major regulator of intracellular iron, is composed of heavy-chain ferritin (H-ferritin) and light chain ferritin (L-ferritin).[108] H-ferritin has ferroxidase activity converting Fe^{2+} to Fe^{3+}, which permits the incorporation of reactive iron into ferritin to avoid iron-induced ROS generation. Mice with proximal tubules deficient of H-ferritin had higher mortality, worse tissue injury and renal function, and increased apoptosis in both cisplatin and rhabdomyolysis models of AKI. Notably, there was disrupted iron trafficking, with altered distribution of ferroportin (iron export transporter) in the proximal tubule of H-ferritin–deficient mice. Iron deficiency exacerbated cisplatin-induced AKI and rhabdomyolysis by markedly increasing nonheme catalytic iron and Nox4 protein, which together catalyze production of hydroxyl radicals followed by protein and DNA oxidation, apoptosis, and ferroptosis.[408] Hemopexin, a heme-scavenging protein, and hemoglobin accumulated in PTC in the kidneys during AKI and correlated with increased KIM-1 and HO-1 that were downregulated in hemopexin$^{-/-}$ mice in AKI. Deferoxamine, an iron chelator, and ferrostatin-1, a ferroptosis inhibitor, inhibited the deleterious effects of hemoglobin and hemopexin in PTC, implicating hemopexin as a mediator of iron toxicity in AKI.[409] These results support an important role of proximal iron homeostasis during AKI.[108]

Heme oxygenase-1

The enzyme HO-1 contributes in renal TEC protection.[410-413] The biologic actions of HO-1 include antiinflammatory, vasodilatory, cytoprotective, and antiapoptotic effects and regulation of cellular proliferation in the setting of AKI. HO-1 is arguably one of the most readily inducible genes, responding to numerous stressors including, but not limited to, hypoxia, hyperthermia, oxidative stress, and exposure to LPS. HO-1 is induced in various forms of AKI, including ischemic, endotoxic, and nephrotoxic models. Following ischemia-reperfusion, aged mice exhibit reduced induction of renal medullary HO-1 and worse renal injury than younger mice. Prior induction of HO-1 by hemoglobin can reduce LPS-induced renal dysfunction and mortality. Inhibition of HO-1 activity in the intact, disease-free kidney reduces medullary blood flow without exerting any effect on cortical blood flow, perhaps reflecting changes in endogenous carbon monoxide production. Overexpression of HO-1, induced by hemin, results in a significant reduction in cisplatin-induced cytotoxicity,[414,415] and TNF-α–induced apoptosis in endothelial cells is attenuated by HO-1 induction.

HO-1–deficient mice revealed marked exacerbation of renal insufficiency and mortality in glycerol-induced AKI.[416] HO-1 in cultured renal epithelial cells induces upregulation of the cell cycle inhibitory protein p21 and confers resistance to apoptosis and ferroptosis.[417,187] Induction of ferroptosis indicators (erastin and RSL3) was less profound in HO-1$^{+/+}$ compared with HO-1$^{-/-}$ PTCs. Treatment with ferrostatin-1, a ferroptosis inhibitor, significantly improved viability. Macrophages in which HO-1 is upregulated by adenoviral strategies also protect against ischemic AKI. Fibroblasts from organs of transgenic pigs expressing HO-1 are resistant to proapoptotic stressors and exhibit a blunted proinflammatory response to LPS or TNF-α. Pretreatment with hemin augments glomerular HO-1 expression and renal expression of thrombomodulin and endothelial cell protein C receptor (EPCR) while reducing LPS-induced renal dysfunction, glomerular thrombotic microangiopathy, and the procoagulant state. Hemin also increases plasma levels of activated protein C in this model, suggesting its important role in the endothelial-epithelial axis in AKI. The connection between ROS and HO-1 has prompted studies on the use of antioxidants in AKI. Treatment with antioxidants such as oxypurinol,[418] an active metabolite of allopurinol, or 6-Shogaol,[419] a component of ginger, restored renal function and attenuated inflammation and tubular damage after renal IRI.

Mice deficient in Pannexin 1 (Panx1), a channel-forming protein involved in ATP release, were protected from IRI,[420,421] which also correlated with induction of HO-1 expression and inhibition of ferroptinophagy via the MAPK/ERK pathway.[421] Mechanistically, PANX1 disrupts mitophagy by influencing the ATP-P2Y-mTOR signal pathway.[422]

HO-1 might contribute to the repair and regeneration of tubular cells.[111,423] Following an acute insult, HO-1 is rapidly induced, but its expression subsides before renal recovery fully occurs; such abatement in HO-1 expression may allow the continued expression of proinflammatory and fibrogenic genes. In this regard, HO-1 deficiency promotes epithelial-mesenchymal transition, a process that may underlie the transition of AKI to CKD.[411] HO-1 gene expression is regulated differently in mice and humans; however, a study with a novel humanized transgenic mouse confirmed the rescue of the pathologic phenotypes observed in $HO^{-/-}$ mice by the human HO-1 gene.[410] These mice offer an important tool to study the mechanisms of regulation of human HO-1 gene. In addition, the protective effects of HO-1 promoter polymorphisms in AKI could allow us to understand its significance better in clinical contexts and enable the identification of potential novel therapeutic targets.

REPAIR AND REGENERATION IN ACUTE KIDNEY INJURY

Human kidneys subjected to mild injury have the ability to recover renal function. Experimentally, the renal tubule has the capacity to undergo regeneration within a few days after AKI.[424] Early studies that focused on progenitor-stem cells in TEC injury found that different types of stem cells may reside in the renal architecture. In the human kidney, $CD133^+$ progenitor-stem cells with regenerative potential have been identified.[425-429] Hematopoietic cells do not contribute significantly to the repair of epithelium following injury, as shown by chimeric studies using enhanced GFP-labeled donor bone marrow.[430]

Currently, there are likely two major hypotheses to explain the recovery of renal function by tubule regeneration[431] (Fig. 27.7). The first hypothesis suggests that tubules regenerate from any surviving TECs without the contribution of a preexistent intratubular stem cell or progenitor population. Genetic fate-mapping studies have demonstrated that regeneration following injury is achieved primarily by surviving epithelial cells.[432,433] Lineage-tracing studies have demonstrated that fully differentiated epithelial cells that survive an AKI episode undergo a process of reversible dedifferentiation and proliferation

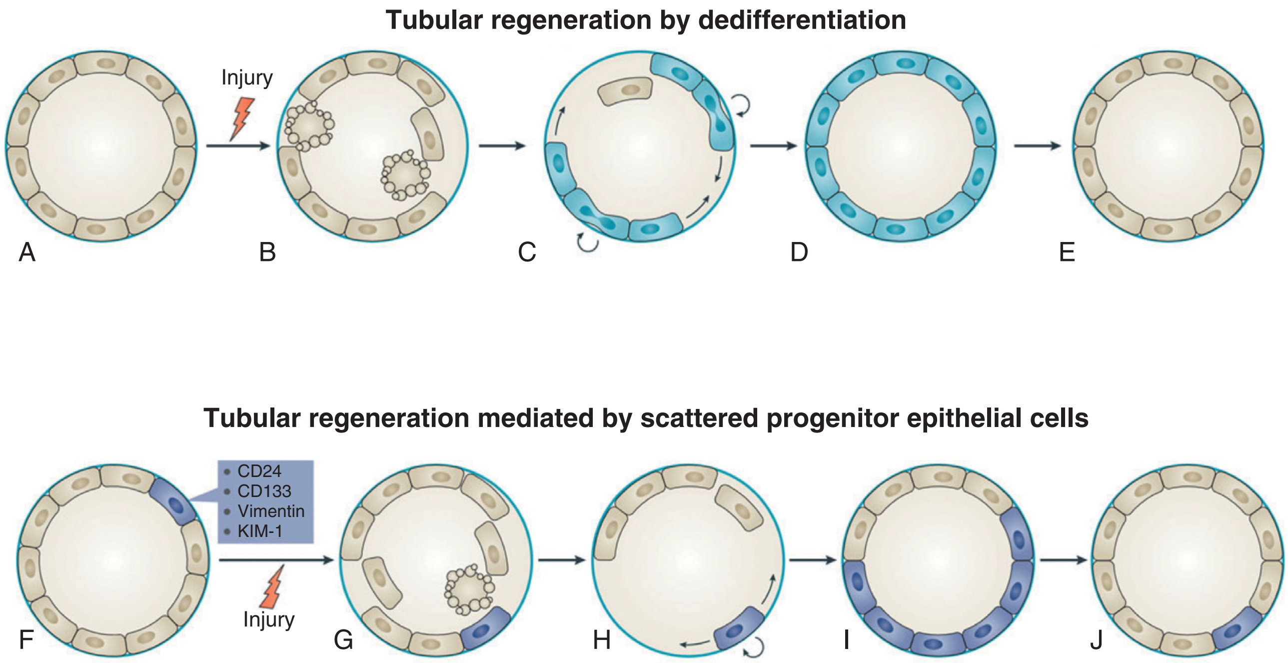


Fig. 27.7 Tubular regeneration following acute kidney injury. *Top panel,* Tubular regeneration by dedifferentiation. (A) Healthy tubules consist of nonproliferative mature epithelial cells that express markers of differentiation. (B) After injury, the epithelium is lost through apoptosis and necrosis. (C) Surviving epithelial cells dedifferentiate, either in response to sublethal injury signals or owing to signals from other injured cells, and acquire a proliferative phenotype. (D) The surviving dedifferentiated cells reconstitute the nephron epithelium. (E) Ultimately, most dedifferentiated cells redifferentiate and downregulate the expression of dedifferentiation genes. *Lower panel,* Tubular regeneration mediated by scattered progenitor epithelial cells. (F) Scattered tubular epithelial cells express signal transducer CD24, prominin-1 (CD133), and other genes that are characteristic of proximal tubule dedifferentiation, such as vimentin and KIM-1. (G) After injury, mature tubular cells, but not scattered cells, undergo apoptosis. (H, I) Scattered tubular cells expand in response to injury and their progeny reconstitute the tubule. (J) The small subpopulation of scattered tubule cells is preserved after regeneration. (From Chang-Panesso M, Humphreys BD. Cellular plasticity in kidney injury and repair. *Nat Rev Nephrol.* 2017;13[1]:39–46.)

during repair.[434] Another independent group used the same methodology of genetic cell mapping and doxycycline-inducible parietal epithelial cell mapping (PEC)–specific transgenic mouse to label proximal tubules in the model of IRI. They found a significant increase in genetic labeling of proximal tubules, with increased expression of scattered tubular cell biomarkers such as CD24 and CD133, during ischemia and subsequent recovery, suggesting that scattered tubular cells arise from any surviving tubular cell (upper panel, see Fig. 27.7).[435]

The second hypothesis suggests that tubules regenerate from a specific tubular cell subpopulation with high regenerative potential or so-called "scattered tubular cells"[434,436-438] (lower panel, see Fig. 27.7). Another study supports this hypothesis and challenges the current paradigm that functional recovery after AKI relates to a regenerative capacity of all TECs. These investigators applied a lineage tracing approach using conditional Pax8-Confetti mice and the models of ischemia-reperfusion and glycerol-induced AKI to demonstrate the following: 1. AKI involves a permanent loss of TECs, even when GFR recovery occurs; 2. Pax2-positive TECs located in the S3 segment of the proximal tubule are endowed with higher resistance to death and are responsible for the spontaneous regeneration of necrotic tubules after AKI; 3. only Pax2-positive cells complete mitosis while other TECs go through a process of endoreplication-mediated hypertrophy that accounts for the recovery of renal function; and 4. this process of endoreplication is the dominant cell response on AKI in mice and can be detected in renal biopsy tissue of patients who developed CKD after AKI.[439] The therapeutic targeting of tubule progenitors carrying specific biomarkers of scattered tubular cell phenotype could be a promising new strategy to improve long-term outcomes after AKI.

Expression of transcriptional factor FoxM1 that is downstream to glycogen synthase kinase-3 β (GSK3β) attenuates renal repair and promotes fibrosis. Inhibition of FoxM1 with thiostrepton or Gskβ improved renal repair and function by increased β-catenin, Cyclin D1, and c-Myc, and reduced cell cycle inhibitors p21 and p27.[440,441] Tubule repair involves activation of a PIK3C3-mediated mechanism for sensing of amino acid availability and PTC growth via the MTORC1-S6K1-rpS6 signaling pathway leading to the clonal expansion of the tubular progenitor cells.[439,441]

Several factors regulate a productive versus maladaptive repair,[442] as shown by transcriptomic studies.[195,443-445] Maladaptive cells continued to express inflammatory transcription factors and cytokines (IL-1b, IL-34, and Cxcl2). Ferroptosis and pyroptosis are key druggable targets, as treatment of mice with VX765 or liproxstatin protected from maladaptive repair and fibrosis.[195] Tubules that failed to repair properly were enriched for proinflammatory genes (*Vcam1*, *Dcdc2a*, *Sema5a*, *Relb,* and *NFkB*) and had downregulation of terminal differentiation markers (*Slc5a12, Slc22a30,* and *Slc7a13*),[443] similar to findings showing Vcam1$^+$/Ccl2$^+$ Slc22a7$^-$ PTC as the maladaptive PTCs.[444] On the other hand, successful repair tubules were enriched for expression of transcription factors that drive differentiation (*Hnf4a, Hnf1b,* and *Pbx1*), as well as solute carriers (*Plxdc2, Gas2,* and *Dab2*). Other studies[446] identified nephrogenic features of the repairing juvenile kidneys as compared with failed-repair aged kidneys (*Sox4, Cd24a, Npnt, Lhx1, Osr2, Foxc1, Hes1, Pou3f3,* and *Sox9*). The productive repair was associated with pericyte and vascular smooth muscle cells with strong expression of Notch (Notch3 and Jag1), consistent with their role in development and angiogenesis.[443] Meanwhile, in the failed-repair tubules, glucose *(Slc5a2, Hmox1)* and amino acid metabolism (*Bcat1, Slc6a6, Slc7a1*, *Bckdha, Bckdhb* and *Ppm1k*) pathways were dysregulated and correlated with upregulation of fibrogenic genes (*Tgfbr1, Map3k1, Stat3,* and *Myh9*) This was accompanied by dysregulation of lactate metabolism (*Ldha* and *Ldhb*) that was independently identified as being related to fibrogenesis by Agarwal and coworkers.[447] AKI leads to a transient lipid accumulation in PTC along with an increase in genes related to uptake and metabolism of lipids (*Cpt1a, Acox1, Hadha,* and *Hadhb*) or lipid handling (*Plin2, Fabp4, Acsl4,* and *Ehd1*), as well as oxygen consumption rate (OCR) for metabolism of the consumed lipids, likely as a consequence of tubule-mediated efferocytosis and autophagy. This was accompanied by upregulation of genes involved in DNA replication, cell cycle, and proliferation, all high-energy demanding processes.[448] These results paint a comprehensive picture of different shared and unique states in injury and repair responses in epithelial cells across the nephron and kidney stromal cells, Such data highlight the importance of metabolic adaptations leading to successful repair in AKI.

Growth factors and signals from injured cells are crucial to promote the timely and appropriate regenerative capacity of viable cells. In animal models, administration of exogenous growth factors accelerates renal recovery from injury. These include epidermal growth factor, IGF-1, α-melanocyte stimulating hormone, erythropoietin, hepatocyte growth factor, and bone morphogenic protein-7 (BMP-7).[449-452] These effects have not yet been validated in human clinical trials of ATN.[453,454] They all likely increase GFR through direct hemodynamic effects and may therefore hasten TEC recovery. Renal recovery is promoted by increased lymphangiogenesis as shown by overexpression of lymphangiogenic protein VEGF-D.[455]

Extracellular vesicles (EVs) derived from bone marrow mesenchymal stromal cells promote the regeneration of kidneys in models of AKI.[456] EVs represent a heterogeneous group of vesicles derived from nearly all mammalian cells. Smaller vesicles or exosomes range in size from 30 to 100 nm and larger microparticles from 100 to 1000 nm.[457] Microvesicles administered to severe combined immunodeficient (SCID) mice subjected to glycerol-induced AKI induced proliferation of tubule cells and accelerated morphologic and functional recovery.[458] Tubule proliferation was mediated by RNA-dependent effects because RNase abolished the effects of microvesicles on this process.[458] When various populations of EVs were isolated by differential centrifugation, transfer of 10 K (isolation at 10,000 g) and 100 K (isolation at 100,000 g) preparations into mice had different effects on AKI. The 100 K EVs contained mRNAs regulating proliferative, antiapoptotic, and growth factors, whereas the 10 K EVs had lower expression of proproliferative molecules and were unable to induce renal regeneration.[456]

Expression of the Sox9 transcription factor regulates renal development and also marks the induction of renal tubular dedifferentiation during repair and regeneration after AKI. Early growth response 1 (EGR1) is a transcription factor that regulates regeneration processes in AKI, increases SOX9 expression in renal TEC by directly binding to Sox9 promoter, and promotes proliferation via the Wnt/β-catenin pathway.[459] Antiinflammatory Tregs also participate in renal repair and regeneration. AT2R activation using the agonist C21 increases the Foxp3+ and IL-10 producing Tregs in kidneys following AKI to participate in the repair process.[460] Tissue-resident IL-33R+ and IL-2Ra+ Tregs increase post injury, and their expansion promoted regenerative and proangiogenic pathways during regeneration.[461] Indeed, in the adriamycin nephrotoxicity model, treatment with the IL233 hybrid cytokine, as late as 2 weeks after the AKI, completely restored renal function and structure by initiating a reparative program in the kidneys via upregulating the expression of renal progenitors SOX9, CD133, LGR4, LGR5, PAX8, and Ki67, as well as the genes from the entire nephron segments.[462] Thus Tregs are antiinflammatory and immunomodulatory enhancement of Tregs has the potential to facilitate tissue homeostasis and repair.[463]

Microvascular function

The microvasculature consists of the endothelium and pericytes. Each of these structures contributes to barrier integrity for normal homeostatic function. During AKI, disruption may occur at each of these sites, leading to increased permeability and cell death. Normally, the microvasculature controls vascular tone, regulation of blood flow to local tissue beds, modulation of coagulation and inflammation, cellular trafficking, and vascular permeability. Both ischemia and sepsis have profound effects on the endothelium and renal vasculature, which are particularly sensitive to these insults. An insult renders the endothelial bed ineffective in performing its function, and the ensuing vascular dysregulation leads to continued ischemic conditions and further injury following the initial insult, which has been termed the “extension phase” of AKI.[464] Histopathologically, this is seen as vascular congestion, edema formation, diminished microvascular blood flow, and margination and adherence of inflammatory cells to endothelial cells.

Vascular and perivascular function

Hemodynamics. Conger and associates were among the first to demonstrate that postischemic rat kidneys manifest vasoconstriction in response to decreased renal perfusion pressure and hence cannot autoregulate blood flow, even when total renal blood flow had returned to baseline values up to 1 week after injury.[465,466] Single-fiber, laser Doppler flow cytometry revealed that blood flow is reduced to 60% and 16% of preischemic values in cortex and medulla, respectively, and increased 125% in inner medulla following ischemia.[467] Selective inhibition, depletion, or deletion of iNOS has renoprotective effects during ischemia.[468,469] NO production from the endothelium (eNOS) may be impaired at the level of enzyme activity or modified by ROS to impair normal vasodilatory activity.[470] In ischemic AKI,[471] there is an imbalance of eNOS and iNOS and a relative decrease in eNOS, secondary to endothelial dysfunction and damage, which leads to a loss of the antithrombogenic properties of the endothelium, as well as increased susceptibility to microvascular thrombosis.[471] Administration of L-arginine, the NO donor molsidomine, or the eNOS cofactor tetrahydrobiopterin can preserve medullary perfusion and attenuate AKI induced by IRI. Conversely, the administration of N^{ω}-nitro-L-arginine methyl ester, a NOS enzymatic blocker, aggravates the course of AKI following IRI.[472,473]

The pattern of renal blood flow in experimental sepsis is inconsistent.[474] Renal blood flow was decreased in 62% of studies and unchanged or increased in 38%.[475] Administering *Escherichia coli* to sheep was shown to induce hyperdynamic sepsis associated with increased cardiac output, decreased mean arterial pressure, and AKI, as evidenced by a rise in creatinine levels.[475] Furthermore, there was marked increase in renal blood flow, decrease in urine output, and decrease in creatinine clearance. When angiotensin was dose titrated to restore blood pressure to presepsis levels, these effects were reversed; there was a significant decrease in renal blood flow, increase in urine output, and increase in creatinine clearance.[476] These studies suggest that improvement of function may be more than simply an improvement in mean arterial pressure and that the increase in glomerular filtration pressure may be through selective constriction of efferent arterioles and/or increase in mesangial cell tone.[476] On the basis of these principles, patients with vasodilatory shock who were receiving another vasopressor were randomized in a clinical trial to Ang II or placebo. The group that received Ang II had higher mean blood pressure[477] and, in a subgroup of patients with AKI who required renal replacement therapy, 28-day survival, mean arterial pressure, and rate of discontinuation of renal replacement were higher in the angiotensin group. These results together show the potential benefit of Ang II in hyperdynamic sepsis, which may in part be due to improved glomerular hemodynamics.

Endothelium. The glycocalyx, which is located on the blood side of the endothelium and has a depth of 1 to 3 µm, is composed of cell-bound proteoglycans, glycosaminoglycan side chains, and sialoproteins.[478] Sandwiched between blood components and the endothelium, the glycocalyx is situated to play a key role in microvascular and endothelial homeostasis and endothelial physiology.[479] The specific functions served by the endothelial glycocalyx include shear stress mechanotransduction to endothelial cells, regulation of vascular permeability, inhibition of intravascular thrombosis, and protection of the endothelium from platelet and leukocyte adhesion.[478,480] The glycocalyx may be damaged by local changes in shear stress, ROS, altered oncotic pressure, altered fluid composition, and inflammatory molecules. Shedding of the glycocalyx occurs, exposing the endothelial adhesion molecules to circulating immune cells and leading to immune cell activation, inflammation, and lipoprotein leakage to the subendothelial space. Coronary ischemia-reperfusion leads to shedding of the glycocalyx and an increase in neutrophil adhesion with subsequent myocardial injury.[481] Similar events are likely to occur in septic AKI and kidney IRI (Fig. 27.8).

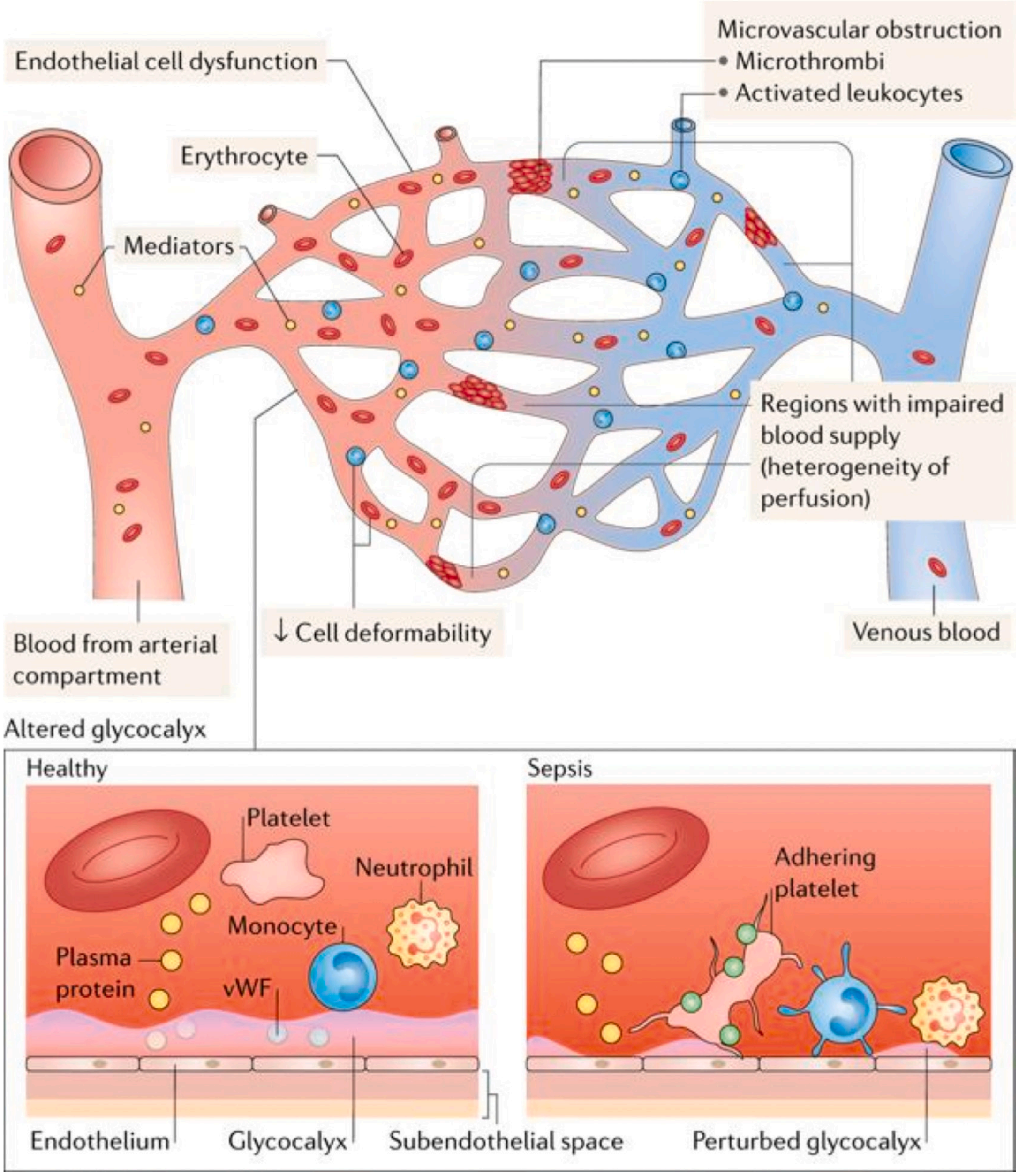


Fig. 27.8 Microvascular dysfunction in septic acute kidney injury. Multiple mechanisms are involved in the development of sepsis-related microvascular dysfunction, among which endothelial dysfunction (related partly to circulating host-derived and pathogen-derived mediators, as well as to reactive oxygen species) and an altered glycocalyx have major roles. The glycocalyx is a thin layer of glycosaminoglycans that covers the endothelial surface, facilitating the flow of red blood cells and limiting the adhesion of leukocytes and platelets to the endothelium. The glycocalyx may be substantially altered during sepsis, so interactions between the vascular endothelium and circulating cells (e.g., leukocytes and platelets) are impaired, and leukocyte rolling and adhesion to the endothelium may occur. Activation of coagulation and the generation of microthrombi might also participate in sepsis-induced microvascular alterations, as well as alterations in erythrocyte deformability and/or their adhesion to the endothelium. All these phenomena might cause heterogeneity in microvascular blood flow, with a decrease in vascular density and nonperfused capillaries, resulting in an increased diffusion distance for oxygen and in alterations in oxygen extraction. ***vWF,* von Willebrand factor.** (From Lelubre C, Vincent JL. Mechanisms and treatment of organ failure in sepsis. *Nat Rev Nephrol.* 2018;14[7]:417–427.)

Within 24 hours after ischemic injury, there is loss of vascular endothelial cadherin immunostaining, suggesting severe alterations in the integrity of the adherens junctions of the renal microvasculature.[482] In vivo two-photon imaging demonstrated a loss of capillary barrier function within 2 hours of reperfusion, as evidenced by leakiness of high-molecular-weight dextrans (>300,000 Da) into the interstitial space.

Critical constituents of the perivascular matrix, including collagen IV, are substrates for matrix metalloproteinase-2 (MMP-2), MMP-7, and MMP-9, which are collectively known as gelatinases. Breakdown of barrier function may also be due to MMP-2 or MMP-9 activation, and this upregulation is temporally correlated with an increase in microvascular permeability.[73,464,482]

MMP-7 has a renoprotective effect by degrading Fas ligand (FasL) and E-cadherin. It mobilizes β-catenin, activates AKT protein kinase, suppresses P53 expression, and proapoptotic protein Bax, and attenuates renal tubule apoptosis.[483,484] MMP-9 gene deletion stabilizes microvascular density following ischemic AKI, in part by preserving tissue VEGF levels.[485] MMP-9 also attenuates renal apoptosis via the soluble stem cell factor (SCF)-c-kit pathway.[486] In contrast, MMP-2 contributes to AKI thought to be due to degradation of peritubular capillary basement membrane leading to hemorrhage and tubule necrosis.[487]

In addition, minocycline, a broad-based MMP inhibitor, and the gelatinase-specific inhibitor ABT-518 both ameliorated the increase in microvascular permeability in this model. Taken together, many studies have indicated that the loss of endothelial cells following ischemic injury is not a major contributor to altered microvascular permeability, although renal microvascular endothelial cells are vulnerable to the initiation of apoptotic mechanisms following ischemic injury, which can ultimately reduce microvascular density.[488]

Krüppel-like factors (KLF) are a family of zinc-finger transcription factors involved in a number of fundamental cellular processes (cell cycling, differentiation, regeneration, and human disease).[489,490] KLF2, KLF4, and KLF11 are expressed in kidney endothelial cells, and KLF2 and KLF4 modulate antithrombotic and antiinflammatory processes.[489] Less studied is the endothelial-specific KLF11. KLF11 may be a novel renal protective member of the KLF family. KLF11-deficient mice were found to be more susceptible to bilateral IRI. They demonstrated more vascular congestion and induction of endothelin 1 and IL-6, suggesting the importance of KLF11's effect on endothelial function.[491]

The angiopoietin (Angpt)/Tie2 system consists of the transmembrane endothelial tyrosine kinase Tie2 and circulating ligands (Angpt-1 and 2).[492,493] In endothelial cells, Tie 2 phosphorylation governs by the mutually antagonistic properties of Angpt-1 and Angpt-2. In inflammatory states, Angpt-1 attenuates vascular inflammation, leakage, and apoptosis, whereas Angpt-2 has the opposite effect. In septic individuals, Angpt-2 increased up to 50 times, whereas Angpt-1 was unchanged.[494,495] Although seemingly harmful, somewhat surprising results were obtained from global Angpt-2 deficient mice in which Angpt-2$^{-/-}$ mice were not protected from sepsis-associated AKI.[496] Additional studies are necessary to determine the role of Angpt/Tie2 in sepsis-associated AKI.

Endothelial cells play a central role in coagulation via interactions with protein C through the EPCR and thrombomodulin. The protein C pathway, which helps maintain normal homeostasis and limits inflammatory responses, is activated by thrombin-mediated cleavage, and the rate of this reaction is further augmented (by 1000-fold) when thrombin binds to the endothelial cell surface receptor protein thrombomodulin. The activation rate of protein C is further increased by approximately 10-fold when EPCR binds protein C and presents it to the thrombin-thrombomodulin complex. Essentially, activated protein C then has antithrombotic actions and profibrinolytic properties and participates in numerous antiinflammatory and cytoprotective pathways to restore normal homeostasis.[497] On the basis of these properties, the endothelial cell plays an absolutely essential and critical role in maintaining a normal and healthy vasculature and endothelial bed. In AKI, microvascular function is ultimately compromised, resulting in disseminated intravascular coagulation and microvascular thrombosis, decreased tissue perfusion, and hypoxemia, leading to organ dysfunction and failure. Both pretreatment and postinjury treatment with soluble thrombomodulin attenuate renal injury, with minimization of vascular permeability defects and improvement in capillary renal blood flow.[498]

Leukocyte and endothelial cells are dynamically involved in the process of adherence of leukocytes to the vascular endothelium. Leukocyte activation and cytokine release require signals through chemokines circulating in the bloodstream or through direct contact with the endothelium. Rolling leukocytes can be activated by chemoattractants such as complement C5a and platelet-activating factor. Once activated, leukocyte integrins bind to endothelial ligands to promote firm adhesion. β_2-Integrin (CD18) seems to be the most important neutrophil ligand for endothelial adherence. These interactions with the endothelium are mediated through endothelial adhesion molecules that are upregulated during ischemic conditions.[499] The initial phase starts with slow neutrophil migration mediated by tethering interactions between adhesion molecules and their endothelial cell ligands. Within 2 to 4 hours of IRI or sepsis, endothelial P-selectin and intercellular adhesion molecule 1 (ICAM1) are expressed on endothelial cells associated with leukocyte trapping in peritubular capillaries.[283] Platelet P-selectin and not endothelial P-selectin is the main determinant in neutrophil-mediated ischemic kidney injury.[499] There is also significant protection from both ischemic injury and mortality by blockade of the shared ligand to all three selectins (E-, P-, and L-selectin), which seems to be dependent on the presence of a key fucosyl sugar on the selectin ligand.[500] In a CLP model of septic azotemia, mice deficient for E-selectin, P-selectin, or both were completely protected. Furthermore, selectin-deficient mice demonstrated similar intraperitoneal leukocyte recruitment but altered cytokine levels when compared with WT mice,[501] in addition to engagement in leukocyte-endothelial cell interactions.[501] Other adhesion molecules such as VAP-1 appear to play a role in AKI.[302]

The critical role that leukocyte integrin CD11b–CD18 contributes to both AKI and CKD was demonstrated in nonhuman primates.[502] In an IRI model, mAb107, a CD11b–CD18 inhibitor, markedly reduced plasma creatinine and ATN, indicating that the leukocyte integrin CD11b–C18 contributes to AKI. Moreover, subsequent assessment up to 9 months after AKI revealed improvement in microvascular perfusion and histology.

Microparticles (MPs) are cell membrane–derived particles of 0.2 to 2 μM in diameter that can promote coagulation and inflammation and may be involved in sepsis-related AKI.[503] In the cecal ligation model of sepsis, overexpression of calpastatin, which limits procoagulant microparticle release, improved survival, organ dysfunction (including lung, kidney, and liver damage), and lymphocyte apoptosis. Increased calpastatin expression decreased the sepsis-induced systemic proinflammatory response and disseminated intravascular coagulation by reducing the number of procoagulant circulating microparticles and therefore delaying thrombin generation.[504] Transferring microparticles from septic WT to septic calpastatin transgenic mice worsened survival and coagulopathy, thus demonstrating the deleterious effect of microparticles in this model. The development of therapeutics aimed to prevent the release of MPs from endothelium or other inflammatory cells may be a reasonable target in microvascular dysfunction during sepsis-induced AKI.

Perivascular cells. Kidney pericytes are extensively branched cells that are scattered on the outer wall of capillaries and embedded in the capillary basement membrane; they serve a homeostatic role in angiogenesis and vessel maturation.[505] Following kidney injury, they detach and migrate into the kidney interstitium[506] and, in some cases, transform into myofibroblasts, leading to progressive kidney fibrosis.[474] Thus the endothelial-pericyte crosstalk, involving VEGF receptors and platelet-derived growth factor receptors, contributes to normal microvascular health and disease. Selective pericyte ablation caused an abrupt decline in renal function and an increase in albuminuria within 96 hours.[479] These results demonstrate that pericytes are essential for normal kidney microvascular function.

ACUTE KIDNEY INJURY TO CHRONIC KIDNEY DISEASE TRANSITION

AKI is one of many initiating events that contribute to chronic, progressive kidney disease.[507,508] Some patients with AKI fully recover kidney function, whereas in others, the development of CKD is accompanied by a progressive decline in kidney function, leading ultimately to end-stage kidney disease (ESKD).[509-511] Regardless of the cause of CKD (e.g., nephrotoxic kidney injury, ischemia, infection, genetics; paraneoplastic, immunologic processes), there is a stereotypical response leading to interstitial fibrosis, tubular atrophy, and peritubular rarefaction and inflammation.[507,512,513] Incomplete repair following AKI contributes to the progression to CKD and ESKD[424,514] (eFig. 27.5).

Role of the Endothelium

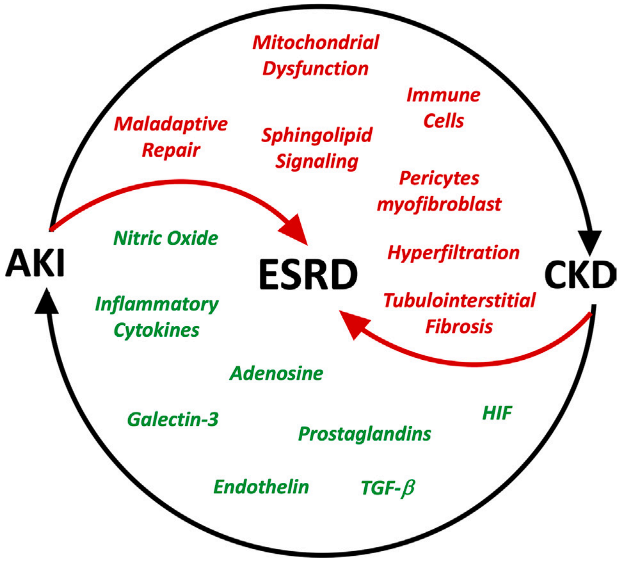
Human biopsies demonstrate microvascular rarefaction associated with progressive kidney failure.[515] In experimental AKI, acute injury to endothelial cells may have long-term implications; a significant reduction in blood vessel density following ischemic injury leads to the phenomenon of vascular dropout.[516] Vascular dropout was revealed by a decrease in vascular density approximating up to 45% at 4 weeks after an ischemic insult.[488] Micro-CT imaging of vascular alterations demonstrated a progressive decline of renal relative blood volume in three models of progressive kidney disease (IRI, unilateral ureteral obstruction [UUO], and Alport) of up to 61% in late disease stages that precede fibrosis.[517] Ex vivo micro-CT imaging and 3D quantification of the renal vasculature showed macrovascular to microvascular alterations in progressive renal disease, as evidenced by a reduction in vessel diameter and branching and increased vessel tortuosity (Fig. 27.9). These findings suggest that unlike the renal epithelial tubular cells, the renal vascular system lacks comparable regenerative potential. Renal endothelial injury can result in capillary rarefaction due to hypoxia and a lack of upregulation of VEGF in response to injury.[518-520] It is not yet clear whether apoptosis and/or necrosis play a major role in endothelial cell dropout. Ischemia inhibits the angiogenic protein VEGF while inducing ADAMTS1, which is argued to be a VEGF inhibitor.[521] Lack of vascular repair could be due to functional deficiency of VEGF, as shown by experiments where administration of the VEGF-121 protein isoform preserved the microvascular density.[521] Reduction of the microvasculature density increases hypoxia-mediated fibrosis and alters usual hemodynamics, which may lead to hypertension. Thus loss of microvasculature density and its consequential effects may play a critical role in the progression of CKD following initial recovery from ischemia-reperfusion–induced AKI.[488,516,517] Delayed deletion of sphingosine 1-phosphate receptor 1 (S1P1)in endothelial cells after IRI, using a tamoxifen-inducible system, prevented kidney recovery, resulting in chronic inflammation and progressive fibrosis. Specifically, S1P1 directly suppressed endothelial activation of leukocyte adhesion molecule expression and inflammation. Taken together, the data indicate that activation of endothelial S1P1 is necessary to protect from IRI and permit recovery from AKI.[522]

Epigenetic modification

AKI is associated with histone modification[523] and DNA methylation,[524] leading to altered transcription of genes thought to contribute to renal injury.[525,526] Epigenetic changes are closely related to renal hypoxia and CKD progression.[518,519] The histone methyltransferase EZH2 was upregulated in septic AKI, especially on the Sox9 promoter that led to abnormal activation of Wnt/β-catenin pathway,[527] as well as induction of an epithelial-mesenchymal transition (EMT) program leading to loss of expression of RTEC transporters OAT1, ATPase, and AQP1.[528] The deletion or silencing of EZH2 reduced apoptosis, tubular injury, and inflammation by upregulating PTEN expression and inhibiting EGFR signaling ERK1/2 and STAT3, to block the EMT, G2/M arrest, and fibrosis.[528] In a TGF-β2 repressor, Ybx2 during IRI-AKI promoted the transition of pericytes to CKD to worsen fibrosis, a function that could be reversed by DNA demethylation induced by 5-azacytidine.[527] Genetic or pharmacologic depletion of EZH2 function in the in vivo and vitro settings promoted the EMT program and induced loss of renal TEC transporters. EZH2 could participate in M2 macrophage polarization involving the STAT6 and PI3K/AKT pathways.

Renal oxidative stress in AKI-to-CKD transition

Proximal tubule mitochondria play a central role in the metabolism necessary for the reabsorption of water and solutes. This reliance on mitochondrial energy production makes kidney cells vulnerable to damage.[529] Mitochondria produce ROS and reactive nitrogen species (RNS) during acute and chronic kidney injury while antioxidants are depleted, further aggravating the course of the kidney injury.[400,530] Ischemic injury upregulates proinflammatory cytokines and recruits immune cells.[225,241,274,353] Although the proximal tubule relies primarily on oxidative metabolism, following AKI, damaged proximal tubule cells that are unable to recover undergo a metabolic switch in the transition to CKD from AKI. There is a downregulation of respiratory enzymes following reperfusion[531] and FAO.[532,533] Adaptive proximal tubule repair is correlated with FAO and oxidative



e-Fig. 27.5 Acute kidney injury *(AKI)* to chronic kidney disease *(CKD)* transition. AKI can lead to CKD and then to end-stage renal disease (ESRD). Conversely, CKD can lead progressively to ESKD or can predispose to AKI, which then enters a vicious circle and predisposes to CKD. A number of factors contribute to AKI-to-CKD transition, including traditional concepts *(green)* and emerging *(red)* concepts.

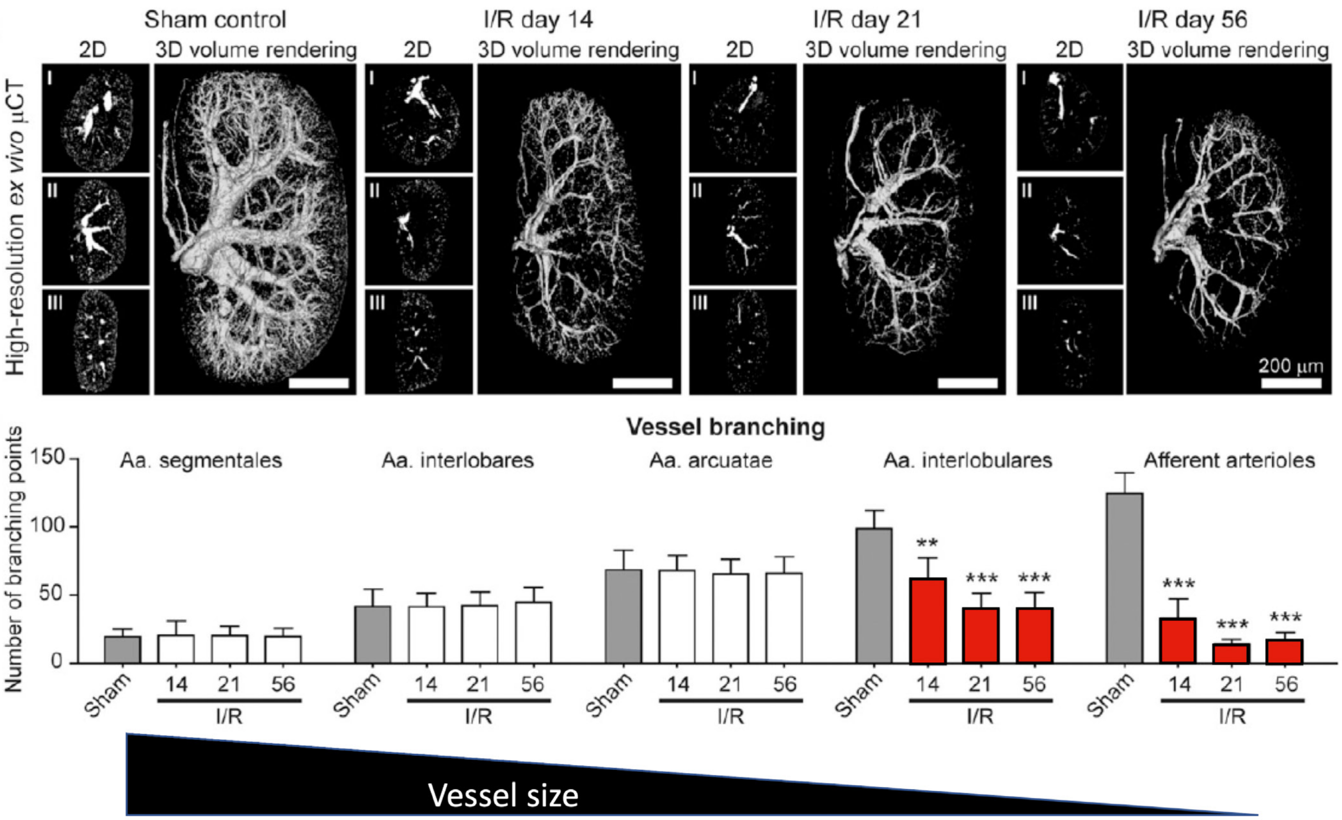


Fig. 27.9 Ex vivo microcomputed tomography *(CT)* imaging and three-dimensional quantification of the renal vasculature reveal significant alterations of renal arteries in progressive ischemia-reperfusion–induced (IRI) renal injury. *(Lower panel)* Representative high-resolution, ex vivo, micro-CT images of sham control and I/R kidneys from days 14, 21, and 56 after Microfil perfusion (two-dimensional cross-sectional images in transverse [I], coronal [II], and sagittal [III] planes, as well as three-dimensional volume renderings). Note the progressive rarefaction of functional vessels besides the continuous shrinkage of fibrotic kidneys over time. Scale bar, 200 μm. *(Upper panel)* Micro-CT–based quantification of vascular branching points per increasing vessel order (from hilus to periphery). A continuous reduction in vessel branching and vessel size was linked to an increased vessel tortuosity during fibrosis progression. **, $P < .01$; ***, $P < .001$. *AKI,* Acute kidney injury. (From Ehling J, Babickova J, Gremse F, et al. Quantitative micro-computed tomography imaging of vascular dysfunction in progressive kidney diseases. *J Am Soc Nephrol.* 2016;27[2]:520–532.)

phosphorylation.[195] Gain of function of FAO through the overexpression of the fatty acid shuttling enzyme carnitine palmitoyl-transferase 1A (CPT1A) in tubule epithelial cells of mice subjected to 3 models of renal fibrosis reduced inflammation, macrophage infiltration, and epithelial cell damage. Furthermore, FAO gain of function restored oxidative metabolism and mitochondrial number.[534] Thus drugs targeting FAO such as metformin (AMPK activator) or baicalin (CPT1A activator) may be useful in AKI-to-CKD transition.[535]

Renal oxidative stress persists for many weeks in rats following recovery from ischemia-reperfusion–induced AKI. Also noted was the association with increased Ang II sensitivity. This sustained renal oxidant stress following recovery from AKI alters the hemodynamic and fibrotic responses to Ang II and may contribute to the transition to CKD following AKI.[536] This also highlights the importance of managing recovering AKI carefully to avoid further insults that might enhance ROS or sensitivity to Ang II.

Cell cycle arrest

As discussed earlier, in successful repair, mature surviving cells dedifferentiate, proliferate, and repopulate proximal tubules in response to kidney injury.[432,433] Upregulation of KIM-1 is antiinflammatory through its engulfment and clearance of dead and apoptotic cells[537] but becomes maladaptive when chronically expressed.[538] Repetitive injury may result in the upregulation of cyclin-dependent kinase (CDK) inhibitors, $p16^{Ink4}$ and P53-p21,$^{Cip1/Waf1}$ leading to cell cycle arrest.[539] An increase in $p16^{Ink4}$ inhibits CDK4 and CDK6 and is associated with tubulointerstitial changes leading to fibrosis; kidneys from $INK4a^{-/-}$ mice develop less fibrosis following kidney IRI.[222] Thus cell cycle arrest and cellular senescence lead to relative resistance to apoptosis and persistent metabolic activity, promoting a senescence-type secretory phenotype with the secretion of TGF-β, cytokines, proteases, and other proinflammatory and profibrotic factors.[424,540] In four models of kidney injury including severe bilateral IRI, unilateral IRI, aristolochic acid, and UUO, G2/M arrest contributes to maladaptive repair and fibrosis by activating c-jun NH(2)-terminal kinase (JNK) signaling, which upregulates profibrotic cytokine production associated with AKI-CKD transition.[541] Similar findings have been reported and, importantly, pharmacologic inhibitors of G2/M arrest reduce fibrosis.[542,543] In models of fibrosis, damaged proximal tubule cells contain rapamycin (TOR)-autophagy spatial coupling compartments (TASCCs), which are cellular compartments resulting from the fusion of late autophagosomes with lysosomes that contain mTOR complex 1 (mTORC1). TASCCs are necessary for the paracrine production of profibrotic factors and subsequent fibrosis.[544]

Inflammation in AKI-to-CKD transition

Inflammation in acute kidney injury–chronic kidney disease transition. During kidney recovery, renal and extrarenal cells participate in the wound-healing response and can initiate fibrosis. Immune cells of the mononuclear phagocyte system, including macrophages and dendritic cells, contribute to injury and have also emerged as important cells in the recovery of kidney function during adaptive repair or fibrosis during maladaptive repair. The balance between wound healing and progressive fibrosis dictates the final outcome. The intrinsic plasticity of monocytes-macrophages and dendritic cells, as well as attempts to relate in vitro studies to in vivo findings, makes the functional definition and phenotype of this myeloid population in kidney pathophysiology complex.[545-548] In vitro studies have led to two well-defined mononuclear phagocytes. Classically activated macrophages—M1 mononuclear phagocytes consisting of macrophages and dendritic cells—are produced by exposure to LPS or INF-γ and are largely thought to be proinflammatory and contribute to initial kidney injury. Alternatively activated macrophages (M2 mononuclear phagocytes) are induced by IL-4 and IL-10 and appear later, after AKI has been established, and have a genetic signature associated with wound healing and/or fibrosis.[549] These mononuclear phagocyte phenotypes depend on the complex local tissue microenvironment, which may induce phenotype switching.[276]

A key feature of fibrosis is the activation of extracellular matrix–producing myofibroblasts.[550,551] Other factors relevant in CKD progression include endothelial cell damage and vascular damage in AKI,[516] hypoxia-inducible factor (HIF),[552] innate and adaptive immunity,[553] cell cycle arrest,[541] and epigenetic mechanisms.[554,555]

Although some injured tubules undergo repair and regeneration, injury may also be accompanied by inflammation, maturation, and proliferation of fibroblasts, and extracellular matrix deposition as part of the process of fibrosis. Tissue fibrosis is a common component of inflammatory diseases, including various forms of CKD. Collagen deposition, characteristically formed during normal wound healing, is reversible, but progressive irreversible fibrosis may occur through repeated injury due to dysregulated inflammation.[556] Some studies have focused on the role of pericytes or resident fibroblasts that contribute to normal homeostatic microvascular function but transform into myofibroblasts on activation.[557-559] Myofibroblasts are the primary collagen-producing cells.[560]

The source of fibroblasts in the injured kidney has been controversial.[557,560] Although the myofibroblast is the cell type responsible for depositing collagen, investigators have been searching for their precursors. Other than interstitial fibroblasts that are known to transition, it has been suggested that pericytes, dendritic, endothelial, and epithelial cells may also represent potential contributors. In vivo and in vitro studies have demonstrated that endothelial cells can develop a myofibroblast phenotype.[561,562] Epithelial cells grown in culture can express genes that are typically expressed in myofibroblasts, suggesting EMT.[563-565] Fate-mapping studies indicate that although epithelial cells can express mesenchymal markers in vitro, they appear not to penetrate the basement membrane to enter into the interstitial space and differentiate into myofibroblasts in vivo.[557] Furthermore, lineage analysis demonstrated that platelet-derived growth factor receptor (PDGFR)-β–positive pericytes differentiate into myofibroblasts in a UUO model. Following AKI, a potential step in the AKI-CKD transition is the dissociation of pericytes from endothelial cells. This was revealed by using bigenic Gli1-CreERt2; R26tdTomato reporter mice.[506] Pericyte ablation resulted in endothelial damage, and pericyte loss led to a significantly reduced capillary number, a hallmark of CKD.

Two studies have shed light on this controversy. *Snai1* (encodes snail family zinc finger 1, Snail1) expressed in mouse renal epithelial cells is required for kidney fibrosis.[566] The Snail family of transcription factors plays a key role in repressing the adhesion molecule E-cadherin, thereby regulating the EMT during embryonic development. Moreover, *Twist* (encodes twist family bHLH transcription factor 1) genes are also essential regulators of EMT.[567] Inhibition or deletion of these genes or overexpression includes or inhibits EMT, respectively.[568,569] Lineage tracing studies have demonstrated that during activation, labeled epithelial cells do not contribute to myofibroblasts or interstitial cells; however, there was downregulation of epithelial markers and loss of epithelial polarity and cell differentiation.[569] Thus partial EMT leads to cell cycle arrest, blocks proliferation and repair, and blocks secretion of growth factors, such as TGF-β, that drive proliferation of myofibroblasts[570] (Fig. 27.10). Genes and pathways that are associated with successful or failed repair are shown in Fig 27.11.

Signaling pathways Wnt, hedgehog (Hh), and Notch play important roles during renal fibrosis. In CKD, numerous Wnt ligands are upregulated, resulting in prolonged activation of the Wnt/β-catenin pathway.[571,572] Epithelial β-catenin activation leads to epithelial dedifferentiation and interstitial fibrosis.[573] Wnt derived from tubules is necessary for myofibroblast activation and interstitial fibrosis.[574] The sphingolipid pathway is important in progressive fibrosis.[575,576,353] Sphingosine 1-phosphate (S1P), a pleiotropic lysophospholipid that is involved in diverse functions such as cell growth and survival, lymphocyte trafficking, and vascular stability,[577-582] is the product of sphingosine phosphorylation by two sphingosine kinase isoforms (SphK1 and SphK2). SphK2 is localized to the nucleus and produces nuclear S1P, which acts as a histone deacetylase (HDAC) inhibitor,[583] thereby allowing gene expression to be induced. IFN-γ may mediate reduced fibrosis in *Sphk2*$^{-/-}$ mice or mice treated with SpHK2 inhibitor at day 14 .[575,576] Perivascular cell S1P transported via spinster homolog 2 (Spns2) bound to S1P1 in an autocrine fashion to enhance production of proinflammatory cytokines/chemokines upon injury, leading to immune cell infiltration and subsequent fibrosis. Inhibition of S1P transport through perivascular deficient Spns2$^{-/-}$ or the use of small molecule inhibitors ameliorated kidney fibrosis in mice.[584] These studies demonstrated that targeting the perivascular

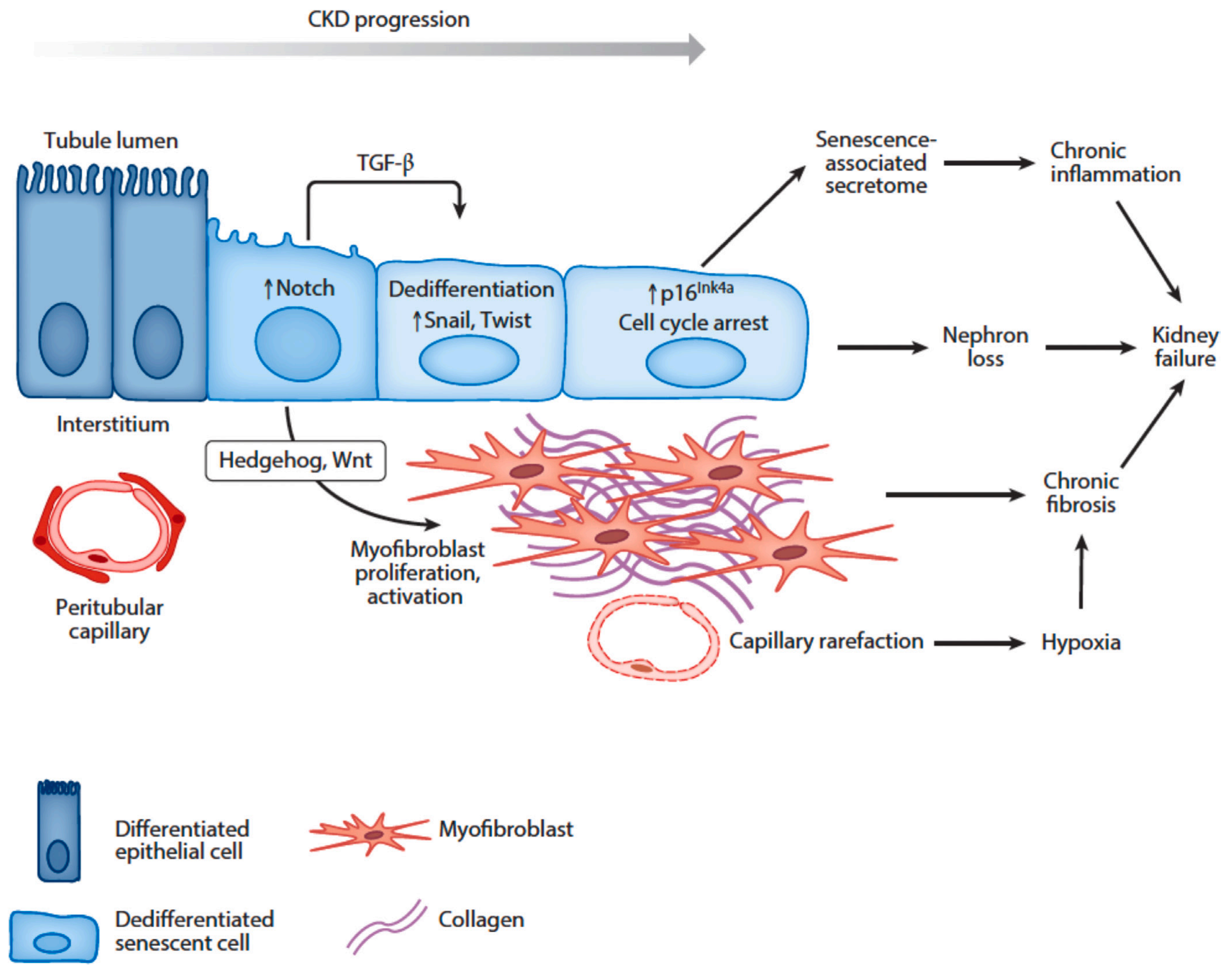


Fig. 27.10 Critical cells and signaling pathways activated in progressive chronic kidney disease *(CKD)*. CKD is initiated by cellular injury, either to the epithelial or endothelial compartment. In the tubulointerstitium, injury leads to epithelial dedifferentiation that may be induced by transforming growth factor-β (TGF-β) and Notch pathway upregulation. Dedifferentiated epithelial cells secrete paracrine signaling factors, such as hedgehog and Wnt ligands, that act on interstitial pericytes and mesenchymal stem cell–like cells to activate myofibroblast differentiation, proliferation, and matrix secretion. This in turn causes peritubular capillary rarefaction and ongoing hypoxia. Chronic tubular injury leads to epithelial cell cycle arrest and senescence, with accompanying secretion of proinflammatory cytokines that amplify inflammation. These interrelated events ultimately drive nephron loss, ongoing interstitial fibrosis, and kidney failure. (From Humphreys BD. Mechanisms of renal fibrosis. *Annu Rev Physiol.* 2018;80:309–326.)

SPHK2–SPNS2–S1P1 axis may serve as a novel therapeutic approach for attenuating kidney fibrosis.[353,575,576]

Activation of myofibroblasts is also mediated through paracrine signals from immune cells (lymphocytes and antigen-presenting cells), including cytokines, chemokines, angiogenic factors, growth factors, peroxisome proliferator-activated receptors (PPARs), and other molecules.[585] Although Th2 cytokines may induce fibrosis, Th1 cytokines such as IL-12 or IFN-γ inhibit pulmonary[586,587] and experimental renal fibrosis.[588]

An unbiased approach to identify upregulated and downregulated pericyte genes in injury revealed increased activation and expression of ADAMTS1 (a disintegrin-like metalloprotease with thrombospondin type 1) and downregulation of MMP-3 inhibitor TIMP Metallopeptidase Inhibitor 3 (TIMP-3).[589] TIMP-3–stabilized pericytes maintained collagen capillary tube networks, whereas ADAMTS1-treated pericytes led to enhanced destabilization. TIMP-3 has many functions including regulating VEGF signaling, pericyte migration, and MMP-2 and MMP-9 activity.[590] Furthermore, TGF-β1 activates the pericyte-myofibroblast transition, thus adding to the stimuli for pericyte involvement in fibrosis.[591] Injured epithelial cells reduce production of VEGF, a trophic factor for endothelial maintenance, and increase production of TGF-β and PDGFR, which enhance pericyte dedifferentiation into myofibroblasts. Finally, PDGFR blockade on pericytes or VEGF receptor 2 (VEGFR-2) on endothelial cells led to reduced fibrosis and stabilized the microvasculature in the UUO model.[592] Thus numerous factors favor microvascular maladaptation after injury, including TGF-β production and lack of VEGF production by epithelial cells, reductions

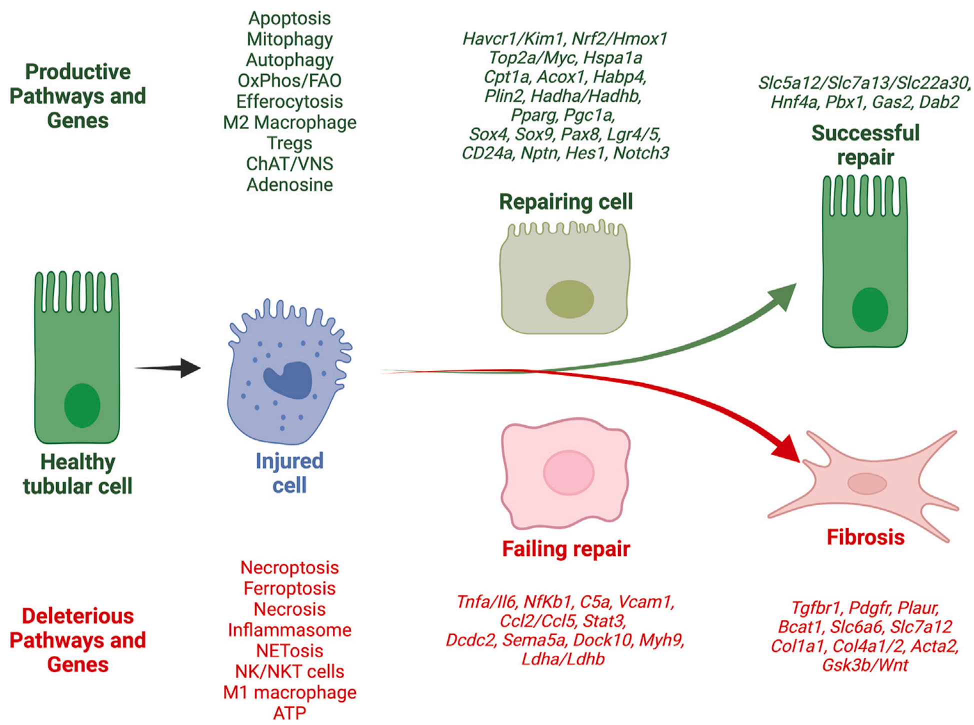


Fig. 27.11 Successful and failed-repair postacute kidney injury. Multiomics studies have revealed the productive and deleterious genes and pathways that regulate successful and maladaptive (failed) repair, respectively. The injured tubular cells release several mediators that activate tubular cell-intrinsic and extrinsic pathways. Activation of uncontrolled cell death pathways, such as necroptosis, ferroptosis, inflammasome activation, and release of ATP lead to recruitment of proinflammatory cells such as neutrophils, which further cause NETosis, as well as NK/NKT cells and M1 macrophages. Early activation of pathways, such as apoptosis, which promotes mitophagy, autophagy, and efferocytosis, promotes clearance of cell debris. Recruitment of antiinflammatory cells such as Tregs and M2 macrophages, which may be mobilized from the spleen by vagus nerve stimulation (VNS) and Choline-acetyl transferase (ChAT) pathway, induce anti-inflammatory mediators including adenosine. Transcriptomic and metabolomic studies showed that the tubular and phagocytic cells that engulf apoptotic debris and upregulate the pathways regulating the uptake and oxidative metabolism of lipid cargo (repairing cell) transition toward successful repair as indicated by enhanced expression of genes related to cell cycle, dedifferentiation, tubular progenitors, and solute transporters related to function. On the other hand, cells that are enriched for expression of proinflammatory genes (failing repair) and genes related to dysregulation of glucose, lactate, and amino acid metabolism eventually activate profibrotic genes. (Created in BioRender. Sharma R. 2025. https://BioRender.com/idu28is.)

in PDGF production by endothelial cells, and increased ADAMTS1 expression, with reduced TIMP-3 production by pericytes. These events also suggest several therapeutic targets that could limit microvascular dropout and loss of kidney function secondary to the fibrotic process. Normal pericyte function therefore becomes deranged following transformation to activated myofibroblasts, which leads to loss of microvascular stability, ineffective angiogenesis, increased permeability, and capillary rarefaction, which are all processes leading to fibrosis.[505,507]

As discussed earlier, proximal tubules are highly metabolically active, relying on aerobic metabolism. Mitochondrial FAO is a major source of energy, and mitochondrial dysfunction leads to lipid accumulation, ATP depletion, and fibrosis in proximal tubule cells.[532] A number of therapeutic approaches are in development, including some agents in clinical trials for diabetic nephropathy and other kidney diseases,[593,594] but as yet there are no approved treatments to prevent the development of kidney fibrosis or accelerate repair. The development of effective treatments requires a better understanding of the inflammatory, injury, wound healing, matrix deposition, and cellular repair processes that accompany fibrosis.[540,556,570,595-597] For example, mitochondria may be an important therapeutic target in AKI-to-CKD progression.[598] One approach is a novel mitoprotective drug, SS31.[599] SS peptides bind selectively to cardiolipin in the mitochondrial inner membrane to protect mitochondrial cristae. SS31 allows better function of the respiratory chain complexes, prevents peroxidase activity, and enhances electron transfer through the cytochrome complex to improve oxidative phosphorylation and ATP production.[599-601] SS31 administered 1 month after renal IRI attenuated glomerulosclerosis and senescence and reduced parietal epithelial cell activation and changes in podocyte and endothelial structure.[602] Dynamin-1-like protein (DRP1) is a GTPase enzyme responsible for regulating mitochondrial fission. Proximal tubule–specific deletion of DRP1 prevented renal IRI, inflammation, and programmed cell death and promoted epithelial recovery, which was associated with activation of the renoprotective β-hydroxybutyrate signaling pathway.[603] Importantly, delayed deletion of proximal tubule *Drp1* using a tamoxifen-inducible system after IRI attenuated kidney fibrosis. These results highlight DRP1 and mitochondrial dynamics as an important mediator of AKI and progression to fibrosis and suggest that DRP1 may serve as a therapeutic target for AKI.

Effects of organ crosstalk

AKI is associated with significant morbidity and mortality. In the intensive care unit, where patients with AKI sustain multiorgan dysfunction, the mortality rate is around 60%.[604] Although AKI is associated with consequences such as accumulation of uremic toxins, metabolic acidosis, fluid and electrolyte imbalance, and fluid overload, these factors alone do not explain the mortality associated with AKI.[605,606] Often presenting as a systemic condition, AKI induces dysfunction in distant organs, including lung, heart, brain, liver, and intestines (eFig. 27.6). Moreover, distant organ function can alter kidney function.

Renal ischemia can affect cardiac tissues[607] including induction of IL-1, TNF-α, and ICAM-1 mRNA was seen in cardiac tissues as early as 6 hours after renal ischemic injury that remained elevated up to 48 hours. At 48 hours, echocardiography also revealed increases in left ventricular end-systolic and diastolic diameter and decreased fractional shortening.

AKI can lead to acute lung injury (ALI) and vice versa. It is well known that patients with ALI frequently have AKI, in part due to mechanical ventilation. It is thought that mechanical ventilation induces AKI through the effects of changes in arterial blood gases, barotrauma-induced systemic release of inflammatory agents, and the influence on systemic and renal blood flow.[608,609] Kidney ischemic injury leads to an increase in pulmonary vascular permeability defects that are mediated through macrophages.[610] When combined with splenectomy, AKI resulted in increased serum IL-6 and worse lung injury as evidenced by an increased lung capillary leak, higher lung myeloperoxidase activity, and higher lung CXCL1 when compared with AKI alone. The absence of splenic IL-10 production was thought to lead to an enhanced proinflammatory response and lung injury.[611] Kidney and lung single-cell RNA sequencing after AKI identified osteopontin (OPN) as a novel AKI-ALI mediator. OPN release from kidney tubule cells triggered lung endothelial leakage, inflammation, respiratory failure, and pharmacologic blockade or OPN deletion protected from AKI-ALI.[612]

Abnormal salt and water handling of cells following AKI contributes to lung injury. The lung epithelial Na channel, Na^+-K^+-ATPase, and aquaporin-5 expression were downregulated after kidney ischemic injury or nephrectomy, but not in unilateral ischemic models, suggesting the role of uremic toxins in modulating these effects in the lung.[613]

AKI produces functional changes in the brain including blood-brain barrier disruption.[614] Mice with AKI had increased neuronal pyknosis and microgliosis in the brain; there were increased levels of the proinflammatory chemokines keratinocyte-derived chemoattractant and granulocyte colony–stimulating factor in the cerebral cortex and hippocampus and increased expression of glial fibrillary acidic protein in astrocytes in the cortex and corpus callosum.

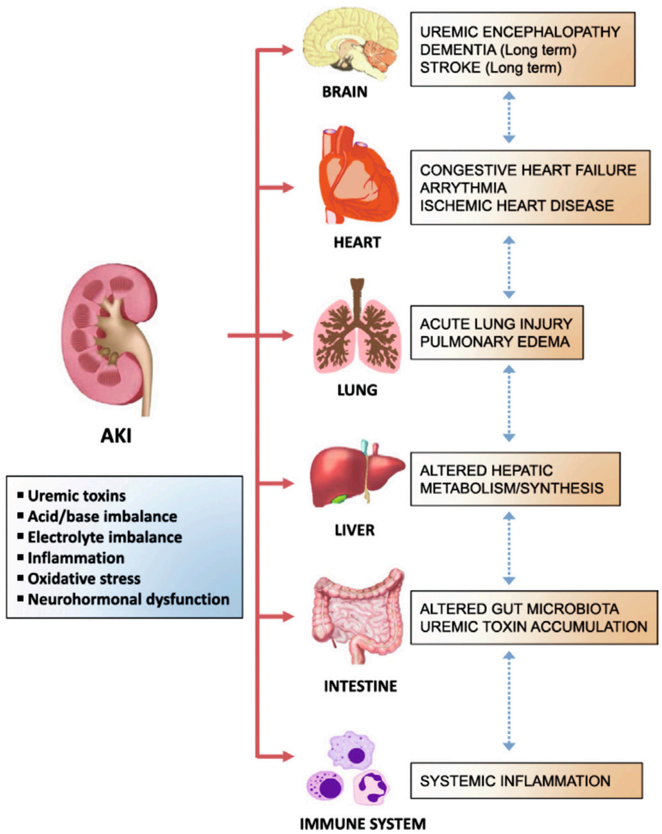
Many of the same processes involved in kidney-lung, kidney-heart, and kidney-brain interactions have been observed in the liver—increased neutrophil infiltration, vascular congestion, and vascular permeability after AKI. Following experimental AKI, levels of hepatic TNF-α, IL-6, IL-17A, ICM-1, keratinocyte-derived chemoattractant, IL-10, and MCP-1 are increased. The presence of AKI before a second-hit ischemic hepatic injury exacerbates liver injury as well, suggesting ongoing crosstalk between the kidney and liver[605] (Table 27.1).

An important example of organ crosstalk is the one that occurs between the nervous and immune systems. These two systems are linked to maintain normal homeostasis and to respond to stress and pathophysiologic disorders. An interesting example is the effect of brain death on ischemic AKI in kidney allografts. Traumatic brain injury elicits a cytokine and inflammatory response. These cytokines result in renal inflammation in kidney transplants from brain-dead donors, distinct from living or cardiac-death donors.[615] As an example, preimplantation biopsies of brain-dead donor kidneys reveal more infiltrating T lymphocytes and macrophages than normal kidneys. Furthermore, reperfusion of kidneys from brain-dead donors is associated with the instantaneous release of inflammatory cytokines, such as granulocyte colony–stimulating factor, IL-6, IL-9, IL-16, and MCP-1. In contrast, kidneys from living and cardiac-death donors show a more modest cytokine response, with the release of IL-6 and small amounts of MCP-1.[616]

Numerous studies have demonstrated that neural control of inflammation and AKI is partially mediated by a pathway referred to as the cholinergic antiinflammatory pathway (CAP), which requires an intact spleen.[289,292,617-619] Studies have identified neural pathways that regulate immunity and inflammation via the inflammatory reflex pathway, thus identifying specific molecular targets that can be modulated by stimulating neurons electrically,[617,620,621] by ultrasound,[289,622] or optically.[618] The CAP is the efferent limb of the so-called inflammatory reflex pathway mediated through the vagus nerve.[623-625] The afferent limb is activated by bacterial products (PAMPs), DAMPs, proinflammatory cytokines, immunoglobulins, and ATP.[626,627] Following detection of these inflammatory molecules at receptors, afferent signals from injured tissue and immune cells are transmitted to the brain,[625,628] which then activates the vagus efferent nerve. The subsequent inflammatory reflex controls peripheral cytokine levels and inflammation through key cellular components such as macrophages and $CD4^+$ T cells. Indeed, vagus nerve stimulation also shows antiinflammatory and organ-protective effects in varied disorders, such as arthritis,[629] colitis,[630] ileus,[630] and AKI.[292] The spleen plays an integral role in linking the nervous and immune systems via the CAP. Prior splenectomy worsens AKI,[622] blocks the antiinflammatory effect of the CAP,[292,618] and worsens lung inflammation following kidney IRI.[611] Because AKI is associated with high mortality and morbidity, these studies indicate that multiorgan crosstalk that occurs in the setting of AKI is likely to be a major contributor to nonrenal organ dysfunction and may mediate clinically observable events, such as cardiac, pulmonary, and central nervous system events.

Supplement

Acute phosphate nephropathy. AKI has been described as a complication following the administration of oral sodium phosphate solution as a bowel cathartic in preparation for colonoscopy and bowel surgery.[651-653] Although the mechanism linking oral sodium phosphate administration with AKI remains incompletely understood, the pathogenesis likely relates to a transient and significant rise in the serum



eFig. 27.6 The impact of acute kidney injury *(AKI)* on distant organs. AKI causes hemodynamic, humoral, and immunologic changes, which lead to dysfunction of distant organs, including lung, heart, brain, liver, intestine, and the immune system. (From Lee SA, Cozzi M, Bush EL, Rabb H. Distant organ dysfunction in acute kidney injury: a review. *Am J Kidney Dis*. 2018;72[6]:846–856.)

Table 27.1 Acute Kidney Injury–Induced Distant Organ Injury

Organ	Mechanism	Species	Reference
Lung	CXCL1	Mouse	638
	TNF-α	Mouse	639
	IL-10	Mouse	611,640
	Neutrophil elastase	Mouse	641
	Lung permeability	Mouse	610
	Osteopontin and lung injury	Mouse	612
	Pulmonary edema, inflammatory cytokines	Rat	642
	Oxidant stress	Rat	643
	Systemic cytokines and HMGB1	Human (ex vivo)	644
Heart	TNF-α	Rat	607
	Mitochondrial fission protein	Mouse	645
Brain	RAAS	Mouse	646
Liver	Administration of glutathione	Rat	647
	Hepatic oxidant stress	Mouse	648,649
Intestine	Hepatic oxidant stress	Mouse	650

AKI, Acute kidney injury; *CXCL1,* chemokine (C-X-C motif) ligand 1; *HMGB1,* high-mobility group box protein B1; *IL,* interleukin; *RAAS,* renin-angiotensin-aldosterone system; *TNF-α,* tumor necrosis factor–α.
Modified from Lee SA, Cozzi M, Bush EL, Rabb H. Distant organ dysfunction in acute kidney injury: a review. *Am J Kidney Dis*.

phosphate concentration that occurs simultaneously with intravascular volume depletion.

When the urine is oversaturated and buffering factors such as pH, citrate, and pyrophosphate are overwhelmed, renal phosphorus excretion becomes compromised, especially when flow rates are reduced. This may lead to the intratubular precipitation of calcium phosphate salts when the solubility coefficient is exceeded, leading to obstruction of the tubular lumen and direct tubular damage. ROS generated by the binding of calcium phosphate crystals further promotes tubular damage. Risk factors for acute phosphate nephropathy include preexisting volume depletion, the use of ACE inhibitors and angiotensin receptor blockers (ARBs), NSAIDs, CKD, older age, female gender, and higher dosage of oral sodium phosphate.[652,653] Patients who develop acute phosphate nephropathy typically present with elevated serum creatinine concentrations days to months after the administration of oral sodium phosphate solution and can experience progression to CKD and ESKD. As in other conditions associated with hypercalcemia, hyperphosphatemia, or hyperphosphaturia, calcium phosphate precipitation in renal tubules is seen on renal biopsy as bluish-purple crystals that are nonpolarizable.[654]

EXPERIMENTAL MODELS

The goal of preclinical AKI research is to translate basic scientific knowledge of AKI to clinical practice. Despite extensive research in AKI focusing on standardized definitions of AKI (from *R*isk, *I*njury, *F*ailure, *L*oss of kidney function, and *E*nd-stage kidney disease (RIFLE), *A*cute *K*idney *I*njury *N*etwork (AKIN), and Kidney Disease: Improving Global Outcomes [KDIGO]), biomarkers of AKI, novel drug targets, and improved clinical trial design, our knowledge remains incomplete and effective therapies are lacking.[132,133] To advance the field, it is important that we identify relevant targets through the analysis of human tissue biopsy and necropsy specimens; develop relevant disease models of AKI; include proper pharmacokinetic, pharmacodynamic, and dose-response studies; and, finally, improve preclinical and human clinical trial design.[133,135,136,655] The National Institutes of Health (NIH) and the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) have initiated a bold project, the Kidney Precision Medicine Project (KPMP). Its objectives are to obtain human kidney biopsies ethically from participants with AKI and CKD with a goal to identify critical cells, pathways, and targets for therapy. The following section focuses on current animal models.

IN VIVO MODELS OF ACUTE KIDNEY INJURY

Current investigations into the pathophysiology of AKI include a combination of animal and cell culture models of AKI designed to understand the pathophysiology of AKI better and investigate novel therapeutic agents. However, there remains a need to develop in vivo experimental models of AKI that more closely resemble clinical AKI for the development of effective therapies.[133,136,656,657] Some of the important principles in studying the pathophysiology of AKI in various models include outcome measures at multiple time points and the ability to control physiologic functions known to affect kidney function (e.g., temperature, blood pressure, anesthesia, and fluid status). Current models of AKI (e.g., warm ischemia-reperfusion using a pedicle clamp) test fundamental proof-of-principle concepts that identify potential molecular targets. A simplified model limits confounding variables but can clearly identify potential therapeutic targets. However, additional models of AKI are necessary that should reflect human disease (e.g., aged animals, impaired kidney function, multiorgan failure, preexisting vascular changes, multiple renal insults) that often coexist in human AKI. We briefly describe the pros and cons of using presently characterized experimental models.[605,607,610,638-650]

The warm ischemia-reperfusion renal pedicle clamp model is one of the most widely used experimental models in rats and mice because of its simplicity and reproducibility. However, the inflammatory response differs greatly between mice and rats. It is important to realize that there is considerable variability between mice and rats, mouse strains (C57BL/6 vs. BALB/c), and the same strain from different vendors.[658] Although mice are the primary species used experimentally for examining the immune response to AKI, there are significant differences when compared with the human immune system.[659] Additionally, there are structural differences between rodents and humans, although pig kidneys are most similar to humans.[657]

These differences need to be taken into consideration when using mice in preclinical models to mimic human AKI. Furthermore, tubular injury and repair and medullary congestion are difficult to compare with human ischemic ATN. In human AKI, pure ischemia alone is seen in the minority of cases, and there is rarely complete cessation of blood flow to the kidneys. The parenteral delivery of prophylactic therapeutic agents is impossible in complete occlusion models. Because oxygen and metabolic substrates are unable to reach the kidney, the generation of ROS and peroxynitrite species, considered to be important mediators of injury, might have a different or delayed role as compared with low-oxygen states in hypoperfusion models. Total blood flow cessation also prevents the degradative products of the ischemic kidney from being washed out. Other factors playing a role in the pathophysiology of AKI, such as inflammatory mediators released from the spleen, ischemic gut, endothelium, and vascular smooth muscle cells, need to be taken into consideration in any experimental model. Bowel proteins released into the circulation can act as inflammatory mediators and increase the susceptibility to AKI.[660] Others have shown that short-chain fatty acids derived from gut bacteria prevent AKI[661] or germ-free conditions render mice more susceptible to AKI.[662,663] The S3 segment of the proximal tubule is characterized by severe necrosis in clamp models, a finding seen rarely in human AKI. Human biopsies, however, rarely sample the outer medulla, where most of the injury is thought to occur. In contrast to animal models, human AKI histologic biopsy data are lacking at early time points from the onset of insult.[656] Thus the NIH/NIDDK KPMP will be vital in this regard.

The cold ischemia–warm reperfusion model resembles human kidney transplantation but is inadequately studied and experimentally difficult to perform. In the isolated perfused kidney model, the kidney is perfused ex vivo using perfusates with and without erythrocytes, and the model uses ischemic (stopping perfusate) or hypoxic (reduced oxygen tension of erythrocytes) to induce functional impairment. The morphologic patterns are different in erythrocyte-free and erythrocyte-rich perfusates. The latter system is more comparable with what is observed histologically in animal models. Additionally, limitations include exclusion of various inflammatory mediators, neuroendocrine hemodynamic regulation, and systemic cytokine and growth factor interactions known to be present and that play a pathophysiologic role in animal models and likely in human ischemia.

Cardiac arrest commonly leads to AKI. Burne-Taney and colleagues[664] have described a whole-body ischemia-reperfusion injury (IRI) model induced by 10 minutes of cardiac arrest, followed by cardiac compression resuscitation, ventilation, epinephrine, and fluids that lead to a significant rise in serum creatinine level and renal tubular injury at 24 hours. One of the unique advantages of this model is the crosstalk among vital organs such as the brain, heart, and lung and renal hemodynamics.[607] A hypoperfusion model of AKI using partial aortic clamping, first described by Zager,[665] may be more representative of human AKI, reflecting a state of reduced blood flow to the kidney, with systolic blood pressure of approximately 20 mm Hg, resulting in reproducible AKI.[665]

Toxic models of kidney failure use various known toxins, such as radiocontrast media, gentamicin, cisplatin, glycerol, and pigments including myoglobin, folic acid, and hemoglobin. Septic models to study AKI include CLP, endotoxin infusion, and bacterial infusion into the peritoneal cavity. The endotoxin model, which is simple, inexpensive, and suitable for studying new pharmacologic agents, has certain drawbacks as well. There is variability among sources of lipopolysaccharide (LPS) endotoxin, the rates and methods of administration vary, and it is usually of short duration due to the high mortality associated with the doses required to induce AKI. It also tends to be a vasoconstrictive model and does not recapitulate the early hemodynamics or inflammation of human sepsis.[666] Wichterman and colleagues[667] were the first to describe a sepsis model in the early 1980s using the CLP laboratory model. In the CLP model, there is considerable similarity to sepsis in humans, with ALI, metabolic derangement, and systemic vasodilation, accompanied initially by increased cardiac output. However, there is some variability depending on the mode and size of cecal perforation. Doi and coworkers[668] have developed a sepsis model that considers the following: 1. animals should receive the same supportive therapy that is standard for ICU patients (fluid resuscitation and antibiotics); and 2. age, chronic comorbid conditions, and genetic heterogeneity vary. Another example is a hyperdynamic form of sepsis established in Merino sheep. These studies have provided important new information that cannot be derived from rodent models of sepsis.[669] Complex animal models of human sepsis that introduce these disease-modifying factors are likely more relevant and may be more pharmacologically relevant than simple animal models.[668]

This description is intended to remind the reader of the potential pitfalls in each model when evaluating experimental studies or therapeutic interventions using these models. The lack of ability to demonstrate the effectiveness of an agent in humans shown to be efficacious in animal models does not necessarily reflect a flaw with the model or the agent in question. Most often, the agent is administered late in the course of the human disease; patient heterogeneity and the difficulty in stratifying patients by severity of injury make it even more difficult to establish efficacy.[133,139] Further studies have led to the development of zebrafish, 3D microfluidic, and human kidney organoid models that are improving our understanding of AKI or facilitating pharmacologic studies for the treatment of AKI.

Zebrafish

Given the capacity of zebrafish adult nephrons to undergo robust epithelial regeneration and to form nephrons de novo, investigations using zebrafish models of AKI have allowed a better understanding of the cellular mechanisms associated with kidney regeneration after AKI. Using chemical genetics, investigators have discovered that histone deacetylase inhibitors (HDACi) are capable of ameliorating gentamicin-induced AKI in the zebrafish embryo with an expansion of renal progenitors that express the genes *Lhx1a, Pax2a,* and *Pax8*.[670] Zebrafish models have been utilized to validate other mechanisms in AKI that include use of mitochondrial-targeted nanoparticles to target mitophagy,[671] the role of telomerase in regeneration post-AKI using telomerase-deficient zebrafish in aristolochic acid–induced nephropathy[672] or the inhibitors of zebrafish GSDMD (GSDMEb) to downregulate pyroptosis in LPS-induced AKI.[673] Future studies using zebrafish transgenic lines, in which injury can be induced in tubular cells via the nitroreductase-metronidazole system, are likely to help characterize how individual populations of cells in the nephron respond to kidney damage.[674]

Three-dimensional microfluidic models and organoids

Side effects, especially drug-induced nephrotoxicity, can often be an important limiting factor in the development of new pharmacotherapy. In addition to each of the current animal models of AKI having limitations in fully recapitulating human AKI, the lack of ability to predict drug-induced AKI has led to failed drug trials. Early nephrotoxicity screening studies included two-dimensional standard well plates with semipermeable filter cups. There were a number of limitations to two-dimensional standard well plates, including the following: 1. use of cell lines of nonhuman origin (MDCK; LLC-PK 1); 2. use of cell lines with limited proximal tubule characteristics (human kidney 2 [HK 2]); 3. cell lines that have features of epithelial-to-mesenchymal transition (HK 2); and 4. static conditions. By contrast, proximal tubule cells are subject to continuous tubule fluid flow and fluid shear stress that modulate cellular signal transduction through mechanosensing receptors, organization of the cytoskeleton, actin filaments, and adherens junctions.[675-679] More physiologically relevant models, such as 3D microfluidic models, have been developed to bridge the step from standard two-dimensional systems to animal models or as a total replacement for costly, inefficient animal models. These three-dimensional microfluidic models, such as parallel flow-plate models and early 3D perfusion models, include chip technology with cell monolayers or tubular structures that are sandwiched between two microfluidic channels. Recently, Qu and coworkers[680] developed a multilayer microfluidic device to simulate glomerulus, Bowman capsule, proximal tubular lumen, and peritubular endothelial cells to investigate the pathophysiology of drug-induced AKI. The authors were able to demonstrate time- and dose-dependent induction of cell death by cisplatin and doxorubicin. The use of this biomimetic device has yielded useful information about drug-induced AKI at the preclinical stage.[680]

More advanced microphysiologic systems have been developed utilizing a triculture of immortalized podocytes, umbilical vein endothelial cells, and proximal tubule cells that have a recirculation circuit and filtrate output for functional testing. The system mimicked retaining of blood serum protein and reabsorption of glucose and could secrete creatinine, as well as expressed nephron-specific proteins (VE-cadherin, nephrin, and ZO-1). The system also recapitulated drug-induced kidney injuries using cisplatin and doxorubicin (Adriamycin) perfusion to demonstrate serum albumin filtration, glucose clearance, and lactate dehydrogenase release.[681] Another such system using 3D vascularized proximal tubule embedded in a permeable ECM was used to simulate albumin uptake and glucose reabsorption in hyperglycemic conditions. The endothelial cell dysfunction that occurred was shown to be attenuated using a glucose transport inhibitor, thus providing a closed-loop system.[682]

Human kidney organoids are 3D clusters of cells that are functionally and genetically similar to kidney. Human-induced pluripotent stem cells (hiPSCs) are ideally suited for human kidney organoids because of their unlimited self-renewal and ability to generate cells of all three embryonic germ layers. Major challenges to the generation of functional bioengineered kidneys are the incorporation of adequate vascularization to kidney organoids and the establishment of an effective drainage system for the removal of blood filtrate after passage and processing through the tubule system.[683] Protocols that add combinations of growth factors to mimic in vivo conditions and growing cells in 3D cultures have generated nephron progenitor cells and kidney organoids from human embryonic stem cells and hiPSCs.[684-687,688] Such organoid systems could be successfully employed to demonstrate the presence of SARS-CoV-2 tropism in PTC via ACE2-mediated internalization and its blockade using an angiotensin receptor blocker.[689] Kidney organoids have been also adapted in microfluidics as organoids-on-chip to study cellular damage to nephrotoxins and their effects on proximal tubular ion transporters.[690-693]

Cytoskeletal and intracellular structural changes

The cytoskeleton in eukaryotic cells consists of intermediate filaments, microtubules, and actin filaments.[694] The microtubule cytoskeleton is composed of α-tubulin and β-tubulin heterodimers that serve to regulate the shape, motility, and division of TECs. One study has demonstrated that IRI to the kidney causes α-tubular deacetylation in microtubules, and inhibition of microtubule dynamics induced by changes in tubulin acetylation during the recovery phase retards TEC regeneration.[695] Epithelial cellular structure and function are mediated in part by the actin cytoskeleton, which plays an integral role in surface membrane structure and function, cell polarity, endocytosis, signal transduction, cell motility, movement of organelles, exocytosis, cell division, cell migration, barrier function of the junctional complexes, cell-matrix adhesion, and signal transduction.[696] On the basis of its role in this multitude of processes, any disruption of the actin cytoskeleton results in changes and/or disruption of the functions mentioned earlier. This is especially important for PTCs, where amplification of the apical membrane by microvilli is essential for normal cell function.

Ischemic insult results in cellular ATP depletion, which in turn leads to a rapid disruption of the apical actin and disruption and redistribution of the cytoskeleton F-actin core, resulting in the formation of membrane-bound EVs or blebs.[697] These can be exfoliated into the tubular lumen or internalized with the capability of being recycled. The core mechanism of disruption is the depolymerization mediated by actin-binding protein known as actin depolymerizing factor (ADF) or cofilin.[698] This protein family is normally maintained in the inactive phosphorylated form, which cannot bind to actin. Ischemia results in ATP depletion, which has been shown to cause Rho GTPase inactivation.[699] This can lead to activation and relocalization of ADF (cofilin) to the apical membranes, where it can mediate different effects, including depolymerization, severing, capping, and nucleation of F-actin. This destroys the actin filament core structure of microvilli and results in surface membrane instability and blebbing[700,701] (Fig. 27.3).

One study has used human primary TECs in culture to examine the role of hypoxic injury on epithelial cell cytoskeletal organization. Given previous information that stabilization of HIF simulates hypoxic injury during AKI, the authors found that HIF stabilization in human TECs reduces tubular cell migration and induces the rearrangement of actin filaments and cell adhesion molecules, including paxillin and focal adhesion kinase. These data support the role of HIF stabilization during epithelial migration, which underlies a potential mechanism of renal regeneration in response to AKI.[702] Similarly, tropomyosins physiologically bind to and stabilize the F-actin microfilament core in the terminal web by preventing access to ADF. After ischemia, there is dissociation of tropomyosins from the microfilament core, resulting in access of the microfilaments in the terminal web to the binding, severing, and depolymerizing actions of ADF/cofilin.[703,704]

Another important consequence of disruption of the actin cytoskeleton is the loss of tight junctions and adherens junctions. These junctional complexes actively participate in numerous functions including paracellular transport, cell polarity, and cellular shape. The tight junctions, also known as zonula occludens (ZO), are composed of proteins such as occludin, claudin, ZO-1, and protein kinase C with numerous barrier functions, such as adhesion, permeability, and transport. The actin present in the terminal web is linked to ZO, and hence any disruption of the terminal web results in disruption of the tight junctions. Early ischemic injury results in opening of these tight junctions, leading to increased paracellular permeability and causing further back leak of the glomerular filtrate into the interstitium.[696] In the glomerulus, ischemia also induces a rapid loss of interaction between slit diaphragm junctional proteins NEPH1 and ZO-1,[144] leading to podocyte damage, effacement, and proteinuria.

The molecular mechanisms underlying these changes have been studied as well and show that ATP depletion that results in actin polymerization is followed by a reduction in cellular adhesion ability. Pretreatment with an actin stabilizer prevented ATP depletion–induced actin polymerization and reduction of cell adhesion, indicating that the cytoskeleton reorganization decreased the cellular adhesion ability. Furthermore, the ATP depletion markedly increased the levels of p38 mitogen-activated protein (MAP) kinase and heat shock protein 27 (hsp27) phosphorylation, with enhanced translocation of phosphorylated hsp27 from cytoskeleton to cytoplasm. The inhibition of p38 MAP kinase by specific inhibitor SB203580 blocked the ATP depletion to induce hsp27 phosphorylation and actin polymerization. These findings suggest that ischemia remodels F-actin, leading to desquamation of proximal TECs through p38 MAP kinase–hsp27 signaling.[705]

Actin cytoskeleton alterations and dysfunction during ischemia result in changes in cell polarity and function. Normally, sodium-potassium adenosine triphosphatase (Na^+-K^+-ATPase) pumps reside in the basolateral membrane of the TEC, but under conditions of ischemia, they redistribute to the apical membrane as early as within 10 minutes.[706] This process occurs due to the disruption of the pumps' attachment to the membrane via the spectrin/actin cytoskeleton. Postulated mediated mechanisms include hyperphosphorylation of the protein ankyrin, with consequent loss of the binding protein spectrin and cleavage of spectrin by the activation of proteases such as calpain. This redistribution of the Na^+-K^+-ATPase pump results in bidirectional transport of sodium and water across the epithelial cell apical membrane, as well as the basolateral membrane. This results in transport of cellular Na back into the tubular lumen, one of the major mechanisms of the high FE_{Na} seen in patients with ischemic ATN,[707] and the inefficient use of cellular ATP. ATP is consumed but effective, and vectorial Na transport is lost.

AUTOPHAGY

- Macroautophagy starts with the de novo formation of a cup-shaped isolation double membrane that engulfs a portion of cytoplasm.
- Microautophagy involves the engulfment of cytoplasm instantly at the lysosomal membrane by invagination, protrusion, and separation.
- Chaperone-mediated autophagy is a process of direct transport of unfolded proteins via the lysosomal chaperonin hsc70 and the receptor for chaperone-mediated autophagy, LAMP-2A.

Autophagy is highly conserved and is a lysosomal degradation pathway for the salvation and reuse of degraded proteins, lipids, and organelles to generate basic structural components and energy. It also removes obsolete and dysfunctional organelles and cytotoxic protein aggregates for cellular homeostasis. Under pathogenic conditions, autophagy plays a key role in ischemic, toxic, immunologic, and oxidative insults that lead to the induction of autophagy in renal epithelial cells, which plays a key role in altering the course of disease.[202] Autophagy-mediated leukocyte clearance is an important mechanism for resolving inflammation; it is a function performed by macrophages.[203] Autophagy is characterized by the formation of autophagosomes, double membrane vesicles wrapping cytosolic components and organelles that are destined for degradation. Autophagosomes fuse with lysosomes, the contents are degraded, and the products are salvaged and serve as building blocks for cellular function.[201] There are a number of genes and proteins associated with autophagy (autophagy-related; Atg).[201]

COMPLEMENT

The immunomodulatory functions of complement are mediated through three canonical pathways of activation—classic pathway, alternative pathway, and lectin pathway. The classic pathway is initiated by binding of PRPs to immune complexes, whereas the lectin pathway is initiated by binding PRPs to carbohydrate structures exposed in injured cells. The alternative pathway amplifies the initial response and maintains a low level of activity via a tick-over mechanism. Complement activation leads to cleavage and deposition of C3b in tissues, which cleaves C5 to generate C5b that initiates the formation of the pore-forming membrane attack complex (MAC) that lyses susceptible microorganisms or damages cells. The C3a and C5a fragments are anaphylatoxins that cause recruitment of immune cells in addition to smooth muscle contraction, vasodilation, histamine release, enhanced vascular permeability, and generation of ROS including in AKI.[376,708,709] C3a and C5a receptors are expressed on leukocytes, endothelial cells, mesangial cells, and TECs.[710]

Heat shock proteins

Much of the previous discussion has been on proteins or mechanisms that promote injury. However, there are protective mechanisms that allow cells to defend against numerous stresses. The complex heat shock protein (HSP) system is induced to exceptionally high levels during stress conditions. HSPs are believed to facilitate the restoration of normal function by assisting in the refolding of denatured proteins and the appropriate folding of newly synthesized proteins and in the degradation of irreparable proteins and toxins to limit their accumulation. Overexpression of HSPs before injury is protective.[711-713] The proteins HSP90, HSP72, and HSP25 in particular have been extensively studied (e.g., overexpression of HSP25 is protective against actin cytoskeleton disruption).[714] After renal ischemia, cytosolic HSP90 is rapidly induced in PTCs, particularly in late stages, leading to the conclusion that HSP90 may be crucial for the disposition of damaged proteins and the assembly of newly formed peptides. Intrarenal transfection with HSP90 protects against IRI with the restoration of endothelial nitric oxide synthase (eNOS)–HSP90 coupling, eNOS activating phosphorylation, and Rho kinase levels, suggesting that HSPs can regulate the NO–eNOS pathway and intrarenal vascular tone.[715] In nephrotoxic models, HSP72 limits apoptosis through an increased Bcl-2/Bax ratio, implicating HSP72 in cell death as well.[714]

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